Part VI

Interest Rate Derivatives (Advanced)

The book's quantitative core. Chapters 28, 30 and 32 carry the dagger: they are aimed at readers who will implement or validate a rate model. Chapters 29 and 31 do not, and are needed by anyone who prices a rate option at all. Chapter 33 returns to market practice and stands on its own: a course that skips the rest of this part can take it directly after the swap material in Chapter 7. One idea runs through all six: choose the numeraire that makes the awkward part of a payoff disappear.

28 Martingales and Measures
the change-of-numeraire machinery, and why the choice is the valuation

29 Standard Market Models for Rate Derivatives
Black's formulas for caps and swaptions, and where they hold

30 Convexity, Timing, and Quanto Adjustments
three corrections, and how large each one actually is

31 Short-Rate Models
calibration, trees, and the Bermudan swaption

32 HJM, LMM, and Multi-Curve Term Structure Models
the drift condition, factor reduction, and calibration

33 Swaps Revisited
the non-standard book: amortizing, compounding, in-arrears, and inflation

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Chapter 28
Martingales and Measures †

Placed here deliberately: everything in Part VI takes it for granted.

Chapter 29 prices a swaption by treating the annuity as a numeraire, Chapter 30 changes measure three times to derive three adjustments, Chapter 32 derives drift conditions from no-arbitrage, and Chapter 33 needs all of it at once. Reading it first means none of those chapters has to borrow machinery it has not been given.

A reader who wants only the working rules can take Section 28.1 and Section 28.6 and move on. The rest is the justification, and it is here because the vocabulary in Section 28.1 is used by every practitioner in the field and is almost never written down precisely. Anyone who has traded a derivative has heard "risk-neutral probability" and "equivalent martingale measure"; comparatively few could state what either one is, and the terms are not hard—they are merely never defined.

The chapter runs from the vocabulary to what it buys. Sections 28.1 to 28.3 fix the terms: what a measure is, what the market price of risk is the price of, and what changes when uncertainty has more than one source. Sections 28.4 to 28.6 are the machinery: martingales, the change of measure written out explicitly, and the numeraire that names each measure. Sections 28.7 and 28.8 put it to work on why a forward price is an expectation and on the annuity measure from which Chapter 30's CMS adjustment comes. Sections 28.9 to 28.11 restate the familiar results in this language, including what $N(d_1)$ is; Sections 28.12 and 28.13 are where a numeraire fails and what the theory is for.

28.1 THE VOCABULARY, STATED PRECISELY

Seven terms carry the whole subject. Each is stated here in the form a practitioner needs, which is less than a probabilist would want and more than the phrases usually get.

A probability measure P assigns a probability to each event, that is, each set of possible paths the market might follow. Two measures can disagree about probabilities while agreeing about what is possible, and that agreement is what the next definition captures.

Equivalent measures. Two measures P and Q are **equivalent**, written $P \sim Q$, if they agree on which events have probability zero:

$$\text{for every event } A: \quad P(A) = 0 \iff Q(A) = 0. \tag{28.1}$$

They may disagree about how likely anything is, but not about what can happen. This is the precise sense in which a change of measure reweights the world without inventing or destroying scenarios, and it is exactly what a no-arbitrage argument needs, because an arbitrage is a payoff that is impossible to lose on, and impossibility must survive the change.

A self-financing portfolio is one whose value changes *only* because the prices of its holdings change: no cash is added and none is withdrawn after it is set up. Rebalancing is permitted, provided every purchase is paid for by a sale. The condition is what makes a replication argument an argument about prices, rather than about someone's willingness to keep funding a position, and it is why a payoff can be replicated at a cost that is known today. It is also the condition hidden in the next definition: an asset qualifies as a numeraire only if its quoted price is the value of a self-financing position, which is why Section 28.12 has to disqualify a stock that pays dividends.

A numeraire is any traded asset with a strictly positive price, used as the unit in which other prices are quoted. Which one is chosen is not innocuous, and the freedom to choose is the technique that Part VI lives on: pick the numeraire that makes the payoff you are pricing simple, and the hard part of the calculation disappears into the choice.

A martingale is a process with zero drift: a process X is a martingale under Q if

$$E^{Q}[X(T) \mid \mathcal{F}_t] = X(t) \qquad \text{for all } t \le T, \tag{28.2}$$

where $\mathcal{F}_t$ is everything observable by time t, the **information** available then. The best forecast of a martingale's future value is its present value. A martingale is not a process that cannot move; it is one whose movements are unforecastable.

An equivalent martingale measure for a numeraire N is a measure $Q^N \sim P$ under which *every* traded asset price, divided by N, is a martingale:

$$\boxed{\frac{V(t)}{N(t)} = E^{N}\left[\frac{V(T)}{N(T)}\,\middle|\,\mathcal{F}_t\right]} \qquad \text{for every traded } V. \tag{28.3}$$

In a market free of arbitrage every numeraire has one, which is what makes the choice above a choice of measure. The word "equivalent" carries a condition and is usually skipped. It is what forbids the measure from assigning zero probability to a state the real world admits—without it, one could price away a genuine risk by declaring it impossible.

The risk-neutral measure is the equivalent martingale measure associated with one specific numeraire: the money market account $B(t)$, which grows at the instantaneous short rate. Under it,

$$V(0) = E^{Q}\left[e^{-\int_0^T r_s\,ds}\,V(T)\right],$$

and every asset has expected return r. The name misleads, so state what it is not. It is not a claim that investors are indifferent to risk, and the probabilities in it are not anyone's forecast. It is a pricing tool rather than a description of the world: a set of weights under which discounted prices have no drift, so that expectation becomes a valid pricing operation.

One sentence connects all seven. A price is a numeraire times an expectation, and the measure is whatever makes that true. Everything after this section proves that statement, draws a consequence from it, or chooses the numeraire well.

28.2 THE MARKET PRICE OF RISK

Start with a quantity that unifies everything after it.

Suppose two assets depend on the same single source of uncertainty dW:

$$\frac{dg}{g} = \mu_g\,dt + \sigma_g\,dW, \qquad \frac{dh}{h} = \mu_h\,dt + \sigma_h\,dW$$

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No-arbitrage forces the two to offer the same compensation per unit of risk:

$$\frac{\mu_g - r}{\sigma_g} = \frac{\mu_h - r}{\sigma_h} \tag{28.4}$$

The common value is the **market price of risk** λ carried by the source dW, and the proof that the two must agree is a portfolio.

Example 28.1 What happens if they differ

Let $r = 3\%$, and take

$$\text{A} : \mu = 8\%,\ \sigma = 20\% \Rightarrow \lambda_A = 0.25, \qquad \text{B} : \mu = 6\%,\ \sigma = 15\% \Rightarrow \lambda_B = 0.20$$

Hold one unit of A and $-\sigma_A/\sigma_B = -1.3333$ units of B. The dW terms cancel exactly, so the portfolio is riskless. Its return is

$$1.0000 \times 8\% - 1.3333 \times 6\% = 0.00\%$$

and its cost is

$$1.0000 - 1.3333 = -0.3333$$

A portfolio with a negative cost (a net borrowing of 0.3333) and a zero return. Financing it at 3% costs $0.03 \times (-0.3333) = -0.0100$, so the position earns **1.00% of the notional of A, riskless, per year**.

Scale it and the profit is unbounded, which is what an arbitrage means. The requirement $\lambda_A = \lambda_B$ then pins B's expected return: at $\lambda = 0.25$ and $\sigma_B = 15\%$, μ_B must be $3\% + 0.25 \times 15\% = \mathbf{6.75}\%$.

This is the whole of asset pricing theory in one calculation, and everything in this chapter is a restatement of it. Chapter 13's risk-neutral valuation is the case $\lambda = 0$; the measure changes of Sections 28.6 and 28.9 are the general case.

The market price of risk is a property of the risk, not of the asset. Two assets exposed to the same dW must carry the same λ whatever else differs about them: their maturity, their issuer, their payoff structure. In Chapter 31's one-factor models every bond carries the market price of interest rate risk, which is why a single λ suffices there and why the whole curve is determined by one process.

28.3 SEVERAL SOURCES OF UNCERTAINTY

Let $W_1, \ldots, W_n$ be independent Wiener processes, and let the state of the economy be carried by n variables $\theta_1, \ldots, \theta_n$ (**factors**, in the asset pricing literature) following

$$\frac{d\theta_i}{\theta_i} = m_i\, dt + s_i\, dW_i, \qquad i = 1, \ldots, n$$

where m_i and s_i may depend on the factors and on time. The factors need not be traded, and usually are not: a short rate, an inflation index, a volatility, a default intensity. The force of the assumption is that nothing else moves: every price in this economy is a function of t and of those n numbers.

So let f be the price of a traded asset, $f(t) = F(t, \theta_1(t), \ldots, \theta_n(t))$. Itô's lemma in n variables (Section 14.11 states the two-variable case) then delivers the dynamics rather than assuming them,

$$\frac{df}{f} = \mu\, dt + \sum_{i=1}^{n} \sigma_i\, dW_i, \qquad \sigma_i = \frac{\theta_i}{F}\frac{\partial F}{\partial \theta_i}\, s_i$$

where μ collects the time and second-order terms, and no cross terms survive because the W_i are independent. Read σ_i as the asset's exposure to factor i: the elasticity of its price in that

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factor, times that factor's own volatility. Such an asset must satisfy

$$\mu - r = \sum_{i=1}^{n} \lambda_i \sigma_i \tag{28.5}$$

where λ_i is the market price of risk for dW_i. The claim being made is that the λ_i are the same numbers for every asset. Applied to one asset in isolation the equation restricts nothing; it merely names n quantities. Applied to all of them at once it is the whole of the theory.

The proof is Example 28.1 with more bookkeeping. There, one asset hedged another and the riskless residual was forced to earn r. Here n assets hedge one.

Begin with n traded assets that are **non-redundant** (no one of them is a portfolio of the others) and whose exposures **span** the risks,

$$\frac{df_j}{f_j} = \mu_j \, dt + \sum_{i=1}^{n} \sigma_{ji} \, dW_i, \qquad j = 1, \ldots, n$$

spanning meaning that the matrix of loadings $[\sigma_{ji}]$ has rank n, hence is invertible: no risk is unhedgeable, and no asset duplicates the others. Define $\lambda_1, \ldots, \lambda_n$ as the solution of

$$\mu_j - r = \sum_{i=1}^{n} \lambda_i \, \sigma_{ji}, \qquad j = 1, \ldots, n$$

n linear equations in n unknowns, solvable and uniquely so because the loadings are invertible. Nothing has been assumed at this point. The λ_i are simply being read off the market: they record what these n assets currently pay per unit of each risk.

Now take *any* other asset in the economy, with drift μ and exposures $\sigma_1, \ldots, \sigma_n$, and hedge it. Hold one dollar of it, and $-w_j$ dollars of each f_j, with the w_j chosen so that every exposure is matched:

$$\sum_{j=1}^{n} w_j \, \sigma_{ji} = \sigma_i, \qquad i = 1, \ldots, n$$

solvable, and uniquely, for the same reason as before. Every dW_i term now cancels and the portfolio is riskless. It costs $1 - \sum_j w_j$ to establish and drifts at $\mu - \sum_j w_j \mu_j$, so no-arbitrage requires the drift to be exactly r times the cost:

$$\mu - \sum_j w_j \, \mu_j = r \left(1 - \sum_j w_j \right)$$

Collecting the interest terms leaves a statement about excess returns alone,

$$\mu - r = \sum_j w_j \, (\mu_j - r)$$

and the remainder is substitution. Replace each $\mu_j - r$ by the defining relation above, exchange the order of summation, and the inner sum is the hedge condition:

$$\mu - r = \sum_j w_j \sum_i \lambda_i \, \sigma_{ji} = \sum_i \lambda_i \sum_j w_j \, \sigma_{ji} = \sum_i \lambda_i \, \sigma_i$$

which is (28.5).

Notice what the argument never did. The λ_i were pinned down by the n spanning assets and then never revisited, so they cannot depend on which asset was subsequently tested against them. The one-factor conclusion, that λ belongs to the risk and not to the asset, survives intact, and it survives for the same reason.

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Suppose (28.5) fails. If some asset's drift exceeds $r + \sum_i \lambda_i \sigma_i$ by ε, the portfolio just built is riskless, costs nothing once the interest leg is financed, and earns ε per dollar per year. Reverse it if the drift falls short. Scale it and the profit is unbounded, so no asset can sit off the hyperplane (28.5) traces out.

Now count the assets against the risks. The argument used n non-redundant traded assets against n independent Wiener processes; with only $k < n$ of them non-redundant, the defining relations become k equations in n unknowns. Solutions still exist (no-arbitrage puts excess returns in the span of the loadings whatever the market's structure), but they form an $(n - k)$-dimensional family, every member of which prices the traded assets correctly, so no observed price singles one out. The rule is that *the market prices of risk are unique exactly when the traded assets are at least as numerous as the sources of uncertainty they must hedge*, counting only the non-redundant ones. Below that count, one member of that family is chosen rather than read off the market; at or above it, every further asset must price consistently with the λ_i already pinned down, or be an arbitrage. That is the incomplete-market case. A market is complete (Section 13.4) when every payoff can be replicated by trading the assets it has, and incomplete when some cannot, and the arbitrariness that the **second fundamental theorem** of asset pricing describes (an arbitrage-free market is complete exactly when its equivalent martingale measure is unique) enters here, at the market price of risk. [TN28] states both theorems and proves them.

Two of the model families in this book sit below that count. Chapter 21's stochastic volatility models add a second Wiener process driving the variance, and variance is not traded: two risks, one traded underlying, so the price of variance risk cannot be read off the market. It is chosen. Heston takes it proportional to the variance v, which keeps the variance in the same square-root class under Q, with Chapter 21's mean-reversion speed κ and long-run variance θ becoming $\kappa + \lambda$ and $\kappa\theta/(\kappa + \lambda)$. This is a modeling decision made for tractability, not a defect. It costs little: calibrating to option prices delivers the risk-neutral parameters directly, so the split between them and λ never has to be made, which is why the model is written under Q from the start. The premium itself does not disappear (it is Section 19.10's volatility risk premium), and once an option trades, its price supplies what the count could not. One traded option completes the market.

Chapter 31's short-rate models are the same case, and the same kind of choice is made there. The short rate drives everything and is not itself traded, so the price of rate risk is specified rather than observed: constant in Vasicek, proportional to $\sqrt{r}$ in Cox–Ingersoll–Ross, each form picked so that the process keeps its class under Q, Gaussian staying Gaussian and square root staying square root. Writing the model under Q from the start, as Chapter 31 does, is that choice already made, and calibrating to today's curve means the real-world parameters and λ never have to be separated.

The economic content of (28.5) is that risks price additively. An asset's excess return is the sum of its exposures times the prices of those exposures, which is the structure of every factor model in asset pricing, arrived at here from no-arbitrage rather than from equilibrium.

And it identifies what a model must supply. Chapter 32's K-factor Heath–Jarrow–Morton (HJM) model needs K market prices of risk to move from the real world to the risk-neutral one. Section 32.2's drift condition is exactly the statement that this move has been made, which is why the drift is not free.

28.4 MARTINGALES

Section 28.1 defined a martingale through its conditional expectation. In the continuous-time notation of Chapter 14 the same condition reads

$$d\theta = \sigma\, dW$$

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a process with no dt term. Zero drift and $E[\theta_T] = \theta_0$ say the same thing under an integrability condition that holds throughout this book, and whose failure [TN28] describes.

The reason martingales matter is that they turn valuation into an expectation. If some ratio V/N is a martingale, then

$$\frac{V_0}{N_0} = E\left[\frac{V_T}{N_T}\right] \quad \Longrightarrow \quad V_0 = N_0\, E\left[\frac{V_T}{N_T}\right]$$

and the pricing problem is solved provided the expectation can be computed. The ratio is a martingale only when V and N are both self-financing, a condition Section 28.12 returns to.

Which ratios are martingales, and under which probability measure, is the entire subject of this chapter.

28.5 THE CHANGE OF MEASURE, MADE EXPLICIT

Before the general theory, it is worth seeing a measure change as an actual list of numbers. A two-period binomial tree is enough: in Figure 28.1 the nodes are fixed, the middle terminal node is reached two ways, and only the branch probabilities change with the measure.

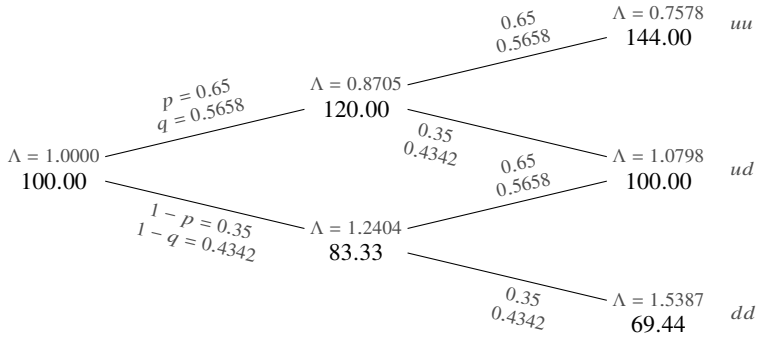


Figure 28.1: Example 28.2's tree: the real-world and risk-neutral probability on each branch, and Λ above the price at each node.

Example 28.2 The Radon–Nikodym derivative in four numbers

$S_0 = 100$, $u = 1.2$, $d = 1/1.2 = 0.8333$, $r = 4\%$, annual steps. The real-world probability of an up move is $p = 0.65$, an assumption about the world and one no one can observe directly.

The risk-neutral probability is not an assumption. It is forced:

$$q = \frac{e^{r\Delta t} - d}{u - d} = 0.565848$$

Both measures assign probabilities to the same three terminal nodes. The ratio of the two is the Radon–Nikodym derivative $\Lambda = dQ/dP$:

Path	$S(2)$	P	Q	Λ
dd	69.4444	0.122500	0.188488	1.538680
ud	100.0000	0.455000	0.491328	1.079842
uu	144.0000	0.422500	0.320183	0.757831
		1.000000	1.000000	$E^P[\Lambda] = 1.000000$

Three facts in that table are the whole of the theory.

$E^P[\Lambda] = 1$ exactly, and every value is positive: no path P allows is impossible under Q, which is what equivalence means. Together these are what make Λ a legitimate change of probability rather than an arbitrary reweighting.

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The reweighting is monotone in the outcome. Good states are weighted *down* (0.7578) and bad states *up* (1.5387). The direction is not automatic: it holds because $p > q$, which is this tree's way of saying the stock carries a risk premium, $e^{-2r}E^P[S(2)] = 106.02$ against $S_0 = 100$. That is what the market price of risk means, expressed as probabilities: a dollar delivered in a bad state is worth more to an investor than a dollar delivered in a good one, so a valuation that discounts every state at r has to make up the difference by treating bad states as more likely than they are. Were λ zero the two measures would coincide, and a negative premium would tilt dQ/dP the other way.

And the expectation comes out right: $e^{-2r}E^Q[S(2)] = 100.0000 = S_0$, so the discounted stock price is a Q-martingale.

Now price a two-year call struck at 100:

$e^{-2r}\,E^P[\text{payoff}]$	17.160733
$e^{-2r}\,E^Q[\text{payoff}]$	13.004930
Error from using the wrong measure	**+32.0**%

Same nodes, same payoffs—only the probabilities differ, and the price is wrong by a third.

In general the object has a name, and a theorem behind it. Equivalence says P and Q agree on which events are impossible, so neither puts weight where the other puts none; the **Radon–Nikodym theorem** says that this is enough for Q to be P reweighted by a single random variable. That variable is the **Radon–Nikodym derivative** $\Lambda = dQ/dP$, also called the **density** of Q with respect to P, positive, with

$$Q(A) = E^P[\Lambda\,\mathbf{1}_A] \quad \text{for every event } A, \qquad E^P[\Lambda] = 1. \tag{28.6}$$

The unit mean is forced by taking A to be everything, and expectations follow the same rule:

$$E^Q[X] = E^P[\Lambda X] \quad \text{for any payoff } X. \tag{28.7}$$

Changing measure is multiplying by Λ before averaging, never a change in which outcomes are possible. On a tree Λ is a list, one number per path (the list above); in continuous time it is a process, $\Lambda_t = E^P[\Lambda \mid \mathcal{F}_t]$, and Section 28.9 identifies which one. That conditional expectation is concrete. Λ is known only at the end, once the path is complete; Λ_t is its P-average over the paths still open at t. On the tree above, it is the probability-weighted average of the Λ values of the paths through the current node.

At $t = 0$ nothing is ruled out, so $\Lambda_0 = E^P[\Lambda] = 1$; at T one path is left and $\Lambda_T = \Lambda$. At the upper $t = 1$ node the open paths are uu and ud, so $\Lambda_1 = 0.65 \times 0.757831 + 0.35 \times 1.079842 = 0.870535$, which is q/p; the lower node gives $(1-q)/(1-p) = 1.240436$. On a tree the density is the running product of one-step ratios. It is a P-martingale by the **law of iterated expectations** (also called the tower property), which says that averaging a later conditional expectation over what is known earlier gives the earlier one, $E[E[X \mid \mathcal{F}_s] \mid \mathcal{F}_t] = E[X \mid \mathcal{F}_t]$ for $t \le s$ (here $0.65 \times 0.870535 + 0.35 \times 1.240436 = 1$ again), and it reweights any event settled by t: $Q(A) = E^P[\Lambda_t\,\mathbf{1}_A]$ for A in $\mathcal{F}_t$.

28.6 NUMERAIRES AND THE MEASURES THAT GO WITH THEM

Prices measured in units of the numeraire are what we make martingales, and the central result is that, in a market free of arbitrage, the measure to do it in always exists: for any numeraire N there is an equivalent measure Q^N under which V/N is a martingale for every traded asset V, which is (28.3). What matters is not that the measures exist but how to move between them, and that is one line.

The change-of-numeraire theorem. Let N and M be two numeraires with associated measures Q^N and Q^M. Their density, restricted to the information $\mathcal{F}_t$

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available at time t (the bar is read that way), is

$$\left.\frac{dQ^M}{dQ^N}\right|_{\mathcal{F}_t} = \frac{M(t)\,/\,M(0)}{N(t)\,/\,N(0)}. \tag{28.8}$$

The left side is the density process Λ_t of Section 28.5, written for this pair of measures. The proof is three lines and worth seeing: it shows the theorem restates the definition rather than adding a fact. For any traded V, the defining property of Q^M gives $V(0)/M(0) = E^M[V(t)/M(t)]$. Rewrite that expectation under Q^N by (28.7), with (28.8) as the density:

$$E^M\left[\frac{V(t)}{M(t)}\right] = E^N\left[\frac{V(t)}{M(t)}\cdot\frac{M(t)/M(0)}{N(t)/N(0)}\right] = \frac{N(0)}{M(0)}\,E^N\left[\frac{V(t)}{N(t)}\right] = \frac{N(0)}{M(0)}\cdot\frac{V(0)}{N(0)},$$

which is $V(0)/M(0)$, as required. The ratio in (28.8) is positive, both numeraires being strictly positive, and it is a Q^N-martingale with unit expectation, so it is a valid density. ■

One specialization generates most of Part VI. Taking $N = P(\cdot, T_2)$ and $M = P(\cdot, T)$, two zero-coupon bonds maturing in either order,

$$\left.\frac{dQ^T}{dQ^{T_2}}\right|_{\mathcal{F}_t} = \frac{P(t,T)\,/\,P(0,T)}{P(t,T_2)\,/\,P(0,T_2)}, \tag{28.9}$$

and because this density is a function of the interest rate path, (28.7) gives, for any X measurable at t,

$$E^T[X] = E^{T_2}\left[X\cdot\frac{dQ^T}{dQ^{T_2}}\right] = E^{T_2}[X] + \mathrm{Cov}^{T_2}\left(X, \frac{dQ^T}{dQ^{T_2}}\right), \tag{28.10}$$

since the density has unit mean. **Every adjustment in Chapter 30 is an application of** (28.10), with the relevant pair of numeraires in place of the two bonds. And the condition for no adjustment falls straight out: if X is uncorrelated with rates, the covariance vanishes and the two expectations agree. That is why a convexity, timing, or quanto correction is always a covariance, and why it is always zero when the variable is independent of the numeraire ratio: the discount rate for a convexity or timing correction, the exchange rate for a quanto.

Four numeraires account for essentially all of practice, and it is worth tabulating what each is for.

Numeraire	Measure	What becomes a martingale	Used for
Money market	Risk-neutral	V/B, $B(t) = \exp\left(\int_0^t r_s\,ds\right)$	Simulation
Zero-coupon bond $P(t,T)$	T-forward	Any T-maturity forward	Single payoffs
Swap annuity $A(t)$	Swap	The forward swap rate	Swaptions, CMS
Foreign money market	Foreign risk-neutral	Foreign assets	Quantos, FX

Two remarks are easy to miss.

The choice of numeraire does not change any price. That is the content of the theorem, and Section 31.7 verifies it numerically, because the invariance is worth seeing rather than accepting.

But it changes the difficulty enormously, and that is the entire practical point. A quantity that requires a simulation under one numeraire may have a closed form under another.

28.7 WHY THE FORWARD PRICE IS AN EXPECTATION

Chapter 5 derived the forward price from arbitrage, in the form $F(t,T) = S_t/P(t,T)$ that holds whatever rates do; Chapter 30 will open by asserting that a forward price is an expectation under the right measure. This section makes "the right measure" precise, and shows what goes wrong under the wrong one.

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Under the T-forward measure (numeraire $P(t,T)$), the forward price, which equals the asset divided by $P(t,T)$, is a martingale. For a stock,

$$\frac{S_0}{P(0,T)} = E^T\left[\frac{S_T}{P(T,T)}\right] = E^T[S_T]$$

since $P(T,T) = 1$. So $F(0,T) = E^T[S_T]$, exactly, which is the result Chapter 30 uses throughout.

While the forward price is a martingale under the T-forward measure, the futures price is a martingale under the risk-neutral measure, where the money market account is the numeraire. This can be shown as follows. The money market account $B(t)$ follows

$$dB = rB\,dt$$

Write $H(t)$ for the futures price for delivery at T. A futures contract costs nothing to enter and pays dH in cash as it accrues, so consider holding $B(t)$ contracts at every t and keeping the proceeds in the account. The strategy starts from nothing, and its value $V(t)$ is the balance in the account, earning interest r. Since a futures position is itself worth zero once it has been marked to market, the change in V can be written as

$$dV = \underbrace{rV\,dt}_{\text{interest on the balance}} + \underbrace{B\,dH}_{\text{settlement on } B(t) \text{ contracts}}$$

Those are the only cash flows there are, which is what makes the strategy self-financing. Dividing through by the account,

$$d\left(\frac{V}{B}\right) = dH, \qquad \text{so} \qquad \frac{V(t)}{B(t)} = H(t) - H(0)$$

Section 28.6 makes V/B a Q-martingale, because V is the value of a self-financing portfolio of traded assets; so the futures price H is a Q-martingale too. At delivery $H(T) = S_T$, which gives the pair

$$\text{futures price} = E^Q[S_T], \qquad \text{forward price} = E^T[S_T] \tag{28.11}$$

Two prices for the same delivery, two measures, and the difference between them is the futures–forward convexity adjustment of Section 30.3. Section 5.12 reaches it without the machinery, by asking when the variation margin arrives.

28.8 THE ANNUITY MEASURE, AND WHERE CMS COMES FROM

The third numeraire in the table is the one that is least intuitive and does the most work.

A swap that starts at T_0 pays its fixed leg on dates $T_1, \ldots, T_N$, and the **annuity**

$$A(t) = \sum_{i=1}^{N} \tau_i\, P(t,T_i)$$

is the present value of the fixed leg per unit of fixed rate on unit notional, τ_i being the year fraction of the ith period. It is a portfolio of zero-coupon bonds, hence a traded asset, hence an admissible numeraire.

The forward swap rate is

$$S(t) = \frac{P(t,T_0) - P(t,T_N)}{A(t)}$$

a traded asset divided by the annuity, so it is a martingale under the annuity measure Q^A. That is the entire justification for Section 29.6.

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Now notice what a swaption's payoff looks like. It is $A(T_0)\max(S(T_0)-K,0)$: the annuity appears in the payoff. Valuing it with A as numeraire,

$$V_0 = A(0)\,E^A[\max(S(T_0)-K,0)]$$

and the annuity cancels. With S a martingale and lognormal, Black's formula follows exactly, which is why the swaption convention works as well as it does.

That cancellation is this chapter's thesis in its cleanest form. Nothing was approximated and no model was chosen—a numeraire was picked so that the awkward part of the payoff divided out. Most of Part VI consists of cases where the cancellation is imperfect, and what fails to cancel is precisely what has to be computed.

The nearest such case differs from a swaption only in how the rate is paid, and it recurs throughout Part VI. In a **constant-maturity swap (CMS)**, one leg pays a long swap rate (the ten-year rate, say) observed afresh on each reset date, against a fixed rate or a short floating rate on the other leg. The floating leg of an ordinary swap references a rate whose own life matches the period it accrues over: the three-month rate set in January covers January to April, and is paid at the end of it. A CMS leg breaks that alignment deliberately. The ten-year rate set in January is paid three months later, on a quarterly accrual, and the ten-year swap it is the rate *of* never happens.

Investors buy the leg for the exposure it isolates: a coupon that follows long rates without the maturity, the duration, or the balance-sheet consequences of holding a ten-year bond, and dealers hedge the resulting book in swaptions. It is also the building block of steepener notes, which pay a multiple of a CMS spread, the difference between two CMS rates of different tenors, such as the ten-year rate less the two-year.

The coupon pays $S(T_0)$ on a *single* date T_p rather than over the annuity, so the numeraire it calls for is the zero-coupon bond $P(\cdot,T_p)$ rather than A. Under that measure S is no longer a martingale, the annuity no longer cancels, and what is left behind is the covariance between S and the ratio of the two numeraires.

That covariance is the CMS convexity adjustment, which Section 30.5 computes, sizes, and draws. What belongs here is where it comes from: it is not a correction for a modeling deficiency and not an empirical fudge, but the price of quoting a rate that is a martingale under one measure in a contract that settles under another.

Even its sign follows from the cancellation that failed. A swap rate that rises raises the coupon linearly but lowers the value of the annuity the coupon would have been paid over, and the CMS contract keeps the first effect while discarding the second—so the adjustment is positive. The holder of a CMS coupon has been handed that convexity, and the adjustment is what it costs.

28.9 GIRSANOV

Changing the measure changes the drift and leaves the volatility alone.

Formally: let dW^N be a Wiener process under Q^N with correlation ρ to the numeraire ratio M/N, whose volatility is $\sigma_{M/N}$. Moving to Q^M,

$$dW^M = dW^N - \rho\,\sigma_{M/N}\,dt$$

is a Wiener process under Q^M. It is a new process and not the old one relabeled (the two differ by that drift), and only the part of dW^N that moves with the numeraire is shifted, which is where ρ comes from. Every process therefore picks up a drift equal to the covariance between its own logarithm and the log of the numeraire ratio, and a process uncorrelated with that ratio keeps the drift it had. That single sentence generates Chapter 30's three adjustments and Chapter 32's two drift conditions, each derived where it is used.

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The simplest case shows the whole mechanism. Take two assets following geometric Brownian motions under the risk-neutral measure, with correlation ρ between their Wiener processes:

$$\frac{dS_1}{S_1} = r\,dt + \sigma_1\,dW_1, \qquad \frac{dS_2}{S_2} = r\,dt + \sigma_2\,dW_2$$

Now elect S_2 as the numeraire. The numeraire ratio is S_2/B, whose volatility is σ_2, so the rule above gives S_1 a drift equal to its covariance with $\ln(S_2/B)$, which is $\rho\sigma_1\sigma_2$, and gives S_2 its own variance σ_2^2:

$$\frac{dS_1}{S_1} = (r + \rho\sigma_1\sigma_2)\,dt + \sigma_1\,dW_1^{S_2}, \qquad \frac{dS_2}{S_2} = \left(r + \sigma_2^2\right)dt + \sigma_2\,dW_2^{S_2}$$

The two Wiener processes are shifted by different amounts, $dW_1^{S_2} = dW_1 - \rho\sigma_2\,dt$ and $dW_2^{S_2} = dW_2 - \sigma_2\,dt$, because S_2 moves with itself perfectly and with S_1 only through ρ.

Both drifts moved and neither volatility did. σ_1 and σ_2 multiply the new Wiener processes unchanged; what changed is the rate at which each asset is expected to grow, and by exactly the covariance the rule names.

The check is that the ratio is now a martingale. Itô's lemma applied to $X = S_1/S_2$ gives it a drift of

$$(r + \rho\sigma_1\sigma_2) - \left(r + \sigma_2^2\right) + \sigma_2^2 - \rho\sigma_1\sigma_2 = 0$$

as it must have, since X is a traded asset divided by the numeraire. Its volatility is $\sqrt{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}$, so the change of numeraire has already produced the volatility that Section 28.11 prices an exchange option on, before any option has been mentioned.

Why volatility is invariant is the part worth understanding rather than memorizing. A change of measure reweights the probability of paths. It can make an upward path more likely and a downward path less likely, which moves the mean. But it cannot change how far apart the paths are, because the paths themselves are unchanged—only their probabilities move.

Two consequences follow directly and are used constantly.

Volatility is the same in every measure, so a volatility estimated from real-world data can be used in a risk-neutral pricing model. Without this, no calibration in this book would work, since historical volatility and implied volatility would not even be comparable quantities.

Expected returns are not. A traded asset drifts at μ under P and at r under Q, and $\mu - r = \lambda\sigma$, the λ of (28.4), reached here by a change of measure rather than by Example 28.1's hedge.

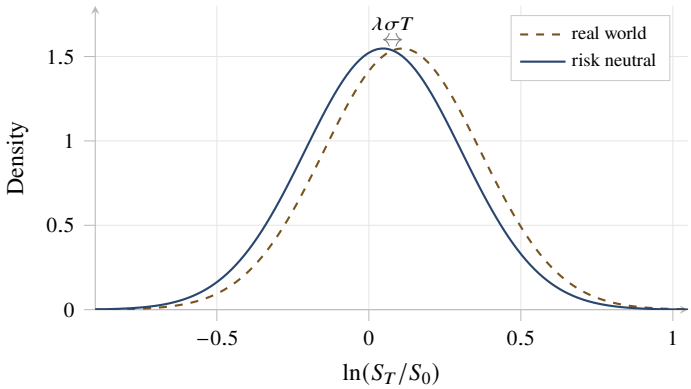


Figure 28.2: Example 28.2's tree in continuous time, at its two-year horizon: one density displaced by $\lambda\sigma T$.

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Figure 28.2 draws the theorem on Example 28.2's tree: one density twice, displaced by $\lambda\sigma T = 0.0626$ and not otherwise altered. What is invariant is the volatility, the standard deviation $\sigma\sqrt{T}$ of the log return, and that is a continuous-time fact, not a general one. Reweighting can change a variance, and in the two-step tree it does, mildly; what rules it out here is that volatility is pathwise, the quadratic variation of $\ln S$, and equivalent measures agree path by path. In the level of S_T the two give lognormals of visibly different shape, which is why the figure is drawn in logs.

28.10 WHAT $N(d_1)$ IS

Every numeraire so far has been the money market account, a bond, or a portfolio of bonds. Nothing in Section 28.6 required that, and dropping the restriction explains a formula the reader has known since Chapter 15.

Split the Black–Scholes call into the two pieces it obviously has:

$$c = e^{-rT}E^Q\left[S_T\mathbf{1}_{\{S_T>K\}}\right] \;-\; Ke^{-rT}\,Q(S_T > K) \tag{28.12}$$

The second piece is the easy one. $Q(S_T > K)$ is a probability under a measure in which $\ln S_T$ is normal with mean $\ln S_0 + (r - \sigma^2/2)T$, and it evaluates to $N(d_2)$. So $N(d_2)$ is the risk-neutral probability that the option finishes in the money, which is the one part of the formula most readers can already interpret.

The first piece is the awkward one, because it is the expectation of a *product*: a terminal price multiplied by an indicator of an event that depends on that same price. Done directly it means integrating S_T against a lognormal density over a half-line and completing a square.

Choose the numeraire to make the product disappear. The payoff $S_T\mathbf{1}_{\{S_T>K\}}$ is the promise to deliver one share if the share finishes above K. Value it using the *stock itself* as the numeraire, which is admissible since a non-dividend-paying stock is traded and strictly positive:

$$\frac{V_0}{S_0} = E^S\left[\frac{S_T\mathbf{1}_{\{S_T>K\}}}{S_T}\right] = E^S\left[\mathbf{1}_{\{S_T>K\}}\right] = Q^S(S_T > K)$$

Dividing by S_T is exactly what cancels the S_T in the payoff, and an expectation of a product has become a probability. So $V_0 = S_0\,Q^S(S_T > K)$.

What remains is that probability, and Girsanov supplies it. Moving from the money-market numeraire to the stock shifts every drift by its covariance with $\ln S$; for the stock itself that covariance is σ^2, so its log-drift under Q^S is $r + \sigma^2/2$ rather than the $r - \sigma^2/2$ it has under Q. *The variance is untouched*, and the same calculation that produced $N(d_2)$ now produces

$$Q^S(S_T > K) = N(d_1), \qquad d_1 = \frac{\ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}$$

Substituting both pieces into (28.12) gives $c = S_0N(d_1) - Ke^{-rT}N(d_2)$.

This says something about the Black–Scholes–Merton formula that the usual derivation hides. $N(d_1)$ and $N(d_2)$ are the probability of *the same event*, the option finishing in the money, computed under *two different measures*. The formula is one event priced in two units of account: shares for the first term, cash for the second. Calling $N(d_1)$ "the delta" is true and unilluminating; it is a probability, and it differs from $N(d_2)$ only because it is measured in a different numeraire. Example 28.3 puts numbers on the gap.

Example 28.3 The same event, two probabilities

A one-and-a-half-year call, $S_0 = 100$, $K = 110$, $r = 4\%$, $\sigma = 28\%$. Then $d_1 = 0.068498$ and $d_2 = -0.274431$, so

$$N(d_1) = 0.527305, \qquad N(d_2) = 0.391877$$

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and the call is worth $100 \times 0.527305 - 110e^{-0.06} \times 0.391877 =$ **12.1344**.

The option is out of the money and the two probabilities disagree by 13.5 percentage points. Under the risk-neutral measure it finishes in the money **39.19%** of the time; under the stock measure, **52.73%**—more likely than not.

Neither number is a forecast, and the gap is not an error: the two $N(\cdot)$ terms are one event measured in two units of account.

The stock measure over-weights the paths that finish high, because it measures value in shares and shares are worth more precisely on those paths. That reweighting is the whole content of d_1 versus d_2, and the gap $\sigma\sqrt{T} = 0.3429$ between them is nothing but the volatility of the numeraire ratio.

28.11 EXCHANGING ONE ASSET FOR ANOTHER

The technique is at its most persuasive on a payoff that looks genuinely harder. An **exchange option** confers the right to give up asset 2 and receive asset 1:

$$\max(S_1(T) - S_2(T),\, 0)$$

Two assets, two volatilities, a correlation, and *a strike that moves*. Attacked directly this is a two-dimensional problem: a partial differential equation in two state variables, or a double integral over a bivariate lognormal.

Use asset 2 as the numeraire and the second dimension disappears. Asset 2 is traded and strictly positive, so it is admissible, and

$$\frac{V_0}{S_2(0)} = E^{S_2}\left[\frac{\max(S_1(T) - S_2(T),\, 0)}{S_2(T)}\right] = E^{S_2}[\max(X_T - 1,\, 0)]\,, \qquad X = \frac{S_1}{S_2}$$

The moving strike has become the number 1, which is what electing an asset as the unit of account buys. And X is a traded asset divided by the numeraire, so it is a martingale under Q^{S_2}, with no drift to estimate.

The volatility of X follows from Itô's lemma applied to a ratio:

$$\sigma = \sqrt{\sigma_1^2 + \sigma_2^2 - 2\rho\,\sigma_1\sigma_2} \tag{28.13}$$

A driftless lognormal struck at 1 is Black's formula, and unwinding the numeraire gives **Margrabe's formula**:

$$V_0 = S_1(0)\,N(d_1) - S_2(0)\,N(d_2) \tag{28.14}$$

where

$$d_1 = \frac{\ln(S_1(0)/S_2(0)) + \sigma^2 T/2}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T}$$

Four features of the result are worth stopping on.

The interest rate is absent. In units of asset 2 there is nothing to finance, so no rate can appear. A trader who asks which curve to discount an exchange option on has misunderstood the instrument.

The Black–Scholes–Merton formula is the special case. Let asset 2 be K zero-coupon bonds maturing at T, so $S_2(0) = Ke^{-rT}$ and $\sigma_2 = 0$. Then (28.13) collapses to σ_1 and (28.14) reads

$$V_0 = S_1(0)\,N(d_1) - Ke^{-rT}\,N(d_2)$$

An ordinary call is an option to exchange cash for stock, and the two formulas agree to fifteen decimal places because they are the same formula.

And the reading of $N(d_1)$ and $N(d_2)$ from Section 28.10 survives the generalization.

$$N(d_2) = Q^{S_2}(S_1(T) > S_2(T))\,, \qquad N(d_1) = Q^{S_1}(S_1(T) > S_2(T))$$

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One event, priced first in units of asset 2 and then in units of asset 1. Which asset is the numeraire is a choice, and the formula is symmetric in a way the Black–Scholes special case conceals.

And the correlation is what the value turns on. Example 34.8 runs it from −0.5 to +0.99 and the value falls sevenfold, while no interest rate enters anywhere—the input a desk can observe does not affect the price, and the input it cannot observe determines it.

Exchange options are rarely labeled as such. An asset manager paid on performance against a benchmark holds one, since the fee is a call on the fund's value with the index as the strike. A stock-for-stock merger is one: a shareholder offered a fixed ratio of the acquirer's shares holds the right to exchange the target for the acquirer, which is why deal terms are quoted with collars and why the arbitrage community prices correlation. A gas-fired power station is a strip of them (Section 36.6's spark spread option is the option to exchange fuel for electricity, exercised hour by hour), one of the largest applications of (28.14) by notional. Chapter 34 develops the multi-asset family and Chapter 36 the energy applications; all of them are one line of change of numeraire.

28.12 WHEN A NUMERAIRE FAILS

Every change of numeraire in this chapter has assumed two things about the asset in the denominator: that it is **traded**, meaning its price is the value of a self-financing portfolio, and that its price is **strictly positive**. Both assumptions fail in practice. The four cases below run in increasing order of what they cost. Each of the first three violates one of the two conditions; the fourth violates neither, and is the most consequential of all.

1. The price is not a self-financing portfolio

A stock paying dividends is not a numeraire, because the holder receives cash along the way, so its price is not the value of a self-financing position. The repair is complete and costs nothing: use the *reinvested* stock instead. Chapter 5's dividend adjustments are that same repair.

2. The price can reach zero

Dividing by a numeraire that can vanish is not permitted, and defaultable assets vanish. An option on a credit default swap takes as its numeraire the forward risky annuity of Section 25.3 (the present value of one basis point paid until default or maturity), under which the forward CDS spread is a martingale. But that annuity is worth nothing in default, so the measure it defines gives probability zero to an event that Q does not, and the two are therefore not equivalent in the sense (28.1) requires.

Single-name convention steps around this by making the option knock out at default. Index options cannot, because they are written not to cancel: the protection buyer is owed the losses accumulated before expiration, which is what **front-end protection** is, and the correction matters most in precisely the state that destroys the numeraire. Morini and Brigo (2011) construct the conditional measure that repairs it. Negative rates are not an instance of this failure, despite the association (the cash annuity of the next subsection *rises* as rates fall), and what broke in the euro and yen swaption markets after 2014 was the lognormal quoting convention, repaired by Section 29.8's shifted and normal conventions.

3. The numeraire is not an asset at all

This is the quiet failure, and it is the one that costs money every day. The cash-settled swaption convention values the contract with $G(S) = \sum_{i=1}^{mn} m^{-1}(1 + S/m)^{-i}$ for an n-year swap paying m times a year, a formula applied to the swap rate rather than a portfolio of bonds, so it is not a traded asset, and it does not cancel against the numeraire the way Section 28.8 required. The

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cash annuity falls as rates rise: on a ten-year annual-pay swap, from 8.98 at a 2% swap rate to 7.36 at 6%, so it is smallest exactly where a payer swaption pays most, and that covariance survives. On an at-the-money five-year option on a ten-year swap, at a 3% forward rate and 25% volatility, it is worth about 9.9 basis points of a 66-basis-point premium (both quoted as a rate, per unit of annuity), or 15%, widening to 28% at a volatility of 40% because it is a convexity term.

The contract pays $G(S_T)(S_T - K)^+$ in cash at expiration, so with the true annuity as numeraire its value is

$$V_0 = A_0 \, E^A\left[\frac{G(S_T)}{A_T}\,(S_T - K)^+\right] \tag{28.15}$$

and the ratio inside is the whole difficulty: A_T is a portfolio of bonds depending on the entire curve at expiration, $G(S_T)$ a formula in one rate, and nothing cancels. As written, the expectation runs over the *joint* law of the swap rate and the curve around it, which no swaption smile supplies.

One step removes the joint law, and it approximates nothing. The payoff depends on S_T alone, so conditioning on S_T and applying the law of iterated expectations gathers every unknown into a single function of the swap rate:

$$V_0 = A_0 \, E^A\left[\alpha(S_T)\, G(S_T)(S_T - K)^+\right], \qquad \alpha(s) \equiv E^A\left[\frac{1}{A_T}\;\middle|\; S_T = s\right]. \tag{28.16}$$

This is an identity, not a model. What remains is a one-dimensional integral against the *marginal* law of S_T, which a smile does supply, with everything unknown pushed into α, the **annuity mapping function**. Nor is α arbitrary: a unit payoff at expiration must be worth the discount bond, forcing $A_0 E^A[\alpha(S_T)] = P(0,T)$. Freezing the annuity at its forward level makes α the constant $1/G(F)$, which satisfies it exactly on a flat curve and yields the figure quoted above; desks run linear terminal-swap-rate models instead, taking α affine in s, and the difference shows in the wings.

A physically settled swaption has a price once the volatility is given. A cash-settled one has a price once the volatility *and* a model of the annuity are given—and a convention that reads as a settlement detail has moved the contract from the first category into the second.

4. Both conditions hold and the numeraire is still wrong

The risk-free bank account is traded and strictly positive, and it is not the numeraire for a collateralized trade. Nothing in the theorem is violated; the wrong admissible asset has been chosen, and that is why this case is the expensive one. It closes a loop opened in Part I.

Chapter 7 argued that a collateral agreement determines the discount curve, and Chapter 8 that the terms of the CSA change the value of a trade. In this chapter's language, those are statements about which numeraire is correct. A trade collateralized in cash earning the overnight rate is replicated by a position financed at that rate, so the correct numeraire is the *collateral account*, not a hypothetical risk-free bank account.

Three chapters' worth of apparatus follows from that one identification. *Multi-curve discounting* is the observation that the numeraire changed and the forecast curve did not. *ColVA* of Section 9.6 is the value of the optionality in *which* numeraire applies, since a CSA permitting several collateral currencies gives the poster a choice, and a choice of numeraire that one party controls has value. *And FVA* of Section 9.4 is what appears when the trade is *not* collateralized, so that no traded numeraire replicates the financing, which is why the FVA debate was, in part, a disagreement about whether a firm's own funding cost can define a numeraire, and why it could not be settled by better data.

The question "what is the numeraire" is not abstract; it is the question of what portfolio actually replicates the trade, and every time the market's answer to that changed (to the collateral

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account once CSAs set the discount rate after 2008, to SOFR for dollar cleared trades when the clearing houses switched discounting in October 2020 (Chapter 7), to the firm's own funding where no collateral was posted), a valuation adjustment appeared.

28.13 WHAT THE THEORY IS FOR

A fair question, at the start of a part that will use this material on every page, is what it buys.

It supplies the drift conditions. Chapter 32's HJM condition and LIBOR market model (LMM) drifts are Girsanov applied twice. Without the theory they are unmotivated formulas, and a desk cannot validate an unmotivated formula.

It makes Black's formulas exact. Section 29.3's caplet and Section 29.6's swaption are not approximations under their own measures—they are exact consequences of the martingale property, and knowing which measure makes each exact is what tells you when to trust them.

It generates Chapter 30's adjustments. Convexity, timing, and quanto adjustments are all the same computation: a rate is a martingale under one measure and the payoff lives under another, so a covariance term appears.

It explains where models are needed and where they are not. A market is complete (every payoff replicable) if and only if its martingale measure is unique, so the second fundamental theorem says in advance which prices will depend on a modeling choice, which is a genuinely predictive statement rather than a retrospective one. [TN28] proves both theorems and works out the range of prices that incompleteness leaves open.

And it disciplines the use of probabilities. The measure that prices correctly is not the one that forecasts correctly—Section 28.1's distinction, missed often enough to matter.

What it does not do is tell you which model to use. The theory says a measure exists; it does not say what the volatility structure is or how many factors the curve has. Chapters 31 and 32 answer those questions with data, and the theory could not.

SUMMARY

The market price of risk $\lambda = (\mu - r)/\sigma$ must be the same for any two assets driven by the same uncertainty; if it is not, a portfolio of the two is riskless and earns more than r. With several sources, $\mu - r = \sum_i \lambda_i \sigma_i$: risks price additively, derived from no-arbitrage rather than equilibrium. Risk-neutral valuation is the case $\lambda = 0$.

A change of measure reweights paths without changing which paths are possible. It moves drifts and leaves volatilities alone, which is why a volatility estimated from history can be used in a risk-neutral model at all, and the error from pricing under the wrong measure is not small: even on a two-period tree, real-world probabilities misprice a call substantially.

Valuation is the search for a ratio that is a martingale. For any numeraire (any traded, strictly positive, self-financing asset) there is a measure under which every price divided by it is driftless, so $V_0 = N_0 E[V_T/N_T]$. The choice changes no price and changes the difficulty enormously.

Four choices cover most of practice. A zero-coupon bond makes the forward rate a martingale, so Black's caplet formula is exact and the forward price is an expectation under that measure, which in general it is not under the risk-neutral one. The annuity does the same for the swap rate, and because a swaption's payoff contains the annuity it cancels; a CMS coupon pays on a single date, the cancellation fails, and what survives is the convexity adjustment. The stock itself turns $E^Q[S_T \mathbf{1}_{\{S_T>K\}}]$ into a probability, which is what $N(d_1)$ is, and a second risky asset prices the right to exchange one asset for another, in which no interest rate appears at all. Two results that look unrelated, the gap between the forward and $E^Q[S_T]$ and the futures–forward adjustment of Section 30.3, are one covariance term under two names, and Chapter 31 shows a third: the error from discounting a caplet at today's zero rate.

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Both conditions on a numeraire fail in practice, and two further failures cost more: the cash-settled swaption annuity is not a traded asset at all, so nothing cancels, and the risk-free bank account is not the numeraire for a collateralized trade, which is what Chapter 7's discount curve, ColVA, and FVA are all statements about.

A market is arbitrage-free if and only if an equivalent martingale measure exists, and complete if and only if that measure is unique; [TN28] proves both and says what has to change in continuous time. The second theorem is where every difficulty in this book lives: when a price depends on a modeling choice, it is because the market is incomplete in a way that can be named.

FURTHER READING

Harrison and Kreps, and Harrison and Pliska, are the papers that established the two fundamental theorems; they are demanding and are the right place to go for anyone who wants the results stated precisely rather than usefully. Geman, El Karoui, and Rochet's paper on change of numeraire is the source for Section 28.6. Baxter and Rennie's short book gives the ideas without measure-theoretic machinery, and Shreve's two volumes are the standard graduate treatment for someone who wants the machinery. Björk's book sits between them and is the one most often used for a first serious course. On the real-world versus risk-neutral distinction, the actuarial literature treats it operationally, because the profession has had to make the distinction workable for regulatory reporting; the papers on economic scenario generators are the useful entry point.

PRACTICE QUESTIONS

28.1 Using Example 28.1, construct the riskless portfolio and compute its excess return, then explain why the market price of risk must be the same for two assets driven by the same source of uncertainty.

28.2 Using Example 28.2, state the three values of $\Lambda = dQ/dP$, explain why they average to one under P rather than under Q, and explain why the reweighting is monotone in the terminal price.

28.3 Using Example 28.2, state the size of the pricing error from using real-world probabilities and explain why it is positive for a call.

28.4 Explain what it means for a ratio to be a martingale, and why finding one is the whole of the valuation procedure.

28.5 Name the four numeraires that cover practice, state the measure each induces, and explain why the choice changes no price but changes the difficulty enormously.

28.6 Explain why Black's swaption formula is exact rather than approximate, being specific about what cancels.

28.7 Explain why a CMS coupon breaks the cancellation that makes Black's swaption formula exact, identify the numeraire its payoff calls for, and say what quantity the resulting adjustment consists of.

28.8 Explain why Girsanov changes drifts and leaves volatilities alone, and what this makes possible throughout the book.

28.9 Explain why $N(d_1)$ and $N(d_2)$ are probabilities of the same event, name the measure each is computed under, and say why they differ.

28.10 Derive Margrabe's formula by taking asset 2 as the numeraire, and explain why no interest rate appears in the result.

28.11 State the two conditions a numeraire must satisfy, and give an instance in which each has

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failed in a real market.

28.12 Explain why the discount curve of Chapter 7, ColVA, and FVA are all statements about the choice of numeraire, and why that made the FVA debate unresolvable by data.

28.13 Using [TN28], state both fundamental theorems, and explain why stochastic volatility, jumps, and credit each produce incompleteness for a different specific reason.

FURTHER QUESTIONS

28.14 Derive the market-price-of-risk condition from the requirement that a riskless portfolio earn r, then show that risk-neutral valuation is the special case $\lambda = 0$ and explain in what sense the risk-neutral measure is a computational device rather than a belief.

28.15 Prove that volatility is invariant under a change of measure, and use the result to explain why a stochastic volatility model calibrated to option prices can be tested against realized volatility at all; then state precisely what such a test can and cannot detect.

28.16 The second fundamental theorem says incompleteness produces a range of arbitrage-free prices. For a stochastic volatility model, characterize that range, explain what determines where inside it the market trades, and connect your answer to Section 19.10's volatility risk premium.

28.17 This chapter is placed first in Part VI, on the argument that four of the part's six chapters would otherwise borrow machinery they have not been given. Construct the opposing case for teaching it last, on the argument that the theory is easier to motivate after seeing what it explains; identify which chapters would become harder and which easier, and give your own judgment with reasons.

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Chapter 29
Standard Market Models for Rate Derivatives

The interest rate options that trade in the largest volume (caps, floors, and swaptions) are priced by the market using a small family of formulas that are, in essence, Black's model applied three times over. All three trade over the counter, dealer to client under ISDA documentation, in the market Section 10.10 described; none is exchange-listed in any size. This chapter sets out those formulas, the conventions attached to them, and the two respects in which the standard apparatus has had to change: once because rates went negative, and once because the reference rate is no longer known at the start of the period it applies to.

It is worth being clear at the outset about what these models are and are not. They are not descriptions of how interest rates behave. No one believes a single forward rate is lognormal with constant volatility, and Chapters 31 and 32 develop models that take term structure dynamics seriously. What the standard market models provide is a *quotation convention*: a one-to-one map between a price and a number called a volatility, agreed by everyone, so that traders can talk about relative value without arguing about models.

That distinction matters more than it might appear. When a trader says three-month expiration swaptions are trading at 90 basis points, they are not asserting that rates are normally distributed. They are quoting a price in units everyone can invert. The model is a language, and its defects matter only where they make the language ambiguous—which, as Section 29.8 explains, is precisely what happened when rates fell below zero.

The chapter runs from the formula to the surface the market quotes on. Sections 29.1 to 29.4 are Black's model and the instruments it prices: bond options, caps and floors, and the backward-looking caplets the transition created. Sections 29.5 to 29.7 are the conventions attached to them: flat against spot volatilities, European swaptions, and the correlation that makes a cap and a swaption different trades. Sections 29.8 and 29.9 are what negative rates did to the quoting convention and the cube that results; Section 29.10 is how the books are hedged.

29.1 BLACK'S MODEL

Chapter 16 introduced Black's model for options on futures. Its natural home turns out to be interest rate derivatives, for a reason Section 28.6 made precise: each of the products in this chapter has a *natural numeraire* under which the relevant forward rate is a martingale, and under that measure the forward rate can be modeled directly without reference to how it got there.

In its general form, consider a payoff at time T equal to

$$\max(V_T - K, 0)$$

where V_T is the value at T of some variable whose forward value today is F_0. At T the forward and the spot coincide, $F_T = V_T$, so the payoff is equally $\max(F_T - K, 0)$, which is what allows the forward, a martingale under the measure that goes with the product's own numeraire, to be

modeled in place of the spot. If V_T is lognormal with volatility σ, the value today is

$$\boxed{c = P(0,T)\,[F_0\,N(d_1) - K\,N(d_2)]} \tag{29.1}$$

with

$$d_1 = \frac{\ln(F_0/K) + \sigma^2 T/2}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T} \tag{29.2}$$

and the corresponding put

$$p = P(0,T)\,[K\,N(-d_2) - F_0\,N(-d_1)] \tag{29.3}$$

Three features make (29.1) the workhorse of this chapter. The discount factor $P(0,T)$ appears *outside* the expectation, so discounting and expectation separate cleanly. The drift of the underlying never appears—only its forward value, which is observable. And σ is the volatility of the *forward* value, not of the spot.

The entire content of the sections that follow is the identification, for each product, of what plays the role of F_0, what plays the role of the discount factor, and over what horizon σ is measured.

29.2 BOND OPTIONS

A bond option confers the right to buy or sell a bond at a specified price on a specified date. Standalone bond options trade, but the economically important ones are embedded: the issuer's right to call a bond before maturity, or the holder's right to put it back.

To apply (29.1), take V_T to be the bond price at option maturity T and F_0 the forward bond price:

$$F_0 = \frac{B_0 - I}{P(0,T)} \tag{29.4}$$

where B_0 is the bond's cash price today and I the present value of coupons paid before T. The volatility σ is that of the forward bond price. Example 29.1 works one through.

Yield volatilities. Bond price volatilities are inconvenient to think about, because a bond's price volatility falls mechanically as it approaches maturity. Traders therefore quote *yield* volatilities, converted using duration. From Section 4.11, $\Delta B \approx -BD^*\Delta y$. Both volatilities are proportional, so put the relation in proportional form:

$$\frac{\Delta B}{B} \;\approx\; -D^* y\,\frac{\Delta y}{y}$$

The price moves by D^*y times the yield's proportional move, and taking standard deviations,

$$\sigma \;\approx\; D^*\,y_0\,\sigma_y \tag{29.5}$$

where σ_y is the forward yield's proportional volatility and D^* the modified duration of the forward bond. A 5% yield with 15% yield volatility on a bond of modified duration 4 implies a price volatility of $4 \times 0.05 \times 0.15 = 3\%$. Example 29.2 shows what skipping the conversion costs.

Example 29.1 A European bond option

A one-year European call struck at 100 is written on a bond whose cash price today is 102.50. The bond pays coupons of 3 at three months and nine months. The relevant discount factors are $P(0, 0.25) = 0.9901$, $P(0, 0.75) = 0.9704$, and $P(0, 1) = 0.9608$. The forward bond price volatility is 4%.

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The present value of the coupons paid before option maturity is

$$I = 3 \times 0.9901 + 3 \times 0.9704 = 5.8815$$

so from (29.4) the forward bond price is

$$F_0 = \frac{102.50 - 5.8815}{0.9608} = 100.5605$$

Then

$$d_1 = \frac{\ln(100.5605/100) + \frac{1}{2} \times 0.04^2}{0.04} = \frac{0.0055893 + 0.0008}{0.04} = 0.15973$$

$$d_2 = 0.15973 - 0.04 = 0.11973$$

giving $N(d_1) = 0.56345$ and $N(d_2) = 0.54765$, so from (29.1)

$$c = 0.9608\,[100.5605 \times 0.56345 - 100 \times 0.54765] = 0.9608 \times 1.89596 = 1.822$$

or \$1.82 per 100 of face value. The coupons are subtracted because the option holder does not receive them.

Example 29.2 A quoted yield volatility, converted

Consider a two-year European call on a bond, struck at 98.00, when the forward bond price is 98.50 and $P(0,2) = 0.9124$. The screen quotes a *yield* volatility of 16%; the forward yield is 4.20% and the modified duration of the forward bond is 6.4.

Convert first. From (29.5),

$$\sigma \approx D^* y_0 \sigma_y = 6.4 \times 0.0420 \times 0.16 = 4.30\%$$

so the total volatility to expiration is $0.0430\sqrt{2} = 0.06082$. Then

$$d_1 = \frac{\ln(98.50/98.00) + \frac{1}{2} \times 0.06082^2}{0.06082} = 0.11408$$

$$d_2 = 0.11408 - 0.06082 = 0.05326$$

giving $N(d_1) = 0.54541$ and $N(d_2) = 0.52124$, so from (29.1)

$$c = 0.9124\,[98.50 \times 0.54541 - 98.00 \times 0.52124] = 0.9124 \times 2.64197 = 2.41$$

Now price the same option with 16% substituted directly into Black's formula, as though a yield volatility were a price volatility: the answer is **8.31**, three and a half times too large.

The conversion is not a refinement. A bond's price volatility and its yield volatility differ by the factor $D^* y_0$, which here is 0.269, and that factor is what the duration is doing in (29.5).

Callable and puttable bonds. A callable bond is an ordinary bond less a call option held by the issuer; a puttable bond is an ordinary bond plus a put held by the investor. Both can be valued by pricing the straight bond and the embedded option separately. The complication is that the embedded options are typically *Bermudan* (exercisable on any of several dates), which puts them beyond Black's model and into the tree methods of Chapter 31 and the simulation methods of Chapter 32. Black's model remains useful for a first approximation and for quoting.

Option-adjusted spread. Comparing callable bonds by yield is meaningless, because a high yield may reflect a wide credit spread or a valuable call the investor has written. The market's solution is the **option-adjusted spread** (OAS): the constant spread which, added to the entire benchmark discount curve, makes a model's price for the callable bond equal its observed market price.

The OAS is what remains of the bond's spread after removing the value of the embedded

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option, and it is the quantity across which callable bonds can legitimately be compared. Two cautions attach to it. First, it is thoroughly model-dependent: the option value depends on an assumed volatility, so a higher volatility assumption produces a larger option value and hence a smaller OAS for the same market price. An OAS quoted without its volatility assumption is not a well-defined number. Second, it is only as good as the exercise rule the model assumes: for a callable bond that rule is optimal exercise by the issuer, while for mortgages, whose prepayment option is the largest embedded option in the market, it has to be a behavioral prepayment model, because borrowers demonstrably do not exercise optimally.

29.3 INTEREST RATE CAPS AND FLOORS

An **interest rate cap** protects a floating-rate borrower against rates rising above a level K. It is written on a notional L over a sequence of accrual periods, and pays, for the period $[T_i, T_{i+1}]$ with accrual factor τ_i, at the period's end T_{i+1},

$$L\,\tau_i \max(R_i - K, 0) \tag{29.6}$$

where R_i is the reference rate for that period. A **floor** is the mirror image, paying $L\tau_i \max(K - R_i, 0)$.

The option covering one period is a **caplet**, or for a floor a floorlet, and the essential observation is that each one settles on its own period's rate, whatever the others do. A cap is a *portfolio* of caplets, not a compound option, so it can be valued one period at a time and summed.

Applying Black's model to a single caplet, with F_i the forward rate for $[T_i, T_{i+1}]$:

$$\text{caplet}_i = L\,\tau_i\,P(0,T_{i+1})\,[F_i\,N(d_1) - K\,N(d_2)] \tag{29.7}$$

with d_1, d_2 from (29.2) using the volatility σ_i of that forward rate over $[0, T_i]$: the rate fixes at T_i, so that is when the option's uncertainty is resolved, even though the payment does not arrive until T_{i+1}. Discounting is done with $P(0, T_{i+1})$, the bond maturing on the payment date, which, as Section 28.6 established, is exactly the numeraire under which F_i is a martingale, so no adjustment is required.

Caplets and bond options. A caplet is also a bond option, and the equivalence is an identity rather than an approximation. The reference rate is read off the bond price prevailing at T_i,

$$R_i = \frac{1}{\tau_i}\left(\frac{1}{P(T_i, T_{i+1})} - 1\right)$$

and the payoff (29.6) is already known at T_i, so discounting it back over the accrual period multiplies it by $P(T_i, T_{i+1})$ and the bond prices cancel:

$$L\,\tau_i\,P(T_i,T_{i+1})\max(R_i - K, 0) = L\,(1 + \tau_i K)\max\left(\frac{1}{1+\tau_i K} - P(T_i,T_{i+1}),\ 0\right)$$

A caplet is therefore $L(1 + \tau_i K)$ European *puts* on the T_{i+1} zero-coupon bond, expiring at T_i and struck at $1/(1 + \tau_i K)$. Puts, because the bond price falls when the rate rises. Nothing has been assumed about how the rate is distributed (the identity is between contracts, not between models), and it is what allows a model of the short rate to price a cap in closed form, since such a model delivers the distribution of $P(T_i, T_{i+1})$ directly. Section 31.7 does exactly that.

Example 29.3 Valuing a caplet

A caplet covers the three-month period beginning in one year, on a notional of \$10 million. The forward rate for that period is $F = 4.00\%$, the strike is $K = 4.25\%$, the volatility is $\sigma = 20\%$, the

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accrual factor is $\tau = 0.25$, and $P(0, 1.25) = 0.9520$.

Taking the horizon as $T = 1$ (the rate is fixed at the start of the period),

$$d_1 = \frac{\ln(0.04/0.0425) + \frac{1}{2} \times 0.20^2 \times 1}{0.20\sqrt{1}} = \frac{-0.060625 + 0.020}{0.20} = -0.2031$$

$$d_2 = -0.2031 - 0.20 = -0.4031$$

so $N(d_1) = 0.41953$ and $N(d_2) = 0.34344$. From (29.7),

$$\text{caplet} = 10{,}000{,}000 \times 0.25 \times 0.9520\,[0.04 \times 0.41953 - 0.0425 \times 0.34344]$$

$$= 2{,}380{,}000 \times (0.016781 - 0.014596) = 2{,}380{,}000 \times 0.002185 = \$5{,}200$$

Cap–floor parity. A useful identity follows from the payoffs: since $\max(R-K, 0) - \max(K-R, 0) = R - K$,

$$\text{cap} - \text{floor} = \text{payer swap struck at } K \tag{29.8}$$

A long cap and short floor at the same strike replicate a swap paying fixed at K and receiving floating. This is a model-free relationship, and it provides an immediate consistency check: cap and floor quotes that violate it are arbitrageable regardless of anyone's view on volatility.

Which caplets are included. Under LIBOR conventions the first caplet of a **spot-starting** cap (one whose first accrual period begins today) was conventionally excluded, because the first period's rate was already fixed and the caplet's payoff therefore known. That convention no longer makes sense, for the reason developed next.

Who buys a cap. The natural buyer is a borrower with floating-rate debt: a company or property owner whose loan resets quarterly buys a cap to hold its funding cost at the strike while keeping the benefit of lower rates, and lenders often require it.

Listed rate options. Caps and floors are over-the-counter. The exchange-traded counterparts are options on SOFR futures (including mid-curve options, which expire before the underlying futures contract) and on Treasury futures. They are priced with Black's model on the futures price, usually quoted in normal volatility, and need no convexity adjustment, because the futures price is already a martingale under the risk-neutral measure (Section 28.7). Asset managers and money-market funds use them to hold rate optionality without a bilateral contract.

29.4 BACKWARD-LOOKING CAPLETS

Section 29.3 quietly assumed the reference rate is fixed at the start of its accrual period, so that a caplet is an option expiring at T_i on a quantity known at T_i. Chapter 4 established that this is no longer how the reference rate works. A compounded risk-free rate accrues across $[T_i, T_{i+1}]$ and is not known until the period ends.

An option on such a rate is a **backward-looking** caplet, and it differs from its forward-looking predecessor in a way that is small in magnitude but conceptually important.

Nothing is exercised by decision here. A caplet settles automatically against the realized rate, so its "expiration" is the date the payoff becomes *known*, not a date on which somebody chooses: T_i when the rate is fixed in advance, T_{i+1} when it is compounded across the period. That is why the backward-looking version is an option at all despite the rate being undetermined at T_i—there is nothing to decide at T_i.

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How much uncertainty is there?

Consider the rate for the period $[T_1, T_2]$, with $\tau = T_2 - T_1$, and write σ for its volatility, in the sense of Section 29.1: the volatility of a forward value, taken as constant. Before T_1 no fixing has entered it yet, so it accumulates uncertainty exactly as a forward-looking rate does: variance $\sigma^2 t$ at time t. Its forward is the same F_0, as the pricing below shows. The two rates part company only once compounding starts.

Once compounding begins at T_1, something new happens. Each day's overnight fixing that has already been observed is locked in. As the period elapses, a growing fraction of the final compounded rate is already determined, so the remaining sensitivity to rate movements declines, linearly to a good approximation, reaching zero at T_2. This is the same decaying-delta effect that Section 30.3 notes for averaging futures.

Modeling that decay as linear, the rate's volatility at a time u inside the period is $\sigma(T_2 - u)/\tau$, falling from σ at T_1 to zero at T_2, and the variance accumulated between T_1 and T_2 is

$$\int_{T_1}^{T_2} \sigma^2 \left(\frac{T_2 - u}{\tau} \right)^2 du \;=\; \frac{\sigma^2 \tau}{3} \tag{29.9}$$

Adding the variance accumulated before T_1 gives the total:

$$\text{Var} = \sigma^2 \left(T_1 + \frac{\tau}{3} \right) \tag{29.10}$$

Figure 29.1 shows the two paths, drawn with $\tau = T_1$ so the effect is visible. The forward-looking rate stops accumulating uncertainty at T_1; the backward-looking rate continues past it, though at a decreasing rate as more of the average is locked in, adding a further $\tau/3$ years' worth of variance.

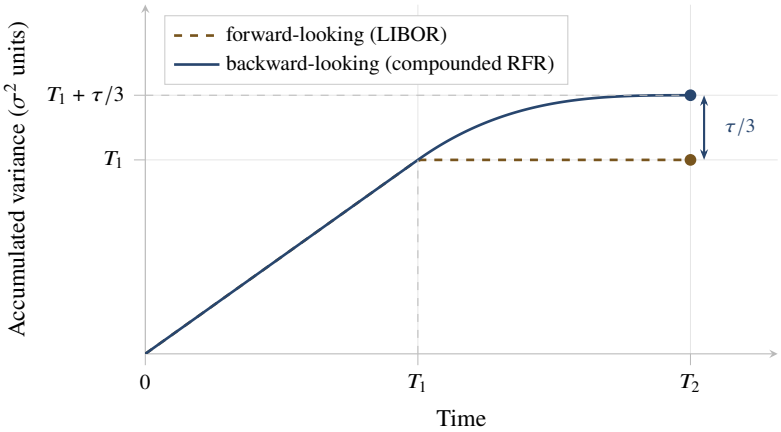


Figure 29.1: Uncertainty accumulated by a forward-looking and a backward-looking rate, drawn with $\tau = T_1$.

Pricing

The measure does not change: a caplet pays at T_2 either way, so $P(0, T_2)$ is the numeraire in both cases. What changes is the process modeled under it. Section 29.3 used the forward rate $F(t)$ for $[T_1, T_2]$, which stops moving at T_1, and a rate still accumulating at T_2 needs an extension of it.

The extension comes from a trading strategy: buy the T_1 bond and, at T_1, roll the proceeds at the overnight rate through to T_2. Its value $C(t)$ is a traded portfolio throughout, so $C(t)/P(t, T_2)$

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is a martingale under the T_2-forward measure, and therefore so is

$$R(t) = \frac{1}{\tau}\left(\frac{C(t)}{P(t,T_2)} - 1\right) \tag{29.11}$$

Before T_1, $C(t) = P(t,T_1)$ and $R(t)$ is the old forward rate unchanged; at T_2 it is the compounded rate the caplet settles against; and $R(0) = F_0$, so both caplets are struck against the same forward. Only the total variance changes: the martingale now runs to T_2 instead of freezing at T_1, and (29.10) is what it accumulates. So price a backward-looking caplet with Black's formula (29.7), but replace $\sigma^2 T_1$ with $\sigma^2(T_1 + \tau/3)$ wherever it appears. Equivalently, use an effective volatility

$$\sigma_{\text{eff}} = \sigma\sqrt{\frac{T_1 + \tau/3}{T_1}} \tag{29.12}$$

over the horizon T_1. Example 29.4 applies it to the caplet of Example 29.3.

Example 29.4 The same caplet, backward-looking

Take Example 29.3 and treat the rate as compounded in arrears rather than fixed in advance. Here $T_1 = 1$ and $\tau = 0.25$, so

$$\sigma^2\left(T_1 + \frac{\tau}{3}\right) = 0.04 \times 1.08333 = 0.043333, \qquad \sqrt{0.043333} = 0.20817$$

Repeating the calculation with $\sigma\sqrt{T}$ replaced by 0.20817:

$$d_1 = \frac{-0.060625 + \frac{1}{2} \times 0.043333}{0.20817} = -0.1871, \qquad d_2 = -0.3953$$

giving $N(d_1) = 0.42577$, $N(d_2) = 0.34631$, and

$$\text{caplet} = 2{,}380{,}000\,[0.04 \times 0.42577 - 0.0425 \times 0.34631] = \$5{,}505$$

against \$5,200 for the forward-looking version—about **5.9% more valuable**.

Two observations follow.

Backward-looking caplets are worth more. Since $T_1 + \tau/3 > T_1$ always, the option on a compounded rate is unambiguously the more valuable. It is exposed to rate movements for longer.

The first caplet is no longer trivial. Under LIBOR the first caplet of a spot-starting cap had $T_1 = 0$ and hence zero variance: the rate was already fixed, and the caplet was a known cash flow, conventionally excluded from the quote. Under a compounded rate, $T_1 = 0$ gives variance $\sigma^2\tau/3$, which is not zero. The first caplet of a spot-starting cap now carries genuine optionality, and the old convention of dropping it silently understates the price.

The effect scales with the accrual length. For quarterly periods $\tau/3$ is a month, negligible against a five-year expiration but material against a three-month one. For annual periods, or for the front of the curve, it is worth having right.

29.5 FLAT AND SPOT VOLATILITIES

A cap is a portfolio of caplets, each with its own volatility. The market, however, quotes a single number per cap. This creates two distinct volatility concepts that are easily confused and must be kept apart.

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- A **flat volatility** is the single volatility which, applied to *every* caplet in a given cap, reproduces the cap's market price. It is a quotation device, not a property of any individual caplet.
- A **spot volatility** (also called a caplet or forward volatility) is the volatility of one specific caplet.

A flat volatility is thus a kind of average of the spot volatilities of the caplets making up that cap. The two coincide only for the shortest cap, the one containing a single caplet. That averaging is visible in Figure 29.2: because a flat volatility summarizes every caplet up to its maturity, its curve peaks later, peaks lower, and decays more slowly than the spot curve underneath it.

Extracting spot volatilities from quoted flat volatilities is a bootstrap, exactly parallel to the zero curve bootstrap of Chapter 4. The price of a two-year cap less the price of a one-year cap is the value of the caplets falling in the second year; those caplets' spot volatilities are whatever reproduces that difference, given the spot volatilities already extracted for year one. Example 29.5 carries out one step.

Example 29.5 Bootstrapping spot volatilities

Annual-reset caps struck at 4.00% are quoted at flat volatilities of 20.0% for the two-year and 21.0% for the three-year. The relevant forward rates are 4.00% for $[1, 2]$ and 4.30% for $[2, 3]$; discount factors are $P(0, 2) = 0.9246$ and $P(0, 3) = 0.8890$. Work per unit of notional.

Two-year cap. It contains a single caplet, covering $[1, 2]$ (the first period is excluded, as under the LIBOR convention of Section 29.3), so its spot volatility is the quoted flat volatility, $\sigma_1 = 20.0\%$. At the money with $T = 1$, $d_1 = 0.1$ and $d_2 = -0.1$, so

$$\text{caplet}_{[1,2]} = 0.9246 \times 0.04\,(0.53983 - 0.46017) = 0.0029460$$

Three-year cap. Valuing *both* caplets at the quoted flat volatility of 21% gives 0.0030928 for $[1, 2]$ and 0.0058161 for $[2, 3]$, a total cap price of

$$0.0030928 + 0.0058161 = 0.0089088$$

The bootstrap step. The $[2, 3]$ caplet must account for the remainder:

$$0.0089088 - 0.0029460 = 0.0059628$$

Solving Black's formula numerically for the volatility that reproduces this value gives

$$\sigma_2 = 21.7\%$$

So a flat volatility curve of 20.0%, 21.0% corresponds to spot volatilities of 20.0%, 21.7%. The second caplet's own volatility exceeds the three-year cap's quote, because that quote is diluted by the lower-volatility first caplet inside it. Using 21% to value the $[2, 3]$ caplet in isolation would understate it by about 2.5%.

The characteristic shape in Figure 29.2, rising to a hump at two to three years and then declining, is remarkably persistent across currencies and decades. The usual explanation is that very short rates are anchored by the central bank's current policy setting and very long rates by expectations about the long run, while the intermediate maturities carry most of the uncertainty about the path of policy in between.

A flat volatility quoted for a five-year cap and a flat volatility quoted for a ten-year cap describe overlapping portfolios; the difference between them is not the volatility of anything. Comparing the two directly, or interpolating between them, is a common and expensive error. Convert to spot volatilities before doing anything analytical.

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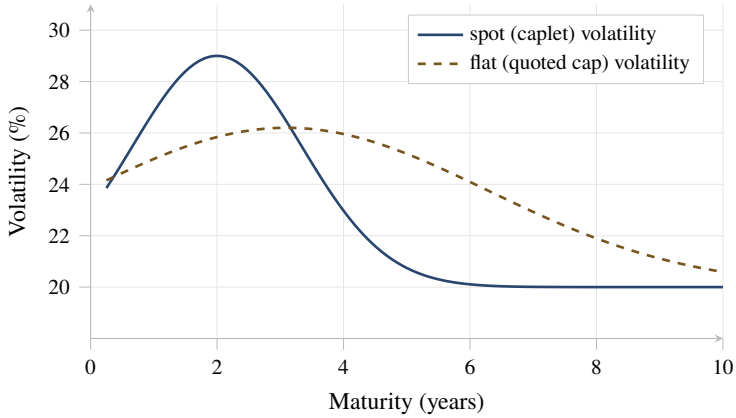


Figure 29.2: Representative shapes of flat and spot volatility curves. Illustrative rather than market data.

29.6 EUROPEAN SWAPTIONS

A **swaption** is an option to enter an interest rate swap. A *payer* swaption confers the right to pay fixed and receive floating; a *receiver* swaption the right to receive fixed. They are described by two maturities (the option expiration and the underlying swap's tenor), so a "5×10" is a five-year option on a ten-year swap.

The key to valuing them is to recognize what the option is really on. At expiration T, the holder of a payer swaption struck at K exercises if the prevailing swap rate S_T exceeds K, gaining $\tau(S_T - K)$ on each of the n payment dates. The total value at exercise is

$$L \sum_{i=1}^{n} \tau_i P(T, t_i) \max(S_T - K, 0) = L\, A(T) \max(S_T - K, 0)$$

where $T < t_1 < \cdots < t_n$ are the payment dates of the fixed leg of the swap the option delivers, τ_i the accrual factor for each, and $A(T)$ the annuity of Section 28.8, the present value at T of one unit paid on each of those dates. So a swaption is an option on the swap rate, with a payoff multiplied by the annuity. In the major currencies that fixed leg now pays annually (Section 7.1), so τ_i is about one and n is the swap's tenor in years, which is what Example 29.6 assumes. Legacy dollar swaps paid fixed semiannually against quarterly floating, and some currencies still do.

That structure points to the right numeraire. Taking the annuity as numeraire (Section 28.8), the swap rate is a martingale, and Black's model applies with the annuity replacing the discount factor:

$$\text{payer} = L\, A(0)\, [S_0\, N(d_1) - K\, N(d_2)] \tag{29.13}$$

$$\text{receiver} = L\, A(0)\, [K\, N(-d_2) - S_0\, N(-d_1)] \tag{29.14}$$

with d_1, d_2 from (29.2) using the forward swap rate S_0 and its volatility, and

$$A(0) = \sum_{i=1}^{n} \tau_i\, P(0, t_i) \tag{29.15}$$

The annuity is not merely a discount factor: it is the numeraire that makes the swap rate driftless, which is what licenses the use of Black's model in the first place. Chapter 30 exploits this directly: the CMS convexity adjustment arises precisely because a CMS coupon is paid at a single date rather than across the annuity.

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Example 29.6 Valuing an at-the-money swaption

Consider a five-year option on a five-year annual-pay swap, notional \$100 million. The forward swap rate is $S_0 = 4.00\%$, the strike is $K = 4.00\%$, and the volatility is $\sigma = 20\%$.

With a flat 4% curve, the discount factors for the five payment dates (years 6 through 10) are 0.790315, 0.759918, 0.730690, 0.702587, and 0.675564, so from (29.15)

$$A(0) = 3.659074$$

Because the option is at the money, $\ln(S_0/K) = 0$ and

$$d_1 = \frac{1}{2}\sigma\sqrt{T} = \frac{1}{2} \times 0.20\sqrt{5} = 0.22361, \qquad d_2 = -0.22361$$

giving $N(d_1) = 0.58847$ and $N(d_2) = 0.41153$. From (29.13),

$$\text{payer} = 100{,}000{,}000 \times 3.659074 \times 0.04 \times (0.58847 - 0.41153) = \$2{,}589{,}700$$

or about 2.59% of notional. At the money the payer and receiver have equal value, as put–call parity requires: their difference is a forward swap struck at the forward rate, which is worth zero.

Swaptions and bond options. A payer swaption is equivalent to a put option on a coupon bond: exercising the right to pay fixed at K is exercising the right to sell, at par, a bond paying coupon K. Receiver swaptions correspond to call options on the same bond, but the swap-rate formulation is the market standard because the swap rate, unlike a bond price, does not drift mechanically toward par as expiration approaches.

Who uses swaptions. A borrower planning to issue fixed-rate debt buys a payer swaption to cap the rate it will pay. Mortgage servicers and insurers with minimum-rate guarantees buy receiver swaptions, because falling rates bring prepayments or deepen the guarantees. Asset managers and pension funds sell swaptions to earn the premium, which is why the market's volatility is not set by hedgers alone.

29.7 CAPS VERSUS SWAPTIONS: THE ROLE OF CORRELATION

Caps and swaptions written on the same schedule and strike look superficially like the same trade. They are not, and the difference between them is one of the more instructive facts in interest rate derivatives.

A cap pays $\sum_i \tau_i \max(F_i - K, 0)$: it is a **portfolio of options**, one on each forward rate, exercised independently. A payer swaption pays $A \max(S - K, 0)$, and since the swap rate is approximately a weighted average of the same forward rates, it is an **option on a portfolio**.

The distinction has a definite direction. For any weights w_i summing to one,

$$\sum_i w_i \max(F_i - K, 0) \;\geq\; \max\left(\sum_i w_i F_i - K,\; 0\right) \tag{29.16}$$

because a sum of positive parts is at least the positive part of the sum. Hence, on comparable terms, **a cap is always worth at least as much as the corresponding swaption**.

The intuition is diversification, running backward. The cap holder collects on whichever rates happen to exceed the strike, and is not penalized for those that do not. The swaption holder owns one option on the average, so a rate that finishes far above the strike is partly canceled by another that finishes below it.

That immediately tells us what governs the size of the gap: the **correlation** between forward rates. If all forward rates started level and moved in perfect lockstep, they would always finish on the same side of the strike, no cancellation could occur, and (29.16) would hold with equality.

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The less correlated they are, the more the averaging suppresses the swap rate's variance, and the cheaper the swaption becomes relative to the cap.

This can be made quantitative. Suppose n forward rates each have volatility σ, enter the swap rate with equal weight, and have uniform pairwise correlation ρ. The variance of the average is then $\sigma^2\,[1/n + (1 - 1/n)\,\rho]$, so the swap rate's volatility is

$$\sigma_{\text{swap}} = \sigma\sqrt{\rho + \frac{1-\rho}{n}} \tag{29.17}$$

At $\rho = 1$ this returns σ, and the two products carry the same volatility. At $\rho = 0$ it falls to $\sigma/\sqrt{n}$. For a ten-period swap with $\rho = 0.7$, the swaption's volatility is $\sigma\sqrt{0.73} = 0.854\,\sigma$—some 15% below the caplet volatility, purely from imperfect correlation.

The practical significance is that this runs in reverse. Correlations between forward rates are not directly observable, but cap and swaption prices are. The ratio between them is therefore the market's main source of information about forward-rate correlation, and it is the primary calibration input for the multi-factor term structure models of Chapter 32. A model calibrated to caps alone will price swaptions badly, because caps contain almost no information about correlation: each caplet depends on one rate at a time.

29.8 NORMAL AND SHIFTED LOGNORMAL MODELS

Every formula so far assumes the underlying rate is lognormal. That assumption carries an implication which went unexamined for thirty years: a lognormal variable cannot be negative or zero. When European rates fell below zero in the middle 2010s, Black's model did not merely become inaccurate—it became *undefined*. A negative forward rate cannot be substituted into $\ln(F_0/K)$.

Case Study 29.1 When Black's model broke

The European Central Bank moved its deposit facility rate below zero in June 2014. Within a year three-month Euribor had fixed below zero for the first time, in April 2015, and short euro swap rates had followed; by 2016 substantial portions of the euro and Swiss curves traded at negative yields. The Bank of Japan followed in January 2016.

For the swaption market this was not an academic inconvenience. Brokers quote volatility, not price, and the standard quotation convention had suddenly stopped being able to represent the market. Screens showed blanks. Risk systems that inverted Black's formula to imply a volatility returned errors on live positions.

The market responded in two ways within a matter of months. Some desks adopted a **shifted lognormal** convention, adding a fixed displacement (typically 1% to 3%) to both rate and strike so that the shifted quantities stayed positive and Black's formula could be applied unchanged. Others moved to quoting **normal** volatilities under the Bachelier model, which places no restriction on sign. Normal volatility eventually became the euro and yen market standard and has since been widely adopted for dollar swaptions as well.

The episode is a useful reminder that a modeling assumption can survive for decades not because it is right but because it has never been tested in the region where it fails.

The normal (Bachelier) model

Assume instead that the rate is normally distributed, with $dF = \sigma_N\,dW$ for a constant **normal volatility** σ_N quoted in basis points per annum rather than as a percentage. The value of a

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payer-type option is

$$c = P(0,T)\left[(F_0 - K)\,N(d) + \sigma_N\sqrt{T}\,\varphi(d)\right], \qquad d = \frac{F_0 - K}{\sigma_N\sqrt{T}} \tag{29.18}$$

where φ is the standard normal density. The formula is defined for any real F_0 and K, positive or negative.

At the money the expression collapses, since $d = 0$, $N(0) = 1/2$, and $\varphi(0) = 1/\sqrt{2\pi}$:

$$c_{\text{ATM}} = P(0,T)\,\sigma_N\sqrt{\frac{T}{2\pi}} \tag{29.19}$$

An at-the-money option's value is proportional to the normal volatility and to the square root of time. Traders use this constantly for order-of-magnitude checks.

Converting between conventions

Because both conventions are one-to-one maps between price and volatility, a price quoted one way can always be restated the other way. Near the money the relationship is approximately

$$\sigma_N \;\approx\; \sigma_{\text{LN}} \times F_0 \tag{29.20}$$

where σ_{LN} is the lognormal (Black) volatility. This is just the statement that a proportional volatility applied to a level gives an absolute volatility. Example 29.7 tests it on the swaption of Example 29.6.

Example 29.7 Lognormal and normal volatility for the same option

The swaption of Example 29.6 has an at-the-money value of 2.59% of notional, with $A(0) = 3.659074$. Dividing out the annuity, the option value per unit of annuity is

$$0.04 \times (0.58847 - 0.41153) = 0.0070776$$

Setting this equal to the at-the-money normal value $\sigma_N\sqrt{T/2\pi}$ from (29.19),

$$\sigma_N = \frac{0.0070776\sqrt{2\pi}}{\sqrt{5}} = \frac{0.017741}{2.23607} = 0.007934$$

or **79.3 basis points**. The rule of thumb (29.20) predicts $0.20 \times 4.00\% = 80$ basis points—within a basis point, which is why the approximation is used so freely.

Which to use

The two models make genuinely different statements about how rates move. Black's σ is a proportional volatility ($dF = \sigma F\,dW$), so the standard deviation of the rate's change, measured in basis points, scales with the level: at a quoted 20%, a 5% rate has an annual standard deviation of 100 basis points against 50 for a 2.5% rate. Bachelier's σ_N is that standard deviation quoted directly, and is therefore the same number of basis points at any level.

The empirical evidence is regime-dependent, and not in one direction: rate moves look close to absolute, independent of the level, while rates sit in a middle band of roughly 2% to 6%, and closer to proportional both below that band and above it, the pattern Deguillaume, Rebonato, and Pogudin (2013) found across currencies and decades. Practice has settled on normal volatilities for quotation, with the understanding that the number will need to be re-marked as the level of rates changes.

Figure 29.3 shows the difference where it matters most, for a forward of 0.5%. The lognormal density is confined to positive values and cannot represent a negative outcome at all; the normal density assigns genuine probability to the shaded region below zero. When

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forwards themselves turned negative, Black's formula ceased to be merely inaccurate and became undefined.

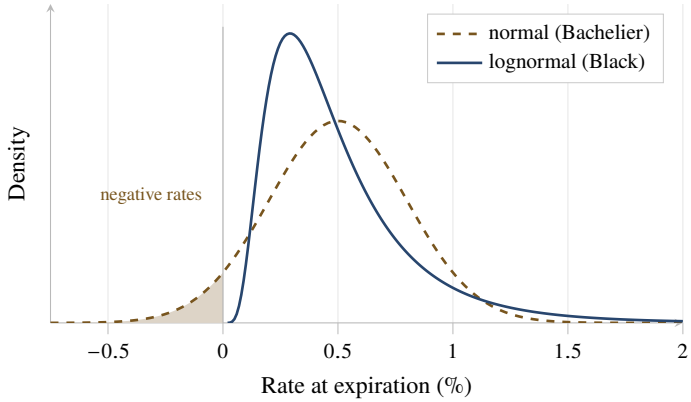


Figure 29.3: Distributions of a rate with a forward of 0.5% under the two conventions.

Shifted lognormal

The shifted lognormal convention is a compromise. Choose a displacement α large enough that $F_0 + \alpha$ and $K + \alpha$ are both comfortably positive, and apply Black's formula to the shifted quantities. It preserves the lognormal machinery, permits negative rates down to $-\alpha$, and interpolates between the two extremes: large α approaches normal behavior, $\alpha = 0$ recovers Black.

Its drawback is that α is an extra convention that must be agreed. A shifted lognormal volatility is meaningless without the shift that goes with it, and quotes carrying different shifts are not comparable. This is the same trap as flat versus spot volatilities, and the same discipline applies: establish the convention before comparing the numbers.

29.9 THE VOLATILITY CUBE

A single volatility number is enough to quote one option. Describing the whole swaption market requires three dimensions:

- **option expiration**, from one month to thirty years;
- **swap tenor**, the length of the underlying swap;
- **strike**, usually expressed as an offset from the at-the-money forward.

The resulting object is the **volatility cube**. Fixing the strike at the money gives the *volatility surface* over expiration and tenor; fixing expiration and tenor and varying strike gives a *smile*, of the kind Chapter 19 discussed for equities.

Two regularities are worth carrying away. Along the expiration dimension, volatility displays the hump of Figure 29.2. Along the tenor dimension, volatility generally *declines*: a thirty-year swap rate is an average of forward rates over thirty years, and averaging suppresses variance (exactly the mechanism of Section 29.7), so long-tenor swap rates move less than short-tenor ones. Figure 29.4 shows the at-the-money slice of the cube with both effects visible at once: the ridge running along two to three years' expiration, where uncertainty about the policy path is greatest, and the steady decline as swap tenor lengthens. Varying strike as well would add the third dimension.

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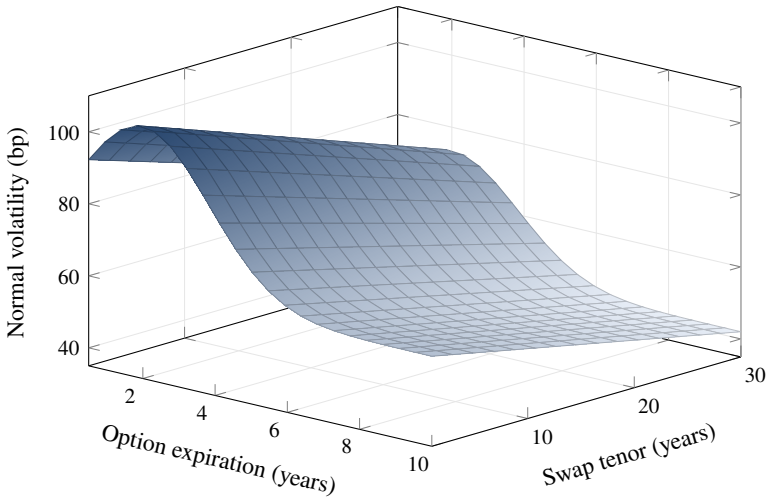


Figure 29.4: Normal volatility against option expiration and swap tenor. Representative rather than market data.

Fitting the cube: SABR

Interpolation within the cube is constrained: an arbitrary interpolation can produce a surface admitting arbitrage: calendar spreads with negative value, or butterfly spreads implying negative probability density. Practitioners therefore fit a parametric model, and for the strike dimension the standard choice is **SABR**, proposed by Hagan and co-authors in 2002. It is a stochastic-volatility model in the sense of Section 21.4, but built around a different requirement from Heston's: it must be fast enough to refit every swaption smile on the grid, many times a day. It posits a forward rate whose own volatility is stochastic:

$$dF = \alpha F^{\beta}\, dW_1, \qquad d\alpha = \nu\, \alpha\, dW_2, \qquad dW_1\, dW_2 = \rho\, dt \tag{29.21}$$

The model has four parameters, each with a distinct job:

- α sets the overall level of volatility, and is what the at-the-money quote pins down.
- β governs the **backbone**, how the at-the-money volatility moves when the forward rate itself moves. At $\beta = 1$ the model is lognormal and at-the-money volatility is unchanged by a shift in F; at $\beta = 0$ it is normal, and lognormal volatility moves inversely with F. This is the same choice as Section 29.8, now made continuously, and at $\beta = 0$ it lets SABR represent negative rates, where a lognormal model has nothing to say.
- ρ produces the skew. A negative correlation between rate and volatility tilts the smile downward to the right, which is the shape rate markets normally display.
- ν, the volatility of volatility, produces the curvature, how much the wings lift relative to a straight line through the middle.

That division is the same one Heston makes in Section 21.4. What made SABR the market standard is not the dynamics, which are unremarkable, but the closed-form approximation Hagan derived for the Black (lognormal) implied volatility $\sigma_B(K)$ they produce at strike K and expiration T, with no integral and no simulation:

$$\sigma_B(K) = \frac{\alpha}{(FK)^{(1-\beta)/2}\, D(K)} \cdot \frac{z}{x(z)} \cdot (1 + C\, T) \tag{29.22}$$

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where, writing $L = \ln(F/K)$,

$$D(K) = 1 + \frac{(1-\beta)^2}{24}L^2 + \frac{(1-\beta)^4}{1920}L^4, \qquad z = \frac{\nu}{\alpha}(FK)^{(1-\beta)/2}L$$

$$x(z) = \ln\left[\frac{\sqrt{1-2\rho z+z^2}+z-\rho}{1-\rho}\right],$$

$$C = \frac{(1-\beta)^2}{24}\frac{\alpha^2}{(FK)^{1-\beta}} + \frac{1}{4}\frac{\rho\beta\nu\alpha}{(FK)^{(1-\beta)/2}} + \frac{2-3\rho^2}{24}\nu^2$$

The formula has three factors, and each does one job: the first is the level, the second bends the smile away from it, and the third is the maturity correction. At $K = F$ the middle factor is 1 and the whole thing collapses to $\alpha F^{\beta-1}(1 + CT)$. A desk can fit a smile in microseconds and quote off it.

β is usually fixed rather than fitted. It is nearly redundant with ρ on a single smile, so fitting both produces unstable parameters that jump between days without the market having moved. [TN29] makes the redundancy exact: the skew depends on β and ρ only through one combination of them, and the curvature is where they separate again. Desks pick β from the observed backbone over weeks and then calibrate the other three daily.

The calibration recipe. Four parameters cannot be identified from a small number of quotes without structure, so desks proceed in a fixed order:

1. **Fix β** by judgment or by regressing historical at-the-money volatility on the level of the forward, never by fitting it to a single day's smile, for the reason just given. Rate desks commonly use $\beta = 0.5$.
2. **Solve α from the at-the-money quote**, given trial values of ρ and ν. This is a cubic in α, and forces the fit through the most liquid point exactly.
3. **Fit ρ and ν to the wings**, re-solving for α at each step so the at-the-money quote is never violated.

Example 29.8 Fitting SABR to a swaption smile

A five-year option on a five-year swap has a forward swap rate of $F = 4.00\%$. Three strikes are quoted, in lognormal volatility:

$$\sigma_B(3.00\%) = 23.6\%, \qquad \sigma_B(4.00\%) = 20.0\%, \qquad \sigma_B(5.00\%) = 18.3\%$$

Fix $\beta = 0.5$. Three quotes then determine the three remaining parameters, and solving (29.22) gives

$$\alpha = 0.0387, \qquad \rho = -0.30, \qquad \nu = 0.35$$

Checking the at-the-money point. With $F^{1-\beta} = \sqrt{0.04} = 0.2$, the leading term is $\alpha/0.2 = 0.1935$, and

$$C = \frac{0.25}{24}\frac{\alpha^2}{0.04} + \frac{1}{4}\frac{(-0.30)\times 0.5\times 0.35\alpha}{0.2} + \frac{2-3\times 0.09}{24}\times 0.35^2 = 0.006681$$

the three terms being 0.000390, −0.002540, and 0.008830. Hence $\sigma_B(F) = 0.1935\,(1+0.006681\times 5) = 0.1935\times 1.03340 = 20.00\%$, matching the quote.

Reproducing the wings. Substituting $K = 3\%$ gives $z = 0.4842$, $x(z) = 0.4403$, hence $z/x(z) = 1.0998$, and (29.22) returns 23.59%. At $K = 5\%$, $z = -0.4268$ and $z/x(z) = 0.9655$, giving 18.26%. Both agree with the quotes to the one decimal place at which they are quoted.

Using the fit. The value of the calibration is what it says about strikes no one quoted:

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Strike	2.5%	3.0%	3.5%	4.0%	4.5%	5.0%	5.5%
σ_B	26.25%	23.59%	21.54%	20.00%	18.92%	18.26%	17.91%

The smile is steeply downward-sloping near the money and flattens in the high-strike wing—the characteristic shape of a rate market, and in its downward slope much like the equity skew of Chapter 19.

Figure 29.5 plots the result. The at-the-money point is matched exactly by construction, since α is re-solved from it at every step of the fit; the wings are what the two remaining parameters must reproduce.



Figure 29.5: The SABR fit of Example 29.8, with $\beta = 0.5$, $\alpha = 0.0387$, $\rho = -0.30$, $\nu = 0.35$.

Where the approximation breaks. Equation (29.22) is an asymptotic expansion, accurate when $\nu^2 T$ is small. For long expirations or high volatility of volatility it degrades, and, more seriously, it can imply a *negative probability density* at low strikes, which is Section 19.5's constraint violated by an approximation rather than by the market. Since rates fell toward and below zero, that region stopped being a remote wing and became where options actually trade.

The responses parallel Section 29.8: a *shifted* SABR displaces F and K so the model operates away from zero, while *no-arbitrage* SABR replaces the expansion with a numerical solution guaranteed to produce a valid density. Both are in use at dealers, and which a desk employs is a material assumption rather than a technical detail.

The general point stands for the whole cube: it is fitted rather than interpolated, and the fitting model is itself a source of model risk. Two desks marking the same quoted points can carry materially different values for everything in between.

Case Study 29.2 The 2022 volatility regime change

For most of the decade following the financial crisis, interest rate volatility was low and falling. Policy rates sat near zero, central banks guided expectations explicitly, and the MOVE index, which tracks implied volatility on Treasury options, spent long stretches well below its historical average.

That ended abruptly in 2022. As inflation forced central banks into the fastest tightening cycle in forty years, rate volatility rose sharply, and the MOVE index climbed through 2022 to a peak during the March 2023 banking stress at its highest level since

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2008. Normal volatilities on short-dated dollar swaptions doubled or tripled from their pre-2022 levels, and longer-dated ones rose by less.

Two lessons follow for this chapter. First, positions that looked like carry trades in the low-volatility regime (selling swaptions, or the convexity trades of Section 30.3) turned out to be short volatility at prices set by a decade of calm. Second, the vol-of-vol embedded in the smile matters: a cube fitted in a quiet regime understates how much its own shape can move.

29.10 HEDGING INTEREST RATE DERIVATIVES

The Greeks of Chapter 18 carry over, with rate-specific names.

Delta for a rate derivative is its sensitivity to a shift in the curve. Because a curve is not a single number, delta is not either: desks compute a **DV01** (or PV01) against a parallel one basis point shift, and a vector of **bucketed deltas** against a one basis point shift in each curve segment separately. The bucketed vector is what actually gets hedged, since a portfolio can be flat to a parallel shift while carrying large offsetting exposures to steepening.

Vega is sensitivity to volatility, and inherits the dimensionality of the cube: a swaption book has vega to each expiration–tenor bucket, and hedging it requires trading options at comparable points.

Gamma measures how fast delta changes as the rate moves, and so how often a hedge must be rebalanced. Backward-looking products add a second source of change. As Section 29.4 showed, a compounded-rate option's sensitivity to the rate decays through the accrual period as fixings are locked in, so its delta changes with the passage of time even when rates do not move. A hedge that is correct at the start of the period is progressively too large, so these products require rebalancing through the accrual period rather than at reset dates alone.

The general caution from Chapter 18 applies with extra force here. These Greeks are computed within a model that is known to be a quotation convention rather than a description of reality. A delta computed under Black and a delta computed under Bachelier differ, and the difference is not small when the smile is steep. Which is right depends on how volatility actually moves when the rate moves—a question the quoting model does not answer.

SUMMARY

Caps, floors, and swaptions are valued with Black's model applied to the appropriate forward quantity under the appropriate numeraire. For a caplet the forward quantity is the forward rate and the numeraire is the zero-coupon bond maturing on the payment date; for a swaption it is the forward swap rate and the annuity. These are quotation conventions rather than descriptions of rate behavior: their function is to provide an agreed, invertible map between prices and volatilities.

A cap is a portfolio of separately settled caplets, so it is valued one period at a time. Cap–floor parity states that a cap less a floor at the same strike equals a payer swap, a model-free relationship useful as a consistency check. Market quotes use flat volatilities (a single number that reprices a whole cap), while analysis requires spot volatilities specific to each caplet, extracted by a bootstrap. The two must never be compared directly.

The move to compounded risk-free rates changed the caplet. A backward-looking rate keeps accumulating uncertainty across its accrual period, with sensitivity decaying linearly as fixings are locked in, giving total variance $\sigma^2(T_1 + \tau/3)$ rather than $\sigma^2 T_1$. Backward-looking caplets are therefore always worth more than their forward-looking equivalents, and the first caplet of a spot-starting cap, formerly a known cash flow, now carries real optionality.

Caps and swaptions on the same schedule are not equivalent: a swaption is a single option

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on a weighted average of the rates a cap options one by one. Because a sum of positive parts is at least the positive part of the sum, the cap is worth at least as much, and the gap is governed by the correlation among forward rates, closing to zero at perfect correlation. Those correlations are not observable while cap and swaption prices are, so the ratio between them is the market's principal source of correlation information and the main calibration input for multi-factor term structure models. Bond options are quoted in yield volatility and converted to price volatility by duration, and an option-adjusted spread means nothing without its volatility assumption.

Negative rates made the lognormal assumption not merely inaccurate but undefined, since $\ln(F_0/K)$ requires a positive forward. The market moved to normal (Bachelier) volatilities, under which an at-the-money option is worth $\sigma_N\sqrt{T/2\pi}$ per unit of numeraire, and to shifted lognormal conventions. Near the money, $\sigma_N \approx \sigma_{\mathrm{LN}}F_0$.

The swaption market is described by a volatility cube over expiration, tenor, and strike. Volatility humps along the expiration dimension and declines along the tenor dimension, since a longer swap rate averages more forwards. The cube is fitted with a parametric model, usually SABR, rather than interpolated, because arbitrary interpolation admits arbitrage; Hagan's expansion can itself imply a negative density at low strikes, which the shifted and no-arbitrage variants repair. Hedging inherits the cube's dimensionality: bucketed delta, vega by expiration and tenor, and a delta that decays through the accrual period of a backward-looking product.

FURTHER READING

Black's "The Pricing of Commodity Contracts" (1976) is the origin of the formula this chapter applies to caps and swaptions. Bachelier's 1900 thesis is the normal model, written five years before Einstein's treatment of Brownian motion and worth reading now that rates have made the normal convention standard again. Hagan, Kumar, Lesniewski, and Woodward's "Managing Smile Risk" (2002) introduced SABR and is the source of the expansion in (29.22). Andersen and Piterbarg's three-volume *Interest Rate Modeling* is the reference to reach for when the market conventions here are not enough, and Lyashenko and Mercurio (2019) is where the $T_1 + \tau/3$ result for backward-looking rates comes from.

PRACTICE QUESTIONS

29.1 Explain why Black's model can be applied to a caplet without any adjustment for the drift of the forward rate. Which discount factor must be used, and why that one?

29.2 A caplet covers a three-month period beginning in two years, on a notional of \$20 million. The forward rate is 3.60%, the strike is 3.75%, the volatility is 18%, and the discount factor to the payment date is 0.9280. Value the caplet, treating the rate as fixed in advance.

29.3 Repeat Question 29.2 treating the rate as compounded in arrears. By what percentage does the value increase?

29.4 State cap–floor parity and prove it from the payoffs. Explain how you would use it to check a set of broker quotes.

29.5 Explain the difference between a flat volatility and a spot volatility. Why is it wrong to compare the flat volatilities quoted for a three-year and a seven-year cap?

29.6 Explain why the first caplet of a spot-starting cap was conventionally excluded under LIBOR, and why that convention is no longer appropriate.

29.7 A five-year option on a three-year annual-pay swap has a forward swap rate of 3.80%, a strike of 3.80%, and a volatility of 22%. The annuity is 2.6540 per unit of notional. Value the payer swaption on a notional of \$50 million, and state the value of the corresponding receiver.

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29.8 Show that at the money the Bachelier formula reduces to $\sigma_N \sqrt{T/2\pi}$ per unit of numeraire.

29.9 An at-the-money swaption is quoted at a lognormal volatility of 24% with a forward swap rate of 3.20%. Estimate the equivalent normal volatility, and explain why the rule of thumb works.

29.10 Explain why the volatility of a swap rate generally declines as the tenor of the underlying swap increases.

29.11 Explain, without algebra, why a cap must be worth at least as much as a swaption written on the same schedule and strike. Under what condition would the two be equal?

29.12 Eight forward rates each have a volatility of 24% and enter a swap rate with equal weight. Using (29.17), find the implied swap rate volatility when the uniform pairwise correlation is 1.0, 0.6, and 0.2. Comment on what the results imply about calibrating a model to caps alone.

FURTHER QUESTIONS

29.13 Derive the result $\text{Var} = \sigma^2(T_1 + \tau/3)$ for a backward-looking rate, stating clearly the assumption made about how sensitivity decays through the accrual period. How would the answer change if the decay were assumed to be quadratic rather than linear?

29.14 A cap consists of four quarterly caplets beginning immediately, with $\tau = 0.25$. Using a flat volatility of 20%, a strike of 4.00%, and a forward rate of 4.00% for every period, and ignoring discounting, compare the total value of the cap when the caplets are treated as forward-looking against the value when they are treated as backward-looking. Comment on where the difference is concentrated.

29.15 A trader observes that three-month expiration swaptions are quoted at 85 basis points of normal volatility while one-year expiration swaptions on the same tenor are quoted at 78. Explain what this implies about the expected path of rate uncertainty, and identify at least two reasons this need not represent an arbitrage.

29.16 Explain why a delta computed under Black's model differs from one computed under Bachelier for the same option at the same price. Under what circumstances is the difference largest, and what additional information would you need to decide which is the better hedge ratio?

29.17 In 2022 the level of rates and the volatility of rates rose together. Explain, using the relationship between normal and lognormal volatility, why a desk quoting in normal volatilities and a desk quoting in lognormal volatilities would have reported very different percentage increases in "volatility" over that year, even though both observed the same market.

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Chapter 30
Convexity, Timing, and Quanto Adjustments †

Almost everything in the preceding chapters rests on the same approach: when the expected future value of a rate or a price is required, today's forward value is used in its place. Forward contracts were valued that way, swaps were valued that way, and Chapter 29 priced caps and swaptions by assuming the relevant forward rate was the mean of a lognormal distribution.

That approach is valid far more widely than one might expect, but it is not universal. It is the consequence of a specific and rather delicate alignment: the forward value of a variable equals its expected future value *only under one particular probability measure*, and which measure that is depends on when the payoff is made and in what currency.

When a contract respects that alignment (a rate observed at the start of a period and paid at the end of it, in its own currency), the forward rate is exactly the right expectation and no adjustment is needed. When a contract breaks the alignment, the forward rate is the wrong expectation, and using it introduces a pricing error that can run from a fraction of a basis point to tens of basis points.

This chapter is about the three ways the alignment breaks and the corrections each one requires:

- a **convexity adjustment**, when the instrument's settlement mechanics change the measure, most importantly when daily margining makes a futures rate behave differently from a forward rate;
- a **timing adjustment**, when a payoff is made at a date other than the one naturally associated with the variable;
- a **quanto adjustment**, when a payoff denominated in one currency is based on a variable belonging to another.

All three are the same phenomenon in a different form, and Section 30.1 explains why. The unifying idea is worth the effort: once you see that each adjustment is a change of measure, you can derive corrections for contracts this chapter never mentions, rather than memorizing a catalog of formulas.

These adjustments are not academic refinements. The convexity adjustment on a ten-year interest rate futures contract can exceed 50 basis points—larger than the bid–ask spread by two orders of magnitude, and large enough that a curve built without it will misprice every long-dated swap on the book. Desks employ people to get these numbers right.

The chapter runs from the principle to the three corrections. Section 30.1 states when a forward value is an expectation and under which measure. Sections 30.2 to 30.5 take the breaks in turn: a payment made at the wrong time, the daily settlement that separates a futures contract from a forward, and the swap rate paid outside its own annuity, which is the largest of the adjustments that come from payment timing. Section 30.6 is the currency break, and

Section 30.7 shows the three as one calculation done three times. Sections 30.2 through 30.6 can be read in any order, and each adjustment is derived where it is stated; those derivations rest on Chapter 28 and can be skipped without losing the results.

30.1 FORWARD VALUES ARE EXPECTATIONS—UNDER THE RIGHT MEASURE

The general principle comes first, because the three adjustments that follow are special cases of it.

The chapter requires a single concept, Chapter 28's numeraire: any asset with a strictly positive price in which all other prices are measured. What follows is the working statement of it, and nothing here requires more. If $N(t)$ is the price of a traded asset that pays no income, then there exists a probability measure (call it Q^N, the measure *associated with* the numeraire N) under which the price of every other traded asset, divided by N, is a martingale: for any traded asset with price $V(t)$,

$$\frac{V(t)}{N(t)} = E^N\left[\frac{V(T)}{N(T)}\,\middle|\,\mathcal{F}_t\right], \qquad t \le T$$

where E^N denotes expectation under Q^N and $\mathcal{F}_t$ the information available at t. This is the fundamental result of arbitrage pricing, and everything in this chapter follows from it. Only the case $t = 0$ is used below; the martingale property is the stronger statement that the equality holds at every intermediate date as well, which is what makes a change of numeraire legitimate at any point in a contract's life.

Two choices of numeraire dominate practice.

The money market account. Take $N(t) = B(t)$, the value of a bank account earning the instantaneous short rate. The associated measure is the familiar risk-neutral measure Q, in which expected returns equal the risk-free rate.

The zero-coupon bond. Take $N(t) = P(t,T)$, the price at time t of a bond paying \$1 at time T. The associated measure is called the **T-forward measure**, written Q^T. It has a property that makes it enormously convenient: since $P(T,T) = 1$,

$$V(0) = P(0,T)\,E^T[V(T)] \tag{30.1}$$

The value today of a payoff received at time T is the discount factor times the expected payoff under the T-forward measure: no stochastic discount factor appears inside the expectation, and discounting and expectation separate cleanly. This is why the T-forward measure is the natural home for any contract with a single payment date T.

Which rates are martingales, and under which measure

Now consider the forward rate. Let $F(t;T_1,T_2)$ be the forward rate observed at time t for the period $[T_1,T_2]$, with accrual factor $\tau = T_2 - T_1$, defined in the usual way from bond prices:

$$F(t;T_1,T_2) = \frac{1}{\tau}\left[\frac{P(t,T_1)}{P(t,T_2)} - 1\right] = \frac{P(t,T_1) - P(t,T_2)}{\tau\,P(t,T_2)} \tag{30.2}$$

Look at the right-hand expression. The numerator, $P(t,T_1) - P(t,T_2)$, is the price of a traded portfolio: long one T_1-bond, short one T_2-bond. The denominator is τ times the price of the T_2-bond. So F is the price of a traded asset divided by τ times a numeraire, and therefore F is a **martingale under** Q^{T_2}:

$$E^{T_2}[F(T_1;T_1,T_2)] = F(0;T_1,T_2) \tag{30.3}$$

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At $t = T_1$ the period begins immediately, so $F(T_1; T_1, T_2)$ is the spot rate for $[T_1, T_2]$, which later sections write $R(T_1)$. Equation (30.3) is therefore the precise statement of the approach relied on throughout: the forward rate is the expected future spot rate, but only under the T_2-forward measure, where T_2 is the date the rate is *naturally* paid.

The two results together give the chapter its subject. Equation (30.1) requires an expectation under Q^T to value a payoff received at time T; equation (30.3) supplies an unbiased expectation under Q^{T_2}. If the payment date T and the natural date T_2 coincide, the two align and the forward rate is used directly. If they do not, because the payment comes early, or late, or in a different currency, or because margining has changed the effective numeraire to the money market account, then the required expectation is under a different measure (Q^T if the payment moves, Q if margining makes the money market account the numeraire), while only $E^{T_2}[F]$ is known, and the difference is the adjustment. Call the measure under which the variable's forward value is an unbiased expectation its **natural measure** (here Q^{T_2}) and the measure the contract's payment calls for the **required measure**; each section below names both.

Why an adjustment is a covariance

Changing measure is done with a Radon–Nikodym derivative, and for forward measures it takes a particularly interpretable form. Moving from Q^{T_2} to Q^T requires reweighting each scenario by the ratio of the two numeraires' growth:

$$\left.\frac{dQ^T}{dQ^{T_2}}\right|_{\mathcal{F}_t} = \frac{P(t,T) \,/\, P(0,T)}{P(t,T_2) \,/\, P(0,T_2)} \tag{30.4}$$

Write Λ_t for this ratio (Chapter 28's density process, with Q^{T_2} in place of P and Q^T in place of Q) and Λ for its value at the date the variable is observed. It is not constant: the two time-zero prices are fixed, so it moves with $P(t,T)/P(t,T_2)$, which is the discount factor between the two dates, not with either bond price on its own, since a rise in rates lowers both. When the required date is the later one, $T > T_2$, that factor falls as rates rise, so high-rate scenarios are downweighted and low-rate scenarios upweighted; when $T < T_2$ the weighting runs the other way. The reweighting is monotone in rates either way, and that is what matters here. Therefore:

> **If the variable being adjusted is correlated with interest rates, reweighting the scenarios changes its mean. If it is uncorrelated, the mean is unchanged and no adjustment is needed.**

Every adjustment in this chapter is a covariance with that density, and at leading order a covariance between the variable and whatever drives it: an interest rate, or an exchange rate. The sign of that covariance gives the sign of the adjustment, which is enough to check any formula in this chapter without re-deriving it.

Section 28.6 derives (30.4) and one consequence that every adjustment below is an instance of. Because the density has unit mean, for any X known at t,

$$E^T[X] = E^{T_2}[X] + \mathrm{Cov}^{T_2}(X, \Lambda_t) \tag{30.5}$$

Every derivation in this chapter is an evaluation of (30.5) for a particular X, and the work each time is finding what Λ_t is a function of.

30.2 TIMING ADJUSTMENTS

Start with the simplest break in the alignment: a payoff based on a variable observed at time T^*, but paid at some later time T rather than at T^*.

Let V be the variable, with known forward value V_0, meaning $E^{T^*}[V(T^*)] = V_0$. What is required is $E^T[V(T^*)]$.

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Write $D = P(T^*, T)$ for the discount factor over the deferral period and $D_0 = P(0,T)/P(0,T^*)$ for its forward value today. Evaluating (30.4) at $t = T^*$ and using $P(T^*, T^*) = 1$, the density is simply the discount factor rescaled,

$$\Lambda \;=\; \frac{P(T^*,T)\,P(0,T^*)}{P(0,T)} \;=\; \frac{D}{D_0},$$

and $E^{T^*}[D] = D_0$ because $P(t,T)/P(t,T^*)$ is a traded asset over the numeraire. So (30.5) reads

$$E^T[V(T^*)] \;=\; V_0 \;+\; \frac{\mathrm{Cov}^{T^*}(V,D)}{D_0} \tag{30.6}$$

exactly. The timing adjustment is the covariance of the payoff with the discount factor that defers it, and nothing has been approximated yet.

What has to be approximated is the translation into quoted terms, because the market quotes the volatility of a rate rather than of a discount factor. Writing $\tau = T - T^*$ and y for the simply-compounded rate covering $[T^*, T]$, the two bond prices are defined by that rate,

$$D = \frac{1}{1+y\tau}, \qquad D_0 = \frac{1}{1+y_0\tau},$$

so $\Lambda = (1+y_0\tau)/(1+y\tau)$, with $\Lambda(y_0) = 1$ and $\Lambda'(y_0) = -\tau/(1+y_0\tau)$. Linearizing Λ about y_0 turns (30.6) into a covariance with the rate,

$$E^T[V] \;\approx\; V_0 - \frac{\tau}{1+y_0\tau}\,\mathrm{Cov}^{T^*}(V,y),$$

and for jointly lognormal V and y that covariance is $V_0 y_0 \rho \sigma_V \sigma_y T^*$, giving:

$$E^T[V(T^*)] \;\approx\; V_0\left[1 - \frac{\rho\,\sigma_V\,\sigma_y\,y_0\,\tau\,T^*}{1+y_0\tau}\right] \tag{30.7}$$

or, in the exponential form the market usually quotes, which agrees with (30.7) to first order and is exact when V and D are jointly lognormal ([TN30] shows why, and how much the linearization costs),

$$E^T[V(T^*)] \;\approx\; V_0 \exp\!\left(-\frac{\rho\,\sigma_V\,\sigma_y\,y_0\,\tau\,T^*}{1+y_0\tau}\right) \tag{30.8}$$

where

- y_0 is today's forward rate for the deferral period $[T^*, T]$, simply compounded over τ;
- σ_y is the volatility of that forward rate (proportional, that is, lognormal);
- σ_V is the volatility of the forward value of V;
- ρ is the correlation between V and y.

Reading the formula. The adjustment is negative when $\rho > 0$. The intuition is exactly the covariance argument of Section 30.1: deferring a payment means discounting it over $[T^*, T]$ at a rate that is not known today. If V tends to be large precisely when that rate is high, then the large payoffs are the ones that get discounted most heavily, and the deferred claim is worth less than a naive calculation suggests. If V and rates are uncorrelated, $\rho = 0$ and no adjustment is needed no matter how long the deferral.

Note also that the adjustment scales with τT^*, the product of the *deferral length* and the *time until observation*. A long deferral of a payoff observed tomorrow needs almost no adjustment, because there is almost no time for V and y to become correlated in their realized movements.

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Example 30.1 Payment deferred by one quarter

A structured note pays the five-year swap rate observed at the end of year 5. Payment, however, is made at the end of year 5.25 rather than immediately on observation.

The forward five-year swap rate is $V_0 = 4.00\%$ with volatility $\sigma_V = 20\%$. The forward rate covering the deferral quarter is $y_0 = 4.00\%$ with volatility $\sigma_y = 20\%$. The correlation between the two is $\rho = 0.7$. Here $T^* = 5$ and $\tau = 0.25$.

The exponent in (30.8) is

$$-\frac{0.7 \times 0.20 \times 0.20 \times 0.04 \times 0.25 \times 5}{1 + 0.04 \times 0.25} = -\frac{0.0014}{1.01} = -0.001386$$

so the adjustment factor is $e^{-0.001386} = 0.998615$ and

$$E^T[V] = 4.00\% \times 0.998615 = 3.9945\%$$

The timing adjustment is -0.55 basis points.

The half basis point of Example 30.1 may look like a rounding error, and on a single trade it is. On a \$5 billion structured note program it is \$275,000 per year of exposure, and it is the kind of error that is systematically in one direction, so it never averages out across the book. This is the general character of timing adjustments: individually small, collectively worth getting right.

30.3 CONVEXITY ADJUSTMENTS I: FUTURES VERSUS FORWARDS

The largest and most economically important convexity adjustment in fixed income arises from a difference in settlement mechanics rather than payment dates. It is the reason a futures rate is not a forward rate.

Why margining changes the measure

Recall Chapter 2. A forward contract settles once, at maturity. A futures contract is marked to market daily: gains and losses are paid in cash as they accrue.

That difference determines the numeraire. Because the cash flows on a futures contract occur continuously and are immediately available for investment (or must be immediately funded), the correct numeraire is the **money market account**, and Section 28.7 showed that the futures price is then a martingale under the risk-neutral measure Q. The forward price is a martingale under Q^T. So for an asset settling at T,

$$\text{Futures price} = E^Q[S_T], \qquad \text{Forward price} = E^T[S_T] \tag{30.9}$$

Two measures, two expectations of the same terminal price, and (30.5) compares them. The density taking Q to Q^T is the T-bond over the money market account, normalized,

$$\Lambda = \frac{1}{B(T)\,P(0,T)}$$

so

$$\text{Forward price} - \text{Futures price} = \text{Cov}^Q(S_T,\, \Lambda) \tag{30.10}$$

and Λ is large exactly when $B(T)$ is small, that is, when rates have been low. An asset positively correlated with interest rates therefore has a futures price above its forward price, and one negatively correlated has a futures price below it. The sign follows from the covariance alone; no model of rates has been assumed.

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Which is larger, and why

Section 5.12 settled the direction from the margin flows alone, before any of the measure language was available:

$$\text{Futures price} \;>\; \text{Forward price} \qquad \text{(asset positively correlated with rates)}$$

with a negative correlation reversing it. Equation (30.10) verifies that argument: the covariance it exhibits carries exactly the sign Chapter 5 reasoned to from the cash flows, and, being an expectation, it can also be evaluated.

An interest rate futures contract is the extreme case, which is why the adjustment is largest here: its underlying is the rate itself, so the correlation is perfect rather than incidental. Writing the futures rate as the rate the quoted price implies (100 less the quotation, for the contracts of Section 6.2), it is an affine function (a constant plus a fixed multiple) of a martingale and a martingale itself, so (30.9) becomes

$$\text{Futures rate} = E^{Q}[R(T_1)], \qquad \text{Forward rate} = E^{T_2}[R(T_1)] \tag{30.11}$$

the second by (30.3). The futures price falls as the rate rises, so the inequality above inverts when read in rates: the futures rate exceeds the forward rate by the convexity adjustment.

The size of the adjustment

Under the Ho–Lee model (the simplest no-arbitrage short-rate model, developed with its mean-reverting extension in Sections 31.5 and 31.6), the short rate follows

$$dr = \theta(t)\,dt + \sigma\,dW$$

with $\theta(t)$ a drift chosen so that the model reproduces today's term structure and σ the standard deviation of the *absolute* (not proportional) change in the short rate per year, and the adjustment takes a memorable form:

$$\text{Forward rate} \;=\; \text{Futures rate} \;-\; \frac{1}{2}\,\sigma^2\,T_1\,T_2 \tag{30.12}$$

where T_1 is the time until the futures contract's rate is determined and T_2 is the time until the end of the period the rate covers.

Strictly, (30.12) and (30.13) adjust the continuously compounded rate; for the simply compounded quarterly rate on which the contract settles, add about $\frac{1}{2}\sigma^2\tau T_1$, which is 0.1 basis points at one year and under 3% of the adjustment at ten. The structure of (30.12) carries the lesson. The adjustment grows with the square of maturity, so it is negligible for near-dated contracts and dominant for far-dated ones. Chapter 6 stated the same formula and worked it at a normal volatility of 100 basis points a year (Section 6.2 and Example 6.12): 0.63 basis points on the one-year contract, 33 at eight years and about 51 at ten, where treating futures rates as forward rates puts an error into the curve two orders of magnitude larger than the bid–ask spread.

Ho–Lee overstates the long end, because it has no mean reversion and mean reversion damps the volatility of long-dated rates. Adding it gives Hull–White,

$$dr = (\theta(t) - ar)\,dt + \sigma\,dW$$

where a is the speed at which the short rate is pulled back toward its mean, and the adjustment becomes

$$\text{Futures} - \text{Forward} = \frac{1}{\tau}\,B(T_1,T_2)\left[B(T_1,T_2)\left(1 - e^{-2aT_1}\right) + 2a\,B(0,T_1)^2\right]\frac{\sigma^2}{4a} \tag{30.13}$$

where

$$B(t_1,t_2) = \frac{1 - e^{-a(t_2-t_1)}}{a}$$

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As $a \to 0$ this must reduce to (30.12), and it does: $B(T_1, T_2) \to \tau$, $B(0, T_1) \to T_1$, and $1 - e^{-2aT_1} \to 2aT_1$, so the bracket becomes $2aT_1T_2$ and the whole expression collapses to $\frac{1}{2}\sigma^2 T_1 T_2$. The check matters because the mean-reverting form is easy to mistype, and a wrong a is invisible at the short end and large at the long one. Equation (30.13) is what desks use.

One caution applies to both formulas. The σ in each is a normal (absolute) volatility. Substituting a lognormal volatility of, say, 20% into (30.12) produces a number four hundred times too large. Volatility quoting conventions are the most common source of error here.

Figure 30.1 plots both versions across the strip. Two features repay attention. The first is the shape: the adjustment is almost invisible for the first two years and then accelerates, because it grows with the square of maturity. A desk can reasonably ignore convexity on the front contracts and cannot possibly ignore it on the back ones. The second is the gap between the curves. Mean reversion pulls long-dated rate volatility down, so the Hull–White adjustment falls progressively below Ho–Lee as maturity extends, by roughly a quarter at ten years in this calibration. Using Ho–Lee at the long end does not merely approximate the adjustment; it systematically overstates it.

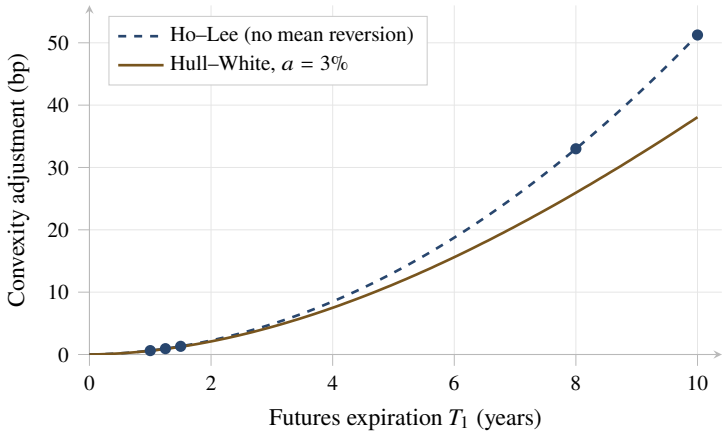


Figure 30.1: The futures–forward convexity adjustment against contract expiration, for the continuously compounded rate, three-month contracts and a short-rate volatility of 100 basis points per year. Markers show the values quoted in Chapter 6.

The modern complication: averaging futures

Equation (30.12) was developed for Eurodollar futures, which settled against a *term* rate: three-month LIBOR, fixed on a single date. The contracts that replaced them work differently, and the difference is worth understanding.

Three-month SOFR futures settle against the **compounded daily average** of the overnight rate over the reference quarter. One-month SOFR futures settle against the **arithmetic average** of the overnight rate over the calendar month. In neither case is there a single fixing date; the settlement rate accrues day by day across the contract's reference period.

This changes the risk profile in a way that matters for hedging more than for pricing:

- **The magnitude of the convexity adjustment is broadly similar.** At the leading order captured by Ho–Lee, an averaging contract and a term contract of the same expiration carry adjustments within a basis point of each other for typical quarterly periods. Equation (30.12) remains a reasonable first approximation.

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- **The delta decays during the reference period.** As each day's overnight rate is observed, that portion of the settlement rate is fixed and its sensitivity disappears. A three-month SOFR futures contract's rate exposure declines roughly linearly to zero across its reference quarter. A Eurodollar futures contract's exposure, by contrast, persisted at full size until the fixing date and then vanished at once.
- **Realized volatility of the settlement rate is lower**, because averaging suppresses the variance of the daily fixings, a point that matters more for options on the averaged rate than for the futures themselves. Chapter 29 computes that variance: the average carries only a third of the accrual period's variance, so it is below that of the period's last fixing, though still above that of a rate fixed when the period begins.

The practical consequence is that convexity *hedging* under SOFR requires continuous rebalancing through the reference period rather than a single roll, but curve construction still requires the same order-of-magnitude correction it always did.

Case Study 30.1 Convexity as a trade

The gap between futures and forward rates is not merely a modeling nuisance; it has long been a trade in its own right. A **convexity trade** buys futures and pays fixed on a swap covering the same period, locking in the higher futures rate on one side and the lower forward rate on the other and collecting the spread between them.

The position is not a riskless profit. The trader collects the convexity adjustment but takes on the exposure it compensates for: the trade loses when realized rate volatility exceeds the volatility implied in the adjustment. It is, in substance, a short volatility position in fixed-income form, with the risk profile that phrase implies. Convexity trades performed poorly through 2022, when realized rate volatility ran far above the levels embedded in convexity adjustments computed from the preceding decade's calm.

The general lesson recurs throughout the book: a pricing adjustment that compensates for a risk can always be harvested as a return, and the return is never free.

30.4 CONVEXITY ADJUSTMENTS II: RATES PAID AT THE WRONG TIME

The second family of convexity adjustments arises when a rate is paid at a date other than the end of the period it covers. Historically this was the domain of the *in-arrears swap*; under risk-free reference rates it has changed character in an instructive way.

The classical case: rates paid in arrears

Under LIBOR conventions, a rate for $[T_1, T_2]$ was *set* at T_1 and *paid* at T_2, the natural alignment. An **in-arrears swap** broke it by paying the rate at T_1, immediately on observation.

What is now required is $E^{T_1}[R(T_1)]$ rather than $E^{T_2}[R(T_1)] = F$. The natural measure is Q^{T_2} and the required measure is Q^{T_1}, so the frame is the same as before; what is different is the density. Since $P(T_1, T_2) = 1/(1 + R(T_1)\tau)$, evaluating (30.4) at $t = T_1$ with $T = T_1$ gives

$$\Lambda = \frac{1}{P(T_1, T_2)} \cdot \frac{P(0, T_2)}{P(0, T_1)} = \frac{1 + R(T_1)\tau}{1 + F\tau}$$

which has unit mean under Q^{T_2} precisely because $E^{T_2}[R] = F$. This density is *affine* in R, and that is what makes the case exact: where Section 30.2 had to linearize a density that was not, here (30.5) can be evaluated as it stands,

$$\mathrm{Cov}^{T_2}(R, \Lambda) = \frac{\tau}{1 + F\tau}\,\mathrm{Cov}^{T_2}(R, R) = \frac{\tau\,\mathrm{Var}^{T_2}(R)}{1 + F\tau}$$

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so that

$$E^{T_1}[R(T_1)] \;=\; F \;+\; \frac{\tau\,\mathrm{Var}^{T_2}(R)}{1+F\tau} \tag{30.14}$$

exactly, with no approximation beyond the model for the variance. If R is lognormal with volatility σ, then $\mathrm{Var}^{T_2}(R) = F^2\left(e^{\sigma^2 T_1}-1\right)$, giving

$$E^{T_1}[R(T_1)] \;=\; F + \frac{F^2\left(e^{\sigma^2 T_1}-1\right)\tau}{1+F\tau} \;\approx\; F + \frac{F^2\sigma^2 T_1\tau}{1+F\tau} \tag{30.15}$$

The adjustment is positive, unlike the timing adjustment of Section 30.2, and the reason is simple: here the payment is being *accelerated*, not deferred. Receiving a rate early is worth more when rates are high, so the scenarios that get upweighted are the favorable ones.

Example 30.2 In-arrears convexity

A swap pays the three-month rate for the period beginning in five years, but pays it at the start of that period rather than the end. The forward rate is $F = 4.00\%$ with volatility $\sigma = 25\%$; $T_1 = 5$ and $\tau = 0.25$.

The exact lognormal variance is

$$\mathrm{Var} = F^2\left(e^{\sigma^2 T_1}-1\right) = 0.0016\times\left(e^{0.3125}-1\right) = 0.0016\times 0.36684 = 5.8694\times 10^{-4}$$

so from (30.14) the adjustment is

$$\frac{0.25\times 5.8694\times 10^{-4}}{1.01} = 1.453\times 10^{-4} = 1.45\text{ bp}$$

The rate to use is 4.0145%. The first-order approximation $F^2\sigma^2 T_1$ in (30.15) gives 1.24 bp—an error of 15% in the adjustment itself. At long maturities and high volatilities the exact form is worth using.

What survives under risk-free rates

A compounded RFR is determined at the end of its accrual period and paid then, so the standard product no longer breaks the alignment and the classical adjustment of Example 30.2 largely dissolves; Section 33.4 prices what is left, the timing adjustment for the few days between the end of observation and payment, and finds it a few hundredths of a basis point even at ten years. The exception is the "last reset" convention of Chapter 4, which pays a rate determined at the end of the *previous* period one period late: that is a deferral, and it carries a timing adjustment in the opposite direction to (30.14), priced with Section 30.2's machinery.

30.5 CONVEXITY ADJUSTMENTS III: CMS AND THE ANNUITY MEASURE

The third convexity adjustment is the largest of the payment-timing family and, unlike Section 30.4's, was unaffected by the LIBOR transition. It concerns constant-maturity swaps.

The product and the problem

Recall from Section 28.8 that a CMS coupon pays a long *swap rate* (the ten-year rate, say) observed on each reset date. Unlike an ordinary floating leg, it is not matched to the period it is paid for.

A related and equally common structure is the **CMS spread**, paying the difference between two swap rates of different tenors (the 10-year less the 2-year, for instance). These are the building blocks of steepener notes, which pay attractive coupons unless the yield curve flattens.

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The measure problem here is structural. A swap rate S is the fixed rate that makes a swap worth zero; it equals the value of the floating leg divided by the annuity of Section 28.8 (the present value of the fixed leg per unit of fixed rate, so that one basis point on the fixed leg is worth 0.0001 A):

$$S(t) = \frac{\text{floating leg}(t)}{A(t)}, \qquad A(t) = \sum_{i=1}^{n} \tau_i \, P(t, t_i)$$

The annuity $A(t)$ is the price of a traded portfolio of bonds, so it is a legitimate numeraire, and S, being a traded asset divided by A, is a martingale under the associated swap measure Q^A. This is what justifies Black's formula for swaptions in Chapter 29, where the annuity appears naturally as the discounting term.

But a CMS coupon is a *single* cash flow at a *single* date T_2. Valuing it requires $E^{T_2}[S]$, and the annuity measure supplies $E^A[S] = S_0$. The numeraires do not match, so an adjustment is required.

In the frame of Section 30.1, the natural measure is Q^A, the required measure is Q^{T_2}, and the density between them is the T_2-bond over the annuity, normalized,

$$\Lambda_t = \frac{P(t, T_2) \,/\, P(0, T_2)}{A(t) \,/\, A(0)}$$

so the adjustment is $\mathrm{Cov}^A(S, \Lambda)$ by (30.5), exactly as in the two sections before this one. Evaluating that covariance needs Λ written as a function of the swap rate, and supplying it is what the next subsection's G does.

The adjustment

The correction depends on how the annuity varies with the swap rate, which is a question about the convexity of a bond price–yield relationship. Define $G(y)$ as the price, per dollar of principal, of a bond paying coupon S_0 with the swap's payment frequency and maturity, valued at flat yield y. Because the coupon equals the forward swap rate, $G(S_0) = 1$.

The trick is to adjust the bond price rather than the rate, because the bond price is the martingale. Its forward price satisfies $E[G(S(T_1))] = G(S_0)$, so expanding G to second order about S_0,

$$G(S_0) = E[G(S(T_1))] \approx G(S_0) + G'(S_0)\,(E[S] - S_0) + \frac{1}{2} G''(S_0)\,\mathrm{Var}(S)$$

the leading terms cancel and solving for $E[S]$ gives the adjustment directly. Then

$$E^{T_2}[S(T_1)] \;\approx\; S_0 \;-\; \frac{1}{2}\, S_0^2\, \sigma^2\, T_1\, \frac{G''(S_0)}{G'(S_0)} \tag{30.16}$$

where σ is the volatility of the forward swap rate.

Since bond prices fall as yields rise, $G' < 0$; since bond prices are convex in yield, $G'' > 0$. The ratio G''/G' is therefore negative and the whole adjustment is positive. The CMS rate to use always exceeds the forward swap rate.

The intuition is the familiar one from Chapter 4's treatment of duration and convexity. The holder of a CMS coupon has exposure to a long-maturity rate but does not hold the corresponding bond, so they capture the rate's movements without the offsetting convexity of a bond position. That asymmetry is worth something, and (30.16) prices it.

Figure 30.2 shows where the adjustment comes from geometrically (at the parameters of Example 30.3 the gap is a fraction of a percent of the price, far smaller than drawn), and the same picture explains every convexity adjustment in this chapter. What is pinned down is the *bond price*: G is a martingale, so $E[G(S)] = G(S_0)$. Now suppose the swap rate

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were distributed symmetrically about S_0. Because G is convex, the average of G over the two outcomes lies *above* G evaluated at the average: the chord sits above the curve, which is Jensen's inequality. That would violate the martingale condition. The only way to restore it is for the distribution of S to sit at a higher level, since G is decreasing. The distance the rate must shift is precisely the convexity adjustment.

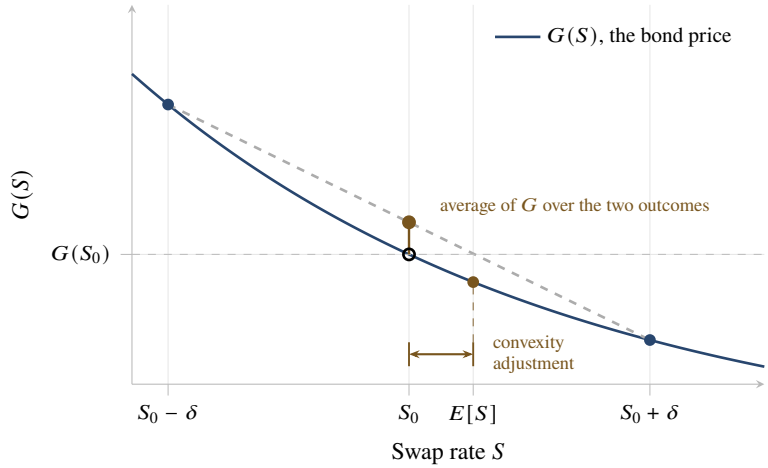


Figure 30.2: Why the expected swap rate exceeds the forward swap rate. Curvature exaggerated.

Example 30.3 CMS convexity adjustment

Consider a coupon paying the five-year swap rate observed in five years' time. The forward five-year swap rate is $S_0 = 4.00\%$, its volatility is $\sigma = 20\%$, payments are annual, and $T_1 = 5$.

$G(y)$ is the price of a five-year annual bond with a 4% coupon. Since the coupon equals S_0, the bond prices at par when $y = 4\%$. Differentiating,

$$G'(0.04) = -4.4518, \qquad G''(0.04) = 25.0125$$

so $G''/G' = -5.6185$ and

$$E^{T_2}[S] = 0.04 - \frac{1}{2} \times 0.04^2 \times 0.20^2 \times 5 \times (-5.6185) = 0.04 + 0.000899$$

The CMS rate is **4.090%** against a forward swap rate of 4.000%—a convexity adjustment of **9.0 basis points**.

Nine basis points on a rate of four percent is a 2.2% relative correction, and it grows with the square of volatility and linearly with time to reset. On a twenty-year CMS structure the adjustment can exceed 50 basis points. Unlike the timing adjustments of Section 30.2, this is not a refinement that can be safely ignored.

CMS spreads

The structures that trade in volume are usually not single CMS coupons but *spreads* between two swap rates. Since the payoff is a difference, it is tempting to assume the adjustments on the two legs largely offset. They do not: the ratio G''/G' that drives the adjustment depends strongly on the underlying swap's maturity. A ten-year bond has far more convexity relative to its duration than a two-year bond, so the long leg carries much the larger correction.

One arithmetic simplification makes these calculations tractable by hand. Because the bond in G is constructed with coupon equal to the forward swap rate, it prices at par, and for a par

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bond the first derivative is exactly the negative of the annuity:

$$G'(S_0) = -A(S_0), \qquad A(S_0) = \sum_{i=1}^{n} \frac{1}{(1+S_0)^i} = \frac{1-(1+S_0)^{-n}}{S_0} \tag{30.17}$$

The 5-year figure $G'(0.04) = -4.4518$ used in Example 30.3 is exactly the five-year annuity at 4%, and can be read off (30.17) without differentiating anything.

Example 30.4 A CMS spread coupon

A note pays the ten-year swap rate less the two-year swap rate, both observed in five years. The forward rates and their volatilities are

$$S_0^{10} = 4.40\%, \ \ \sigma_{10} = 18\%; \qquad S_0^{2} = 3.90\%, \ \ \sigma_{2} = 24\%$$

with annual payments and $T_1 = 5$. The forward spread is $4.40\% - 3.90\% = 50.0$ bp.

Each leg must be adjusted separately, using (30.16) with its own maturity, level, and volatility. Applying (30.17) for G' and the series of Technical Note 30 for G'':

	2-year leg	10-year leg
Forward swap rate S_0	3.90%	4.40%
Volatility σ	24%	18%
Annuity $A = -G'$	1.8888	7.9518
G''	5.4189	78.3883
$-G''/G'$	2.8690	9.8580
Convexity adjustment	6.28 bp	15.46 bp
Adjusted rate	3.9628%	4.5546%

The adjusted spread is therefore

$$4.5546\% - 3.9628\% = 59.18 \text{ basis points}$$

against a raw forward spread of 50.0 basis points—a net convexity effect of **+9.2 basis points**, which is 18% of the spread being valued.

The two-year leg is assumed *more* volatile than the ten-year, as short rates typically are, and this works against the effect. It is nowhere near enough to overcome it: the ratio $-G''/G'$ is 3.4 times larger for the ten-year leg, and that maturity mismatch dominates everything else. Had both legs referenced swaps of the same maturity, the adjustments would have largely canceled, leaving only a small net correction.

Two warnings attach to Example 30.4. First, it values the *linear* spread. Real steepener notes pay something like $\max(\text{spread}, 0)$, or a leveraged multiple of it, and an option on a difference depends on the **correlation** between the two swap rates, a quantity no convexity adjustment captures. Valuing the floored payoff requires a two-factor model or a replication argument, and the correlation involved is neither observable nor stable.

Second, the sign of the effect is systematic. Because the long leg carries the larger adjustment unless the short leg's volatility is far higher than in this example, convexity *widens* the fair spread on a steepener. A dealer who omitted the adjustment would sell every such note too cheaply, and would do so consistently rather than randomly.

Beyond the simple formula

Equation (30.16) assumes a single volatility for the swap rate, which is equivalent to assuming Black's model. Since the adjustment is fundamentally a statement about the *convexity* of the payoff, it is sensitive to the whole shape of the volatility smile, not just the at-the-money level.

Desks therefore price CMS by **static replication**: a CMS coupon decomposes into a

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portfolio of swaptions across all strikes, weighted by the payoff's second derivative. Write $H(s) = s\,\alpha_2(s)$, where $\alpha_2(s) = E^A[P(T_1,T_2)/A(T_1) \mid S(T_1) = s]$ is the annuity mapping function of Section 28.12 for payment at T_2. Then

$$V_0 = A(0)\,H(S_0) + \int_0^{S_0} H''(K)\,\mathrm{Rec}(K)\,dK + \int_{S_0}^{\infty} H''(K)\,\mathrm{Pay}(K)\,dK \tag{30.18}$$

where $\mathrm{Rec}(K)$ and $\mathrm{Pay}(K)$ are today's prices of receiver and payer swaptions struck at K: receivers below the forward swap rate, payers above, each weighted by $H''(K)$. This approach captures the whole volatility surface and is the market standard. Section 12.8 builds an arbitrary payoff from a strip of options and Section 22.3 applies the construction to variance; the CMS case is its most commercially important application in rates. Equation (30.16) remains valuable as a quick estimate and a sanity check on the replication machinery.

Case Study 30.2 Steepeners and the flat curve

CMS spread structures were sold widely to retail and mid-market institutional investors during the 2000s, typically as notes paying a high fixed coupon for a year or two, then switching to a multiple of the 10-year-less-2-year swap spread.

Their appeal rested on the curve remaining upward-sloping. As policy rates rose (the Federal Reserve's until mid-2006, the European Central Bank's into 2008), the curve flattened and in places inverted, and the floating coupons on many of these notes fell to zero. They came back only when the rate cuts of 2008 steepened the curve again. Investors who had understood the note as a high-yielding bond discovered they had bought a leveraged bet on the shape of the yield curve.

The episode shows, obliquely, why this chapter's arithmetic matters. The convexity adjustment is part of the price of the curve exposure embedded in these structures, and the floor on the spread is an option whose value turns on the correlation the adjustment does not capture. An investor who could compute neither could not tell what they had been paid for the risk—and, as it turned out, most had not been paid nearly enough.

30.6 QUANTO ADJUSTMENTS

The final adjustment concerns currency rather than timing. A quanto (Section 16.7) is a derivative whose payoff depends on a variable denominated in one currency but is settled in another, at a *fixed* exchange rate specified in the contract.

Examples are common. A dollar-settled option on the Nikkei 225 that pays \$1 per index point removes the yen exposure entirely: the investor gets the Japanese equity return without the currency return. A euro-settled swap referencing dollar interest rates works the same way.

Why the FX rate matters even though it is fixed

The apparent paradox of a quanto is that the exchange rate is contractually fixed, yet exchange rate volatility affects the price. The resolution lies in hedging.

A dealer short a quanto on a foreign asset hedges by holding that asset, so the hedge's gains and losses arrive in foreign currency and are worth, at home, whatever the exchange rate is when they arrive. The quanto's payoff ignores the exchange rate. If the asset tends to rise when the foreign currency is strong, the hedge's gains are converted at a high rate and its losses at a low one, so in domestic currency the hedge systematically outruns the payoff it replicates. If the asset tends to rise when the foreign currency weakens, it lags. Either way the effect recurs, a benefit or a cost according to the correlation's sign, and it must appear in the price.

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The adjustment

Adopt the convention that S is the exchange rate expressed as **units of domestic currency per unit of foreign currency** (for a US investor holding a Japanese asset: USD per JPY). The convention matters: reversing it reverses the sign of the answer. Let V be the foreign-currency price of the underlying asset, σ_V and σ_S their volatilities, and ρ the correlation between them.

The correction is the same change of numeraire as the rest of the chapter. A foreign asset's natural measure is the *foreign* risk-neutral measure Q^f, the one that goes with the foreign money market account B_f; a quanto settles in domestic currency, so the required measure is the domestic Q^d, whose numeraire is B_d. The density taking one to the other is the domestic account over the domestic value of the foreign one, normalized,

$$\Lambda_t = \frac{B_d(t)}{S(t)\,B_f(t)} \cdot S(0)$$

which moves *against* the exchange rate, so (30.5) charges minus the covariance between the asset and S, the quanto effect in the same form as every other adjustment here, and with the sign that makes a positively correlated asset drift below r_f. [TN30] carries the derivation out in full. Under Q^d the asset's drift is therefore not its foreign risk-free rate r_f but

$$\mu_V = r_f - \rho\,\sigma_V\,\sigma_S \tag{30.19}$$

This can also be checked without the density, in two steps. Under the domestic risk-neutral measure, an investor holding foreign cash owns a traded domestic asset worth $S(t)e^{r_f t}$, whose drift must be the domestic risk-free rate r_d, so $\mu_S = r_d - r_f$, which is covered interest parity. The foreign asset's domestic value, VS, is also a traded domestic asset, and Itô's product rule gives

$$\frac{d(VS)}{VS} = (\mu_V + \mu_S + \rho\,\sigma_V\sigma_S)\,dt + \sigma_V\,dW_V + \sigma_S\,dW_S$$

whose drift must also be r_d; substituting $\mu_S = r_d - r_f$ gives (30.19). The cross-variation term $\rho\sigma_V\sigma_S$ is the entire source of the quanto effect, and it is the covariance the density charges.

The term $-\rho\sigma_V\sigma_S$ is the **quanto drift adjustment**. The measure change is the machinery, the covariance is the economics, and the hedging argument above is what the covariance means on a desk. Hence the quanto forward relates to the ordinary foreign-currency forward F_0^f by

$$\boxed{F_0^{\text{quanto}} \;=\; F_0^f\,e^{-\rho\,\sigma_V\,\sigma_S\,T}} \tag{30.20}$$

Once (30.19) is in hand, quanto derivatives are priced with entirely standard machinery: use the adjusted drift, and otherwise apply the Black–Scholes–Merton model or Black's model as usual. A quanto call, for instance, is valued with the ordinary Black formula using F_0^{quanto} in place of the forward and discounting at the *domestic* rate.

Section 16.7 priced a quanto forward and call at a positive correlation, where the adjustment lowers the forward. For the dollar-settled Nikkei position above it raises it: Japanese equities have tended to rise as the yen weakens ($\rho < 0$ in dollars per yen), so the dealer's hedge earns its index gains in a weakened yen, and at $\rho = -0.3$, $\sigma_V = 20\%$ and $\sigma_S = 10\%$ the one-year quanto forward sits about 0.6% above the yen forward to pay for that shortfall.

Two observations follow. First, the adjustment vanishes when $\rho = 0$: an asset uncorrelated with its own exchange rate needs no quanto correction, however volatile either may be. Second, ρ here is a correlation between an asset and an exchange rate over the life of the contract, which is notoriously unstable: the Nikkei–yen correlation has changed sign across regimes. Quanto positions carry a **correlation risk** that cannot be hedged with liquid instruments, and desks reserve against it instead.

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30.7 A UNIFIED VIEW

Every adjustment here has been the same calculation: name the measure under which the variable's forward value is unbiased, name the measure the contract's payoff requires, write the density between them, and take the covariance. Only the density changes, as the table shows, where each arrow runs from the first measure to the second.

Adjustment	Measure change	Covariance that drives it
Futures/forward convexity (30.12)	$Q^{T_2} \longrightarrow Q$ (money market)	Margin flows with the financing rate
Timing (30.8)	$Q^{T^*} \longrightarrow Q^T$	Variable with the deferral rate
In-arrears (30.14)	$Q^{T_2} \longrightarrow Q^{T_1}$	Rate with itself
CMS (30.16)	Q^A (annuity) $\longrightarrow Q^{T_2}$	Swap rate with the annuity
Quanto (30.20)	Q^f (foreign) $\longrightarrow Q^d$ (domestic)	Asset with the exchange rate

Reading across the table, the recipe for any adjustment is the same:

1. Identify the measure under which the variable's forward value is an unbiased expectation, its **natural measure**, determined by the numeraire that appears in its definition.
2. Identify the measure required by the contract's actual payoff date and currency.
3. Compute the covariance of the variable with whatever drives the numeraire ratio.
4. The adjustment is that covariance, to first order.

Two practical warnings close the chapter.

Adjustments are model-dependent, and the dependence is not small. Every formula here required a distributional assumption: Ho–Lee in (30.12), lognormality in (30.8), (30.15), and (30.16). Each depends on a volatility, and (30.8) and (30.20) additionally depend on a correlation. These are estimates, not observables, and the adjustment inherits their uncertainty. The 9.0 bp CMS adjustment of Example 30.3, computed at 20% volatility, becomes a 20 bp adjustment at 30% volatility—an increase of more than double. Reporting a convexity adjustment to three decimal places implies a precision that the inputs do not support.

Adjustments are systematic, not random. Unlike bid–ask spread or execution slippage, an omitted adjustment is a bias in a known direction that accumulates across every trade of its type on the book. This is why a desk that tolerates a basis point of noise will not tolerate half a basis point of bias, and why these corrections are computed and monitored rather than ignored.

SUMMARY

Forward values equal expected future values only under the measure associated with the contract's natural numeraire. When a contract's settlement mechanics, payment date, or settlement currency depart from that natural arrangement, the forward value is a biased estimate and an adjustment is required. Every such adjustment is exactly a covariance between the variable being valued and the density that changes the numeraire; it can be evaluated exactly where that density is affine in the variable, and to leading order otherwise.

Because futures contracts are settled daily, futures rates are expectations under the risk-neutral measure while forward rates are expectations under the forward measure. The resulting convexity adjustment makes futures rates exceed forward rates, by an amount growing with the square of maturity, negligible at the front of the strip and material at long maturities.

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Averaging-style RFR futures carry adjustments of similar magnitude to the term-rate contracts they replaced, but with delta that decays through the reference period.

Timing adjustments apply when a payoff is deferred or accelerated relative to its natural payment date. They are negative for deferral when the variable is positively correlated with rates, and are typically small. The transition to risk-free rates largely eliminated the classical in-arrears convexity adjustment, since a compounded RFR is naturally determined and paid at the end of its period, but left smaller residual adjustments, from payment delays and from last-reset conventions that pay a stale rate one period late.

CMS convexity is the largest adjustment of the payment-timing family, arising because a swap rate is a martingale under the annuity measure while a CMS coupon is paid at a single date. It is always positive, large enough to matter in pricing, and is priced in practice by static replication with swaptions so that the full volatility smile is captured.

Quanto adjustments arise when a payoff based on a foreign variable is settled in domestic currency at a fixed exchange rate. Exchange rate volatility affects the price despite the fixed rate because it affects the dealer's hedging cost. The adjustment multiplies the foreign forward by $e^{-\rho\sigma_V\sigma_S T}$ and vanishes when the asset and exchange rate are uncorrelated.

All these adjustments depend on volatilities and correlations that are estimated rather than observed. Their importance lies less in their size than in their systematic direction: an omitted adjustment biases every affected trade the same way.

FURTHER READING

Geman, El Karoui, and Rochet (1995) is the foundational reference for the change-of-numeraire machinery this chapter runs on, and the source of the change-of-numeraire formulas used here. Brigo and Mercurio's *Interest Rate Models—Theory and Practice* treats convexity adjustments and CMS pricing in full in its Chapter 13. For the two adjustments computed here in closed form, Kirikos and Novak (1997) derive the Hull–White futures convexity adjustment and Hagan's "Convexity Conundrums" (2003) is the standard practitioner treatment of static replication for CMS.

PRACTICE QUESTIONS

30.1 Explain why a forward rate is a martingale under the T_2-forward measure but not under the risk-neutral measure. Which measure is relevant for valuing a payoff made at T_2?

30.2 Without performing any calculation, state the sign of the required adjustment in each case, and justify your answer in one sentence:

(a) a variable uncorrelated with interest rates, paid three months after its natural date;

(b) a ten-year swap rate paid as a single cash flow in five years;

(c) a quanto forward on a foreign equity index, where the index and the foreign currency are positively correlated.

30.3 A three-month interest rate futures contract has $T_1 = 3$ years and $T_2 = 3.25$ years. The short-rate volatility is 90 basis points per year. The futures rate is 3.85%. Calculate the forward rate using the Ho–Lee adjustment.

30.4 Repeat Question 30.3 with $T_1 = 8$ and $T_2 = 8.25$. By what factor does the adjustment increase, and does that match what equation (30.12) predicts?

30.5 Explain in words why a trader who is long the rate through a futures position benefits from daily settlement, and why this requires the futures rate to exceed the forward rate.

30.6 A payoff equal to the two-year swap rate is observed in three years and paid six months later. The forward two-year swap rate is 3.50% with 22% volatility; the forward rate for

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the deferral period is 3.20% with 25% volatility; the correlation is 0.8. Calculate the timing adjustment in basis points.

30.7 A rate for a six-month period beginning in four years is paid at the start of the period rather than the end. The forward rate is 3.75% with volatility 28%. Calculate the in-arrears convexity adjustment using both the exact form (30.14) and the approximation in (30.15), and comment on the difference.

30.8 Explain why the transition from LIBOR to compounded risk-free rates eliminated most of the in-arrears convexity adjustment. Identify two situations in which a convexity adjustment is still required under RFR conventions.

30.9 A quanto forward on a foreign stock index is settled in domestic currency. The index has 25% volatility, the exchange rate (domestic per foreign) has 12% volatility, and the correlation is 0.4. The one-year foreign-currency forward level is 5,000. Calculate the quanto forward level.

30.10 Continuing Question 30.9, the domestic interest rate is 3%. Value a one-year quanto call on the index, struck at the quanto forward level and paying one unit of domestic currency per index point, and compare it with the same call if the correlation were zero. Why does the correlation matter when the contract's exchange rate is fixed?

30.11 Without using algebra, explain why the convexity adjustments on the two legs of a CMS spread do not cancel. Under what condition would they cancel?

FURTHER QUESTIONS

30.12 A CMS coupon pays the ten-year swap rate observed in three years. The forward ten-year swap rate is 4.20%, its volatility is 18%, and payments are annual. Compute G' and G'' for a ten-year annual bond with a 4.20% coupon at a flat yield of 4.20%, and hence find the convexity adjustment. Compare the size of your answer with Example 30.3 and explain the difference.

30.13 Using the result of Question 30.12, recompute the adjustment at volatilities of 12% and 24%. Comment on what your answers imply about the precision with which convexity adjustments can be quoted.

30.14 A dealer sells a quanto call on a foreign index. Describe the dealer's hedge, explain how the hedge must be rebalanced as the index moves, and use this to give an economic derivation of the sign of the quanto adjustment in equation (30.20). Why can this position not be fully hedged in practice?

30.15 A structured note pays $3 \times$ (10-year swap rate $-$ 2-year swap rate) annually for ten years, subject to a floor of zero. Identify every adjustment from this chapter that its valuation requires, and explain which of them dominates.

30.16 Show that a rate futures contract's convexity adjustment can be interpreted as the value of an option-like exposure to realized rate volatility. Using this interpretation, explain why a convexity trade that appeared profitable on 2019 volatility estimates would have lost money through 2022.

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Chapter 31
Short-Rate Models

Chapter 29 priced a cap and a swaption with Black's model and Chapter 30 corrected the forward when the payoff arrives at the wrong time or in the wrong currency. Both took a quoted volatility and produced one number.

Neither built a model of the term structure, and there is a large class of instruments for which that is fatal. A **Bermudan swaption** can be exercised on any one of a fixed set of dates; its holder decides at year three whether to exercise, and that decision depends on the entire curve at year three—a curve Black's model never constructs. A callable bond's issuer faces the same decision annually for a decade. A structured note pays a coupon that depends on where rates are, conditional on where they have been.

For these, the model has to produce a consistent evolution of the whole curve, not a distribution for one rate at one date. A desk running a book needs more than that: one model that prices every deal on it consistently, so that a swaption, a callable bond and a Bermudan are valued off the same curve dynamics and their risks aggregate. The oldest and still most widely used way to do that is to model the short rate and derive everything else from it.

Section 31.12 is the one that changes how the rest of Part VI reads: it prices the defect that motivates Chapter 32, and it does so with a number rather than an assertion. A one-factor model prices a CMS spread option at 56.2 basis points when, at a realistic correlation between the two rates, it is worth 245.0.

The chapter runs from the models to the defect that motivates the next one. Sections 31.1 to 31.4 are the equilibrium models, Vasicek and Cox–Ingersoll–Ross (CIR), and why a model that cannot reprice today's curve is not used to price. Sections 31.5 to 31.8 are the no-arbitrage answer: Ho–Lee, Hull–White, and the tree that implements it. Sections 31.9 to 31.11 are what the tree is built for: Bermudan swaptions, what calibration cannot pin down, and callable bonds through option-adjusted spread. Sections 31.12 to 31.15 are the repairs and their limits: the one-factor failure priced in basis points, two factors, negative rates, and the transition; Sections 31.16 to 31.18 say where these models are still the right tool and who uses them.

31.1 WHAT A SHORT-RATE MODEL IS

Specify a stochastic process for the instantaneous short rate $r(t)$ (the rate for borrowing over the next instant, the limit of the zero rate as maturity shrinks to zero) under the risk-neutral measure. Every zero-coupon bond price then follows from

$$P(t,T) = \hat{E}_t\left[\exp\left(-\int_t^T r(s)\,ds\right)\right] \tag{31.1}$$

which is Chapter 13's risk-neutral valuation applied to a payoff of \$1.

This is a strong modeling choice. The entire term structure, at every future date, is generated by a single scalar state variable. That is the source of the approach's tractability and of the defect in Section 31.12, and the two cannot be separated.

Two properties make a model usable.

Analytic bond prices. If (31.1) evaluates in closed form, the model is fast enough to calibrate. The widest class for which it does is the **affine** models, in which $\ln P(t,T)$ is affine in r:

$$P(t,T) = A(t,T)\, e^{-B(t,T)\, r(t)} \tag{31.2}$$

Written in rates rather than prices, with $R(t,T)$ the continuously compounded zero rate of Section 4.8,

$$R(t,T) = -\frac{\ln P(t,T)}{T-t} = \frac{B(t,T)\, r(t) - \ln A(t,T)}{T-t}$$

so every rate on the curve is affine in the one state variable, and Example 31.1's curves follow from it.

And the ability to fit today's curve. A model whose output curve differs from the observed one misprices the underlying swaps and bonds themselves, before any option is considered. This requirement, not realism, is what separated the two generations of models, and Section 31.4 makes the separation concrete.

31.2 VASICEK AND THE EQUILIBRIUM APPROACH

The first generation asked what process for r is economically reasonable, and derived the curve from it. Vasicek's model is

$$dr = a(b - r)\, dt + \sigma\, dW \tag{31.3}$$

The drift pulls r toward b at speed a: **mean reversion**, and the property that distinguishes rates from equities. A stock price has no level it returns to; a short rate does, because a central bank and an economy both act on it. Figure 31.1 shows what reversion does to the whole distribution, not just to the expected rate. It is slow at plausible speeds: the half-life, the time over which the expected gap between r and b halves, is $\ln 2/a$, about seven years at $a = 0.10$, so over the life of a ten-year deal the pull toward b is partial rather than complete. Set $a = 0$ and the rate is a driftless random walk whose spread grows as $\sigma\sqrt{t}$ without limit; restore a and the spread converges to $\sigma/\sqrt{2a}$. That second half is what lets the model produce a finite long rate at all, and Section 31.5 returns to it, because the $a = 0$ case is a model in its own right. Note which rate reverts to b: the expected short rate does, as t grows, while the zero curve's long end sits at the lower level R_∞ marked on the figure, whatever the short rate happens to be.

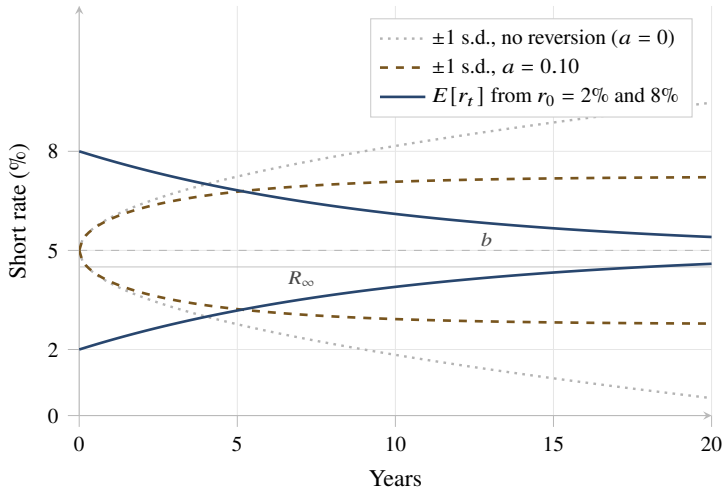


Figure 31.1: Mean reversion at $a = 0.10$, $b = 5\%$, $\sigma = 1\%$, against the same model with $a = 0$; R_∞ is the limiting zero rate of Example 31.1.

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Equation (31.3) is affine, with

$$B(t,T) = \frac{1 - e^{-a(T-t)}}{a}$$

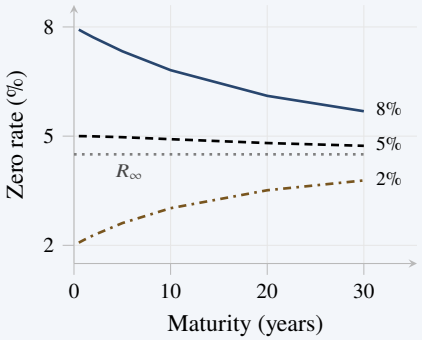
$$\ln A(t,T) = \frac{(B(t,T) - (T-t))\left(a^2 b - \sigma^2/2\right)}{a^2} - \frac{\sigma^2 B(t,T)^2}{4a}$$

Bond prices, and therefore the whole curve, follow from $r(t)$ and three constants. The two terms in σ^2 are the convexity correction. The $-\sigma^2/2$ inside the first is what pulls the long end of the curve below b in Example 31.1; the last term's contribution to the zero rate dies away as maturity grows.

Example 31.1 Three starting rates, one curve

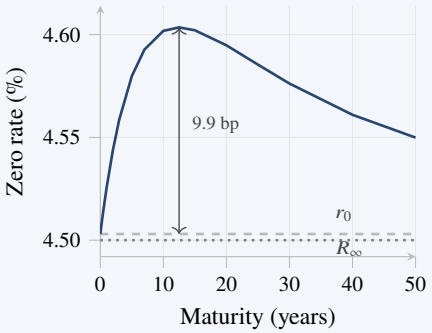
Take $a = 0.10$, $b = 5\%$, $\sigma = 1\%$, and compute the zero curve from three different starting short rates.

Maturity	$r_0 = 2\%$	$r_0 = 5\%$	$r_0 = 8\%$
0.5y	2.073%	5.000%	7.926%
1y	2.144%	4.999%	7.853%
2y	2.275%	4.994%	7.713%
5y	2.610%	4.971%	7.332%
10y	3.020%	4.916%	6.812%
20y	3.513%	4.810%	6.107%
30y	3.783%	4.734%	5.684%



The model produces upward-sloping, flat, and downward-sloping curves from the same parameters: the shape is determined by where r_0 sits relative to b.

A fourth shape is available, but barely. The curve is humped for $b - 3\sigma^2/4a^2 < r_0 < b$, a band 75 basis points wide at these parameters, and the largest hump anywhere in that band is 9.9 basis points, at $r_0 = 4.50\%$ peaking at twelve and a half years. Real curves hump by five to ten times that. *The shape is in the model; the magnitude is not.*



$a = 0.10$, $b = 5\%$, $\sigma = 1\%$, $r_0 = 4.503\%$

All three converge toward the same long rate, and it is not b:

$$R_\infty = b - \frac{\sigma^2}{2a^2} = 5\% - \frac{0.0001}{0.02} = \mathbf{4.50}\%$$

The gap is a convexity effect, and it is the same effect as Chapter 30's: a long bond's price is convex in the rate, so uncertainty raises its expected price and lowers its yield. Here the model produces the adjustment automatically instead of requiring it to be added.

But now count the degrees of freedom. Three parameters and one state variable generate the entire curve. The curve has hundreds of observable points and the model has four numbers, so it cannot match the market. That is not a calibration failure; it is arithmetic.

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31.3 COX–INGERSOLL–ROSS

Vasicek's r is normally distributed and can go negative. The best-known repair is

$$dr = a(b - r)\,dt + \sigma\sqrt{r}\,dW \tag{31.4}$$

The $\sqrt{r}$ has two consequences. Volatility falls as the rate falls, consistent with the pre-2008 data, and the diffusion coefficient vanishes at zero, so the rate remains nonnegative. Strict positivity requires more, namely the **Feller condition** $2ab > \sigma^2$; when it fails, zero is attained and acts as a reflecting boundary.

The model remains affine, so bond prices retain a closed form, and its coefficients are more involved than Vasicek's without being harder to evaluate. With $\tau = T - t$ and $\gamma = \sqrt{a^2 + 2\sigma^2}$,

$$B(t,T) = \frac{2\,(e^{\gamma\tau} - 1)}{(\gamma + a)\,(e^{\gamma\tau} - 1) + 2\gamma}$$

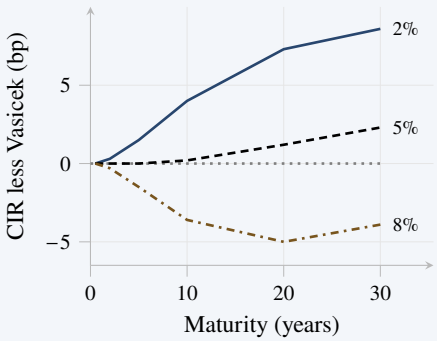
$$A(t,T) = \left[\frac{2\gamma\, e^{(a+\gamma)\tau/2}}{(\gamma + a)\,(e^{\gamma\tau} - 1) + 2\gamma}\right]^{2ab/\sigma^2}$$

The limiting zero rate is $R_\infty = 2ab/(\gamma + a)$, again below b and again by a convexity term, but one that now depends on the level of rates, because the volatility does. Example 31.2 measures what the change of diffusion does to the curves.

Example 31.2 What positivity costs

The two models are comparable only if their volatilities are. Vasicek's diffusion is σ and CIR's is $\sigma\sqrt{r}$, so the two agree at $r = b$ when CIR's σ is set to $0.01/\sqrt{0.05} = 4.47\%$. Take Example 31.1's $a = 0.10$ and $b = 5\%$ with that value, so both models carry a 1% instantaneous volatility at $r = 5\%$. The Feller condition holds comfortably, $2ab = 0.0100$ against $\sigma^2 = 0.0020$.

Maturity	$r_0 = 2\%$	$r_0 = 5\%$	$r_0 = 8\%$
0.5y	2.074%	5.000%	7.926%
1y	2.144%	4.998%	7.852%
2y	2.278%	4.994%	7.710%
5y	2.626%	4.971%	7.317%
10y	3.059%	4.918%	6.777%
20y	3.586%	4.821%	6.057%
30y	3.869%	4.757%	5.645%



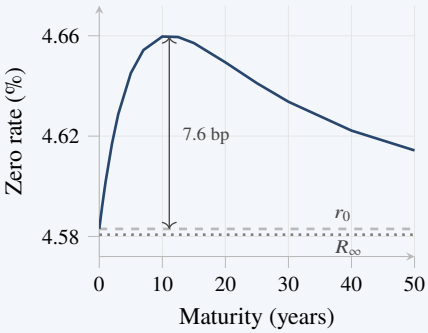
Left: CIR's zero curves. Right: those curves less Example 31.1's, in basis points.

The two sets of curves are nearly the same. The largest gap across thirty years is 8.6 basis points, and inside five years it is under two. Positivity cost almost nothing in the shape of the curve, which is why CIR held the field for two decades.

The gap that does appear has an instructive sign. From $r_0 = 2\%$ the CIR curve sits above Vasicek's, and from 8% below. A convexity correction grows with variance, and in CIR the variance moves with the rate: from a low start there is less of it to depress the long end, from a high start more, while Vasicek applies the same correction either way. The limiting zero rate records the same thing, $R_\infty = 4.58\%$ against Vasicek's 4.50%, a drag below b of 0.42 percentage points rather than 0.50.

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CIR humps too, and over a narrower band: from 4.40% to b, 60 basis points against Vasicek's 75. The largest hump in it is 7.6 basis points, at $r_0 = 4.58\%$ peaking at eleven years. *Both models put their best hump at R_∞ and neither can make it large.* Producing a realistic hump is not something a one-factor model does, whichever diffusion it carries.



$a = 0.10$, $b = 5\%$, $\sigma = 4.47\%$, $r_0 = 4.583\%$

The CIR model was for two decades the standard answer to Vasicek's negative rates, and what happened to that argument should be stated plainly. *Rates went negative*, in several major currencies, for years. A model that cannot produce a negative rate cannot be calibrated to a market that has one—not approximately, but at all, because there is no parameter value under which a process that cannot fall below zero assigns any probability to the observed state.

So the property that made CIR attractive is the property that retired it. Section 31.14 takes this up as a general point, because it recurs.

31.4 WHY EQUILIBRIUM MODELS ARE NOT USED TO PRICE

Example 31.1 contains the whole argument, and it deserves to be stated as a principle rather than a limitation of one model.

An **equilibrium model** treats today's term structure as an output. It asks what curve the assumed process implies, and the answer is a curve with three or four parameters' worth of freedom.

A trading desk needs today's term structure to be an input. If the model says the ten-year zero is 4.92% and the market says 4.10%, then every instrument priced on the model is wrong by an amount that has nothing to do with the option in it. A swaption priced on a curve that is eighty basis points off is not a swaption price.

The distinction is worth being precise about, because it is often stated as though equilibrium models were merely less accurate.

They are answering a different question, and it is a legitimate one. An economist asking what the curve *should* look like, or a risk manager generating scenarios that are internally consistent over decades, wants a model whose curve is derived rather than fitted. Real-world scenario generation for pension and insurance liabilities still uses models of this kind, for exactly that reason.

But a desk quoting a price cannot use one, and the second generation of models exists to remove this problem and nothing else.

31.5 HO–LEE AND THE NO-ARBITRAGE IDEA

Make the drift time-dependent and choose it so the model reproduces the observed curve. A model built this way, taking today's curve as an input rather than producing it, is a **no-arbitrage model**.

Start from what is observable. The input is today's zero curve, the discount factors $P(0,t)$ that Chapter 7 builds from quoted instruments. What a fitted model needs from them is not the zero rate $R(0,t)$ of Section 31.1 but the **initial forward curve**, the instantaneous rate for

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borrowing at t that today's prices imply:

$$f(0,t) = -\frac{\partial \ln P(0,t)}{\partial t} \qquad \text{equivalently} \qquad P(0,t) = \exp\left(-\int_0^t f(0,u)\,du\right)$$

The two descriptions carry the same information, and $R(0,t)$ is the average of $f(0,u)$ over $[0,t]$. The forward curve is the useful one here because a drift is a local quantity, so it has to be matched against a local description of the curve rather than against an average.

Ho–Lee is the simplest case:

$$dr = \theta(t)\,dt + \sigma\,dW \tag{31.5}$$

$\theta(t)$ is not a parameter to estimate; it is solved for:

$$\theta(t) = \frac{\partial f(0,t)}{\partial t} + \sigma^2 t$$

and the model fits the curve exactly by construction.

This is the whole idea of the no-arbitrage generation, and the surprising part is how cheap it is. One function of time, chosen mechanically, converts a model that cannot price into one that can. The cost is that $\theta(t)$ carries no information (it is whatever it has to be), so a model in this class explains nothing about why the curve has the shape it has. It is an interpolation device with a stochastic component, and that is the correct way to understand every model in the rest of this chapter.

Ho–Lee has one defect that rules it out in practice: *no mean reversion.* The volatility of the T-maturity forward rate is σ regardless of T, so the model says a thirty-year forward rate is as volatile as a one-year forward rate. Section 31.6 shows what the data say.

31.6 HULL–WHITE

Add mean reversion to Ho–Lee, or a fitted drift to Vasicek (the same model either way):

$$dr = \big[\theta(t) - ar\big]dt + \sigma\,dW \tag{31.6}$$

As in Ho–Lee, θ is solved for rather than estimated, and in terms of the same initial forward curve $f(0,t)$,

$$\theta(t) = \frac{\partial f(0,t)}{\partial t} + a\,f(0,t) + \frac{\sigma^2}{2a}\left(1 - e^{-2at}\right)$$

which returns Section 31.5's θ as $a \to 0$, the last term tending to $\sigma^2 t$. [TN31] derives this and explains why a single function of time can match every maturity at once. In practice it is never evaluated, because bond prices can be written with today's curve in them directly:

$$\ln A(t,T) = \ln\frac{P(0,T)}{P(0,t)} + B(t,T)\,f(0,t) - \frac{\sigma^2}{4a}\left(1 - e^{-2at}\right)B(t,T)^2$$

so θ disappears from everything that is computed. It was solved for the curve, and the curve is what remains.

This is the workhorse. It fits the initial curve exactly, has closed-form bond and European option prices, and builds on a recombining tree. Almost every desk that uses a short-rate model uses this one or a close relative.

The two constants do different jobs, and confusing them is the most common error in using the model.

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Example 31.3 What a and σ separately control

The instantaneous forward rate follows by differentiating (31.2):

$$f(t,T) = -\frac{\partial \ln P(t,T)}{\partial T} = -\frac{\partial \ln A(t,T)}{\partial T} + e^{-a(T-t)}\, r(t)$$

since $\partial B/\partial T = e^{-a(T-t)}$. Only r is stochastic, so the forward rate inherits its diffusion scaled by that factor, and its volatility seen from today is

$$\sigma\, e^{-aT} \tag{31.7}$$

T	$a = 0.01$	$a = 0.05$	$a = 0.10$	$a = 0.20$
1y	99.0	95.1	90.5	81.9
2y	98.0	90.5	81.9	67.0
5y	95.1	77.9	60.7	36.8
10y	90.5	60.7	36.8	13.5
20y	81.9	36.8	13.5	1.8
30y	74.1	22.3	5.0	0.2

Forward-rate volatility in basis points, at $\sigma = 100$ bp.

σ sets the level of the volatility term structure and a sets its shape. At $a = 0.01$ the curve is nearly flat. That is Ho–Lee, and it says a thirty-year forward is three-quarters as volatile as a one-year forward. At $a = 0.20$ the thirty-year forward has essentially no volatility at all, which is equally wrong in the other direction.

Because the two act on different features, they are separately identified by the data. Section 29.9's volatility cube contains caps and swaptions of many expirations, and the term structure of their volatilities determines a while their level determines σ.

That is the encouraging version. Section 31.10 shows that for the instruments that matter most, it is not true.

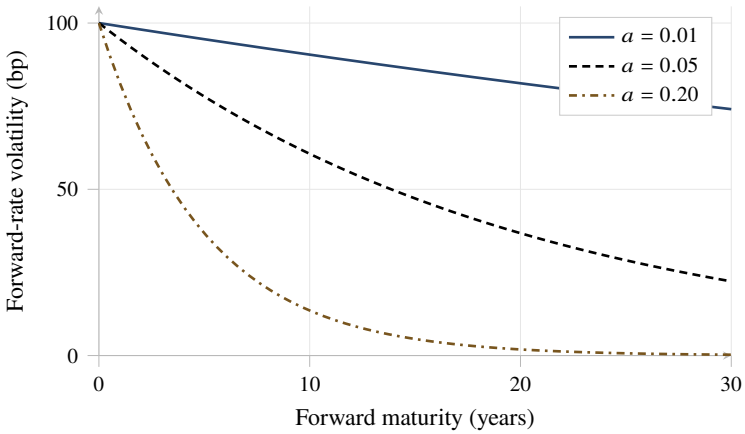


Figure 31.2: Equation (31.7) at $\sigma = 100$ bp.

Figure 31.2 is the parameter that makes Hull–White usable: a alone decides whether a thirty-year forward is nearly as volatile as a one-year forward or barely moves at all.

31.7 EUROPEAN OPTIONS IN CLOSED FORM

The model is usable before any lattice is built: every European rate option has a closed form, and they all come from one of them. Example 31.4 carries these formulas through a cap and a swaption.

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An option on a zero-coupon bond. Under the T-forward measure the forward bond price $P(t,S)/P(t,T)$ is a martingale, and since (31.2) makes $\ln P$ affine in a Gaussian r, it is a lognormal one. Its instantaneous volatility $\sigma e^{-a(T-t)}(1-e^{-a(S-T)})/a$ is deterministic, so the variance accumulates to a number rather than a random variable, and its square root over the option's life is

$$\sigma_P(T,S) = \frac{\sigma}{a}\left(1-e^{-a(S-T)}\right)\sqrt{\frac{1-e^{-2aT}}{2a}} \tag{31.8}$$

Lognormality with a deterministic volatility is the assumption on which Black's formula rests, so for a strike X and an expiration T on the bond maturing at S, the zero-bond call (ZBC) and put (ZBP) are, with Φ the standard normal distribution function,

$$\text{ZBC} = P(0,S)\,\Phi(h) - X\,P(0,T)\,\Phi(h-\sigma_P) \tag{31.9}$$

$$\text{ZBP} = X\,P(0,T)\,\Phi(\sigma_P - h) - P(0,S)\,\Phi(-h) \tag{31.10}$$

$$h = \frac{1}{\sigma_P}\ln\frac{P(0,S)}{X\,P(0,T)} + \frac{\sigma_P}{2}$$

the put (31.10) following from the call (31.9) by parity. Only today's curve and the two parameters appear. Both factors of (31.8) saturate (at σ/a in the bond's remaining life, at $\sigma/\sqrt{2a}$ in the option's), which is (31.7) seen from the other side. [TN31] derives (31.8) and works out both ceilings.

A cap. Section 29.3 showed that a caplet on notional L, struck at K on the rate for the accrual period $[T_i, T_{i+1}]$ of length τ_i, is $L(1+\tau_i K)$ puts on the T_{i+1} bond struck at $1/(1+\tau_i K)$. Substituting (31.10) prices a cap one period at a time, with no approximation anywhere.

A swap needs no formula at all. It is linear in bond prices,

$$V = L\left[P(0,T_0) - P(0,T_n) - K\sum_{i=1}^{n}\tau_i P(0,T_i)\right]$$

and neither a nor σ appears. A swap's value comes entirely from the curve. It could not be otherwise, since the model is fitted to that curve and reproduces it by construction. This locates where a model starts doing work. Its parameters do work only where there is optionality, and a model that misprices a swap has made a fitting error rather than a modeling one. That is Section 31.4's argument in one line.

A European swaption. Exercising a payer swaption is exercising the right to sell, at par, a bond paying $c_i = K\tau_i$ at each t_i, with the principal added to the last payment, $c_n = 1 + K\tau_n$; that is the equivalence of Section 29.6. What makes it tractable here is the single factor. Each $P(T,t_i)$ is a decreasing function of $r(T)$, so the coupon bond is too, and there is a unique r^* at which it is worth par. Above r^* *every* component bond is simultaneously below its value $X_i = P(T,t_i;\,r^*)$ there, and below r^* every one is above it. The exercise regions coincide, so

$$\text{payer swaption} = L\sum_{i=1}^{n} c_i\,\text{ZBP}(T,t_i,X_i) \tag{31.11}$$

This is **Jamshidian's decomposition**, and it is an identity rather than an approximation. It rests on monotonicity in a single state variable, so it survives neither a second factor (where the exercise region is a curve in the plane rather than a threshold) nor a Bermudan schedule, where the decision at one date turns on the value of the rights that remain. Both go to the tree of Section 31.8.

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Example 31.4 A cap and a swaption

Take a flat continuously compounded curve at 4%, so $P(0,T) = e^{-0.04T}$, with $a = 0.08$ and $\sigma = 1\%$ on a notional of 100. Left: a two-year semiannual cap struck at 4%, every period's forward rate being 4.040268%. Right: a one-into-three annual payer swaption struck at the par rate, whose annuity is 2.662181 and par rate 4.081077%, giving $r^* = 3.987895\%$ from (31.11).

Caplet	σ_P	Value
[0.5, 1.0]	33.98	0.142743
[1.0, 1.5]	47.12	0.190218
[1.5, 2.0]	56.60	0.222055
Cap		**0.555017**

Maturity	c_i	X_i	Value
2y	0.040811	0.960860	0.014025
3y	0.040811	0.923177	0.025775
4y	1.040811	0.886915	0.906913
Swaption			**0.946713**

σ_P in basis points.

The caplets rise in value with their expiration although the forward rate is identical in each, because σ_P grows with the option's life, and that growth is bounded, which is mean reversion at work. In the swaption almost all the value sits in the final strip, which carries the principal.

Both figures were also computed a second way, by integrating the payoff directly over the distribution of $r(T)$ with no decomposition and no bond-option formula, and the two routes agree to nine decimal places. That check is only correct under the T-forward measure, where

$$r(T) \sim \mathcal{N}\left(f(0,T),\ \frac{\sigma^2}{2a}\left(1 - e^{-2aT}\right)\right)$$

The mean is the forward rate exactly, while under the risk-neutral measure it carries a further $(\sigma^2/2a^2)(1 - e^{-aT})^2$. Integrating the swaption against that instead returns 0.952366 rather than 0.946713, an error of 0.6% from nothing but the wrong measure in the last step, which is the subject of the two examples that follow.

Chapter 28 proved that the choice of numeraire cannot change a price, and that the right choice can change the work enormously. With a model in hand, both claims can be put to a number.

Example 31.5 Invariance, and what it buys

A caplet on the simple rate over [2.0, 2.5], paid at 2.5, struck at 4%, notional 100. The economy is (31.6) with $a = 0.08$ and $\sigma = 1\%$, on a market curve carrying $P(0,2.0) = 0.932913$ and $P(0,2.5) = 0.914898$. The forward rate follows:

$$F_0 = \frac{1}{\tau}\left(\frac{P(0,T_1)}{P(0,T_2)} - 1\right) = 2.0 \times \left(\frac{0.932913}{0.914898} - 1\right) = 3.9381\%$$

(a) Under the 2.5-year forward measure. The forward rate is a martingale (no drift to model), and by the identity of Section 29.3, with $1 + \tau K = 1 + 0.5 \times 4\% = 1.02$, the caplet is $L(1 + \tau K) = 102$ puts on the 2.5-year bond struck at $1/(1 + \tau K) = 0.980392$. That bond is lognormal here, because (31.2) makes $\ln P$ affine in r and Hull–White's r is Gaussian; its forward price carries a deterministic volatility:

$$\sigma\, B(T_1,T_2)\sqrt{\frac{1 - e^{-2aT_1}}{2a}} = 64.1 \text{ bp}$$

As with the zero-bond options of (31.10), that makes Black's formula exact here rather than an approximation:

$$\text{caplet} = L \cdot P(0,T_2)\left[(1 + \tau F_0)\Phi(d_1) - (1 + \tau K)\Phi(d_2)\right] = \mathbf{0.22478353}$$

A closed form, evaluated in microseconds.

(b) Under the risk-neutral measure. Now the numeraire is the money market account, so the payoff must be discounted along each simulated path of r. Two million antithetic paths, two

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hundred steps:

0.22498411 (standard error 0.00023576)

The two agree to 0.85 standard errors. This is the theorem, confirmed: the numeraire is a choice of units and prices do not depend on it.

(c) The error the measure change exists to prevent. Suppose one computes the expected payoff under the risk-neutral measure and discounts it at today's zero rate, treating the payoff and the discount factor as independent:

0.23044665 an error of **+2.52**%

That 2.52% is exactly the covariance between the payoff and the discount factor. A caplet pays when rates are high, and when rates are high the discount factor is small. Ignoring that correlation overstates the value.

This is what the forward measure does. Using $P(t,T_2)$ as numeraire folds the discounting into the numeraire, so no covariance term survives. The reason Black's formula for a caplet is exact rather than approximate is that the convention already made this choice.

A measure change is not a formality; it is the difference between an exact formula and a simulation, and between a correct answer and one wrong by the covariance you forgot.

The same economy settles a second question. A forward price is an expectation under the T-forward measure, as Section 28.7 shows, and is not one under the risk-neutral measure; the difference is a covariance.

Example 31.6 And it is false under the risk-neutral measure

$S_0 = 100$, equity volatility 25%, $T = 5$, with the curve of Example 31.5 so that $P(0,T) = 0.824639$. The forward price is

$$F(0,T) = \frac{100}{0.824639} = 121.2652$$

and this is a no-arbitrage fact: it does not depend on any correlation.

Now compute $E^{Q}[S_T]$ under the risk-neutral measure, in the Hull–White economy of Example 31.5, for a range of correlations between the stock and the short rate. That expectation has a name. A futures price is a martingale under Q, as Section 28.7 established, so $E^{Q}[S_T]$ *is* the futures price for delivery at T, and the ratio below is the futures–forward basis of Section 30.3, computed inside a model rather than read off a screen:

$\rho(S,r)$	$E^{Q}[S_T]/F(0,T)$	**Gap**
−0.50	0.989442	−1.056%
−0.25	0.996260	−0.374%
0.00	1.003125	+0.313%
+0.25	1.010037	+1.004%
+0.50	1.016997	+1.700%

The forward price is fixed and $E^{Q}[S_T]$ moves with ρ, so with stochastic rates the two agree only at one particular correlation, by coincidence; in general they coincide only when rates are deterministic. Zero correlation is not enough: the $\rho = 0$ row still runs 0.31% high.

The gap has two parts. The part that moves with ρ is the covariance between the asset and the discount factor, the same quantity that produced the 2.52% error in Example 31.5. A caplet mispriced by ignoring that covariance and a futures price standing above a forward are one term under two names. The part that survives at $\rho = 0$ is the discount factor's own convexity:

$$E^{Q}\left[e^{\int_0^T r\,ds}\right] = \frac{e^{\mathrm{Var}\left(\int_0^T r\,ds\right)}}{P(0,T)}$$

This exceeds $1/P(0,T)$ whenever the short rate is stochastic.

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A practical warning follows. A Monte Carlo engine that simulates under the risk-neutral measure and reports "the expected terminal price" is not reporting the forward; it is reporting the futures price. At $\rho = 0.5$ over five years the difference is 1.7%, which is larger than the bid–ask on the forward it is being compared with.

31.8 BUILDING THE TREE

The closed forms of Section 31.7 cover the European cases. *Everything else needs a lattice*, and Hull–White's is built in two stages—which is the construction's central idea, not a computational detail.

Step one: the geometry and the probabilities. Stage one builds a tree for the process with no drift level,

$$dx = -ax\,dt + \sigma\,dW, \qquad x(0) = 0 \tag{31.12}$$

Nodes sit at $x = j\Delta x$. Over one step from level j the move has mean and variance

$$M = -a\,j\Delta x\,\Delta t, \qquad V = \sigma^2\Delta t$$

and the three branching probabilities are fixed by reproducing both, together with the requirement that they sum to one. Writing $\eta = M/\Delta x = -aj\Delta t$ and branching to $j+1$, j, $j-1$,

$$p_u + p_m + p_d = 1, \qquad p_u - p_d = \eta, \qquad p_u + p_d = \frac{V}{\Delta x^2} + \eta^2$$

The spacing is chosen to make this tidy. Setting $\Delta x = \sigma\sqrt{3\Delta t}$ turns $V/\Delta x^2$ into $1/3$, and the three equations solve to

$$p_u = \frac{1}{6} + \frac{\eta^2 + \eta}{2}, \qquad p_m = \frac{2}{3} - \eta^2, \qquad p_d = \frac{1}{6} + \frac{\eta^2 - \eta}{2} \tag{31.13}$$

At the center $\eta = 0$ and these are $1/6$, $2/3$, $1/6$. Nothing about the curve has entered yet, which is why stage one can be built once and reused.

Where symmetric branching breaks down. Write $u = aj\Delta t$, so $\eta = -u$. Both p_u and p_d in (31.13) stay positive for every u (each is a quadratic in u with negative discriminant), but the middle one does not:

$$p_m = \frac{2}{3} - u^2 < 0 \qquad \text{once} \qquad u > \sqrt{2/3} = 0.8165$$

In the node index that reads $j > 0.8165/(a\Delta t)$: far enough above the center, the drift is too strong for symmetric branching to match.

The remedy is to branch downward from the top node, to j, $j-1$, $j-2$. The same three equations then give

$$p_u = \frac{7}{6} + \frac{u^2 - 3u}{2}, \qquad p_m = -\frac{1}{3} - u^2 + 2u, \qquad p_d = \frac{1}{6} + \frac{u^2 - u}{2} \tag{31.14}$$

and again the middle probability is the binding one. Solving $-\frac{1}{3} - u^2 + 2u \geq 0$ gives

$$1 - \sqrt{2/3} \;\leq\; u \;\leq\; 1 + \sqrt{2/3}, \qquad \text{that is} \qquad 0.1835 \leq u \leq 1.8165$$

So each pattern is valid on an interval, and the intervals overlap on $0.1835 \leq u \leq 0.8165$. The switch has to happen somewhere inside that window, and taking it as early as possible keeps the tree narrowest. So $j_{\max}$ is the smallest integer satisfying

$$j_{\max} \;\geq\; \frac{1 - \sqrt{2/3}}{a\,\Delta t} \;=\; \frac{0.1835}{a\,\Delta t} \tag{31.15}$$

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that is, the first level at which the switched branching is legal. *The bottom of the tree mirrors the top:* at $-j_{\max}$ the branching runs up-up-middle, with (31.14) read at $-u$ and p_u and p_d exchanged.

Step two: displacing the tree onto the curve. Stage two shifts every level by an $\alpha(i)$, so that $r = x + \alpha(t)$, with x the process of (31.12), reprices today's zero curve. The curve is the one used throughout this chapter,

$$R(T) = 4.2\% - 1.2\%\, e^{-0.25T}, \qquad P(0,T) = e^{-R(T)\,T}$$

continuously compounded; the discount factors of Examples 31.5 and 31.6, and of every tree example that follows, come from it. The displacements are computed forward, one time step at a time, from the Arrow–Debreu prices. Write $Q(i,j)$ for the price, today, of a security that pays \$1 if the tree reaches node (i,j) and nothing otherwise (Section 13.3's state price, under the name the rates literature uses). The displacements are then *solved*, not fitted, and Example 31.7 carries that out with every number shown.

One feature of the geometry is worth a figure, because it is where the construction stops being obvious.

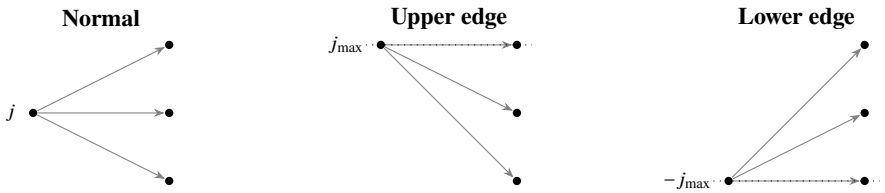


Figure 31.3: Branching switches at $\pm j_{\max}$. Schematic; spacing exaggerated.

Figure 31.3 is (31.15) drawn: below $j_{\max}$ the branching is symmetric, at the top node it runs down-down-middle, and at the bottom node up-up-middle.

The tree therefore stops widening. That is not an approximation—it is what mean reversion means, expressed in the geometry.

Example 31.7 Building the tree, step by step

Take this chapter's curve with $a = 0.08$ and $\sigma = 1\%$, but $\Delta t = 1$ year, so that every number fits on the page. The production tree described after this example runs the identical construction at $\Delta t = 0.25$.

Stage one fixes the geometry and never changes. $\Delta x = \sigma\sqrt{3\Delta t} = 0.017321$ and $j_{\max}$ is the smallest integer at or above $0.1835/(a\Delta t) = 2.294$, so $j_{\max} = 3$. The branching probabilities depend on j alone (not on the curve, not on the step), because stage one knows nothing about either:

j	p_u	p_m	p_d	**branches to**
+3	0.835467	0.089067	0.075467	$j,\ j-1,\ j-2$
+2	0.099467	0.641067	0.259467	$j+1,\ j,\ j-1$
+1	0.129867	0.660267	0.209867	$j+1,\ j,\ j-1$
0	0.166667	0.666667	0.166667	$j+1,\ j,\ j-1$
−1	0.209867	0.660267	0.129867	$j+1,\ j,\ j-1$
−2	0.259467	0.641067	0.099467	$j+1,\ j,\ j-1$
−3	0.075467	0.089067	0.835467	$j+2,\ j+1,\ j$

The drift shows in the asymmetry: above the center the down probability exceeds the up, and the imbalance grows with $|j|$. That is mean reversion, discretized. The two end rows come

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from (31.14), and their probabilities attach to different nodes, which is why p_u is the largest of the three at $j = +3$, where it is the probability of *staying* at j rather than of rising above it. Four steps is enough to reach them: the tree touches $j = \pm 3$ at level 3, and the switched rows are what carry it from there to level 4.

Stage two solves one equation per step. Begin with $Q(0,0) = 1$. At level i the displacement that makes the tree reproduce the next discount factor is

$$\alpha(i) = \frac{1}{\Delta t}\left[\ln \sum_j Q(i,j)\, e^{-j\Delta x\,\Delta t} \;-\; \ln P\big(0,(i+1)\Delta t\big)\right] \tag{31.16}$$

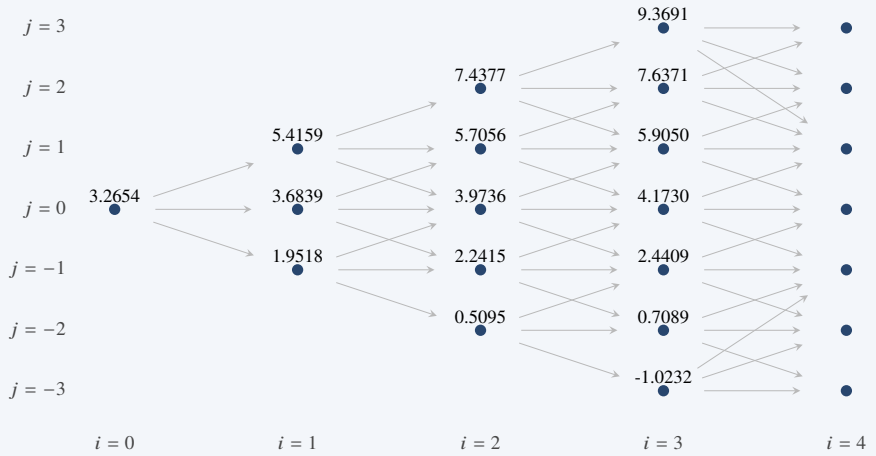
after which the Arrow–Debreu prices roll forward,

$$Q(i+1,k) = \sum_j Q(i,j)\; p(j \to k)\; e^{-(\alpha(i)+j\Delta x)\Delta t}$$

in which $p(j \to k)$ is whichever of p_u, p_m, p_d carries level j to level k, and zero if none of them does, so the *branches to* column above is what decides which. Four steps of that give

	$i=0$	$i=1$	$i=2$	$i=3$	$i=4$
$Q(i,+3)$	—	—	—	0.0018	0.0048
$Q(i,+2)$	—	—	0.0198	0.0369	0.0477
$Q(i,+1)$	—	0.1613	0.2045	0.2092	0.2017
$Q(i,\ 0)$	1.0000	0.6452	0.4799	0.3907	0.3356
$Q(i,-1)$	—	0.1613	0.2081	0.2165	0.2123
$Q(i,-2)$	—	—	0.0205	0.0395	0.0528
$Q(i,-3)$	—	—	—	0.0020	0.0056
$\alpha(i)$	3.2654%	3.6839%	3.9736%	4.1730%	—
$P(0,(i+1)\Delta t)$	0.9679	0.9329	0.8967	0.8604	—

The last column carries no α: level 4 is where this tree stops, and $\alpha(4)$ would be fixed by $P(0,5)$.



Short rates $r(i,j) = \alpha(i) + j\Delta x$, in percent; level 4 is terminal and carries none. *Levels 3 and 4 have the same seven nodes.* That is the tree refusing to widen once it reaches $\pm j_{\max}$.

At each level i the sum $\sum_j Q(i,j)\, e^{-r(i,j)\Delta t}$ returns $P(0,(i+1)\Delta t)$, the bottom row of the table, exactly, to the last bit.

Three features stand out. The Q's are not symmetric (0.0198 against 0.0205 at $i = 2$) because the discounting differs across nodes even though the probabilities at j and $-j$ mirror each other. α rises along the curve, from 3.27% to 4.17%, because it is absorbing the upward slope of the forward rates. And the bottom node at level 3 carries -1.0232%: a Gaussian short rate goes negative, which Section 31.14 takes up as a property rather than a defect.

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Nothing here is iterated. Equation (31.16) is one equation in one unknown with everything on the right already computed, which is why the curve comes back to machine precision rather than to a tolerance.

The production tree of this chapter runs the same construction at $\Delta t = 0.25$, over eleven years: 44 steps, $\Delta x = 0.008660$, $j_{\max} = 10$. Summing the Arrow–Debreu prices at each of those 44 levels returns the input discount factors with errors at or below one unit in the last place: machine precision, not convergence, because $\alpha(i)$ solves one equation exactly rather than being fitted.

That is worth remembering, because it separates two kinds of error. The tree's discretization error affects option values, which is Section 31.9's subject. It does not touch the underlying curve at all, so a mispriced swaption on this tree is never a symptom of a badly fitted curve.

31.9 BERMUDAN SWAPTIONS

Here is what the tree is for.

Example 31.8 One option with five dates, not five options

A receiver swaption on a swap paying 4% fixed and maturing at $t = 6$, notional 100, on the production tree of Section 31.8.

First check the machinery. A European receiver expiring in one year has a closed-form value by Jamshidian's decomposition (a European option on a coupon bond is a portfolio of options on the zeros), and that gives **1.1465**. The tree should converge to it:

Steps/year	Tree	Error
4	1.1637	+0.0172
8	1.1696	+0.0231
12	1.1667	+0.0202
24	1.1586	+0.0121
48	1.1502	+0.0037

The error shrinks as the grid is refined but not monotonically: eight steps a year is worse than four. Section 20.2 established why: a lattice's error depends on where the nodes fall relative to the payoff's kink, and refining the grid moves them. A desk that extrapolates two tree values without checking for this will get a worse answer than either.

Value the Bermudan exercisable annually at $t = 1, 2, 3, 4, 5$, at 24 steps per year, alongside each European:

Exercise right	Value
European, 1y only	1.1586
European, 2y only	1.1953
European, 3y only	1.0559
European, 4y only	0.7974
European, 5y only	0.4412
Sum of all five Europeans	4.6484
Bermudan, any one of the five	**1.6995**

The Bermudan is worth 42.2% more than the best single European and 37% of their sum.

It exceeds any European because the right to choose has value. The holder does not have to guess in advance which date will be best.

It falls far short of the sum because the rights are substitutes, not additions. Exercising at year two extinguishes the option at years three, four, and five. A portfolio of five Europeans is five options; a Bermudan is one option with five dates, and the gap between 1.70 and 4.65 is the entire

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economic content of that sentence.

And Black's model cannot produce the 1.6995. It prices each row of the top block, because each is a European option on a single rate. It has nothing to say about the choice, because making the choice requires knowing the curve at year three, and Black's model never builds one.

31.10 CALIBRATION, AND WHAT IT CANNOT DETERMINE

What calibration does here is narrower than the word suggests. Section 31.6 has already fitted the curve: $\theta(t)$ is solved so that today's $P(0,t)$ comes back exactly, and nothing about it is open to choice. What is left unfixed is the pair (a,σ), and those two govern volatility rather than level. *So calibration in a fitted short-rate model means choosing two constants from option prices* (the caps and swaptions of Section 29.9's volatility cube), and it never means matching the curve, which is automatic.

The procedure is the ordinary one. Take a set of quoted instruments, price each on the model at a trial (a,σ), and minimize a weighted sum of squared differences,

$$\min_{a,\,\sigma} \sum_i w_i \left[V_i^{\text{model}}(a,\sigma) - V_i^{\text{market}}\right]^2$$

with the weights w_i chosen as Section 21.10 describes. The closed forms of Section 31.7 are what make this fast enough to run every morning, which is the practical reason for deriving them.

Which instruments go into that sum decides what the calibration can see. Example 31.3 said a and σ are separately identified. The following example is the most important in this chapter, and it says that where it matters most, they are not.

Example 31.9 Every European matched, the Bermudan unmatched

Hold the one-year-into-five-year European receiver fixed at its tree value of 1.1586. For each mean reversion a, solve for the σ that reproduces it exactly. Then reprice the Bermudan of Example 31.8.

a	σ	**European**	**Bermudan**	**vs.** $a = 0.08$
0.01	0.00823	1.1586	1.6474	−3.1%
0.03	0.00871	1.1586	1.6622	−2.2%
0.08	0.01000	1.1586	1.6995	—
0.15	0.01202	1.1586	1.7541	+3.2%
0.30	0.01724	1.1586	1.8705	+10.1%

Every row prices the calibration instrument identically, to four decimals, and the Bermudan spans 13.1% of its own value.

The mechanism is the reason two-factor models exist. A Bermudan's value depends on the *relationship* between rates at different exercise dates: how much the curve at year three tells you about the curve at year four. That relationship is governed by a, and a single European swaption contains no information about it, because it observes the curve once.

There are three responses, and a desk should know which it is using.

Calibrate to a co-terminal strip. Use the swaptions that expire on each of the Bermudan's exercise dates and mature on its final date. This is the standard practice, and it constrains a far better than a single option because the strip observes several dates.

Fix a from history and calibrate only σ. This is cruder, and defensible when the strip is not liquid. Section 29.9 reaches the same conclusion about SABR's β, which desks fix rather than fit: when the data cannot support a parameter, choose it outside the calibration rather than letting the fit invent it.

Or quote a range. Section 21.11 argued that model risk should be reported rather than hidden, and here the range is computable from a five-row calculation any desk can run.

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What is not defensible is calibrating to one European and reporting one Bermudan price to four decimals, which the table shows to be four decimals of a number determined by an unobserved parameter.

31.11 CALLABLE BONDS AND OPTION-ADJUSTED SPREAD

The largest application of short-rate trees is not swaptions. It is the callable bond market, and the number the model produces there has a name that overstates what it is.

Example 31.10 What the Z-spread hides, and what the OAS assumes

A ten-year 5% annual bond, callable at par each year from year three, priced on the production tree of Section 31.8.

Straight bond, zero spread	106.9013
Callable bond, zero spread	101.6743
Embedded call option	**5.2270**

Now suppose the bond trades at 99.00. Two spreads can be quoted.

The Z-spread is the constant addition to the curve that reproduces the price while ignoring the call: **94.6 basis points**.

The option-adjusted spread (OAS) is the constant addition that reproduces the price with the call modeled: **52.1 basis points**.

The 42.5 basis point difference is the call, expressed as a running yield. A portfolio manager comparing this bond with a non-callable one on Z-spread would see 94.6 and conclude it offered generous credit compensation. Roughly half of that is payment for an option the investor has written, and the OAS exists precisely to strip it out.

Now vary the model's parameters:

a	σ	**OAS**
0.03	0.010	39.5 bp
0.08	0.010	52.1 bp
0.20	0.010	71.5 bp
0.08	0.015	20.9 bp

The OAS ranges from 20.9 to 71.5 basis points across parameter choices no market quotes. The Z-spread, for all its faults, is a fact about the price; *the OAS is a fact about the price and a model*, and the word "adjusted" does not convey how much of it is the second.

The practical rule follows immediately. OAS comparisons are meaningful *within* a book run on one model with one calibration, and close to meaningless across two houses that have not agreed on a and σ. Comparing a dealer's OAS with your own is comparing two models, not two bonds.

31.12 THE ONE-FACTOR DEFECT

Every model so far has one source of uncertainty, and it has a consequence that no amount of calibration can remove.

At any future date, every rate on the curve is a deterministic increasing function of $r(t)$. A two-year rate and a ten-year rate cannot move in opposite directions, cannot move by unrelated amounts, and are perfectly correlated. The model does not merely approximate a correlation of one; it asserts it.

For a swaption or a callable bond this is tolerable, because the payoff depends on one rate or on a monotone function of the state. For anything whose payoff depends on the curve's *shape*, it is fatal, and it is straightforward to price how fatal.

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Example 31.11 What perfect correlation costs

A five-year option on the spread between the ten-year and two-year swap rates is a **CMS spread option**, one of the most widely sold structured coupons. Forward rates 3.40% and 2.80%, normal volatilities 90 and 75 basis points a year, annuity 4.20. The forward spread is 60 basis points, and the option is struck at the money.

The spread's volatility is $\sqrt{\sigma_{10}^2 + \sigma_2^2 - 2\rho\,\sigma_{10}\sigma_2}$:

ρ	Spread vol	Option value	Ratio
1.00	15.0 bp	56.2 bp	1.00
0.95	30.0 bp	112.4 bp	2.00
0.90	39.7 bp	148.7 bp	2.65
0.80	54.1 bp	202.6 bp	3.61
0.70	65.4 bp	245.0 bp	4.36
0.50	83.5 bp	312.9 bp	5.57

A one-factor model is forced to the first row. It returns 56.2 basis points—and the 15.0 is not even a correlation assumption, it is just $|\sigma_{10} - \sigma_2|$, the residual from two rates moving in lockstep at slightly different speeds.

At a market-plausible correlation of 0.70 the option is worth 245.0 basis points, 4.4 times as much.

Three things follow.

No calibration repairs this. One could force a one-factor model to reproduce the 245 by inflating σ, but then every swaption in the book would be mispriced. The error is in the model's structure, not its parameters, which is exactly the distinction Section 21.11 drew for equity models.

And the error has a sign. A one-factor model *always* undervalues a spread option, because $\rho = 1$ minimizes the spread's volatility. A desk short these options on a one-factor model is systematically short at a price it computed to be too low.

The magnitude is the argument for Chapter 32. A factor of 4.4 is not a refinement.

31.13 TWO-FACTOR MODELS

The repair is a second source of uncertainty. The standard construction adds a stochastic mean-reversion level:

$$dr = \big[\theta(t) + u - ar\big]dt + \sigma_1\,dW_1, \qquad du = -b\,u\,dt + \sigma_2\,dW_2 \tag{31.17}$$

with $dW_1\,dW_2 = \rho\,dt$, where b is now the speed at which u reverts to zero rather than a level. Equivalently, r is the sum of two zero-mean mean-reverting factors and a deterministic shift chosen to fit the curve. This is the **G2++** form, whose name is a description of the construction: Gaussian, two factors, and "++" for the shift. That is how the model is usually implemented, because two zero-mean factors discretize more cleanly than a stochastic mean-reversion level does.

Now the short rate and the long rate are driven by different combinations of two shocks, so rates at different maturities are imperfectly correlated and the curve can steepen, flatten, and twist. The model remains affine, so bond prices stay closed-form.

The costs are real and should be weighed.

Five parameters instead of two, and the calibration is correspondingly less stable, a familiar problem from Section 21.4's account of Heston, where no parameter moved one feature alone.

And the lattice becomes two-dimensional. A two-factor tree is a 3×3 branching structure whose node count grows as the square, which is why two-factor implementations usually price by Monte Carlo or by finite differences on a two-dimensional grid rather than on a tree. There is a further obstruction on the tree: the natural correlation correction to the nine branching probabilities is admissible only for $|\rho| \le 1/3$. [TN31] gives that obstruction and the change of

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variable that removes it, together with the two-factor bond and option formulas and the reason Jamshidian's decomposition does not survive the second factor.

Section 32.17 sets out the rule for choosing among all these models; for the number of factors it comes down to whether the payoff depends on the level of rates or on their shape, and Example 31.11's question (what is the payoff worth at $\rho = 1$?) tells which.

31.14 NEGATIVE RATES AND THE REPAIRS

Hull–White's r is normally distributed, so it goes negative with positive probability, and Example 31.12 shows the probability is not small.

Example 31.12 How often the model says rates are below zero

A flat forward curve at 3%, and the probability the model assigns to a negative short rate, with $r(T)$ normal around the forward rate and the standard deviation the model implies:

	1y	2y	5y	10y
$a = 0.05$, $\sigma = 1.0\%$	0.1%	1.3%	6.5%	11.6%
$a = 0.05$, $\sigma = 1.5\%$	2.0%	6.9%	15.7%	21.3%
$a = 0.10$, $\sigma = 1.0\%$	0.1%	1.0%	4.6%	7.5%

At ten years the middle row assigns better than one chance in five to a negative short rate. Before rates went negative (in Denmark in July 2012, in the euro area in June 2014), this was cited as the model's principal flaw and as the reason to prefer CIR or a lognormal short rate.

Then rates went negative and stayed there for years, and the property became the reason Hull–White survived while its competitors did not. A Gaussian model priced the market that existed; a positive model could not be calibrated to it at all.

The general lesson is the one Section 19.9 drew about 1987 and Section 26.6 drew about house prices. A constraint imposed because the unconstrained case "cannot happen" is a statement about the sample, and the sample is always shorter than the future. A model's implausible region is where its next test comes from.

Where positivity or a floor is genuinely wanted, the standard repairs are a **shifted** model (apply a positive model such as CIR or a lognormal short rate to $r + s$ for a constant shift s, which moves the lower bound from zero to $-s$ rather than removing it) or the displaced and normal-versus-lognormal machinery of Section 29.8, which is the same idea applied to the quoting convention rather than to the process.

A lognormal short rate such as Black–Karasinski, in which $\ln r$ mean-reverts, keeps positivity but gives up the closed forms. It has no closed-form bond price, so Jamshidian's decomposition is unavailable, and the tree's displacement $\alpha(i)$ must be found by Newton iteration at each step instead of from (31.16). That is the one genuine difference from Hull–White on the tree.

31.15 AFTER THE LIBOR TRANSITION

Chapter 7 rebuilt the curve framework around risk-free rates and Chapter 29 rewrote the caplet. What did that do to this chapter?

Less than one would expect, and for an instructive reason. A short-rate model models the instantaneous risk-free rate, which is closer to what SOFR and €STR actually are than to what LIBOR ever was. The transition removed an inconsistency rather than creating one: pre-2022, the model's r was a risk-free rate while the products referenced a credit-sensitive rate, and the gap was absorbed by convention.

Two things did change.

Discounting and forecasting are now the same curve for cleared trades in the overnight rates, so the multi-curve complication of Section 7.5 does not reach into the short-rate model at

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all. Where a spread is needed (a credit-sensitive benchmark, or a currency basis), it is added as a deterministic spread rather than modeled, which is a modeling choice worth being aware of and is usually adequate.

And compounded-in-arrears fixings introduce a short averaging window that Section 29.4 handled by adjusting the caplet's variance. On a short-rate tree it needs no adjustment at all: the tree simulates the path of r, so an average over the accrual period is computed directly. That is a small but genuine advantage of this class of model over a forward-rate model, and it is the reverse of the usual ordering.

31.16 WHERE THESE MODELS ARE STILL THE RIGHT TOOL

Chapter 32 introduces models that are more realistic in every respect and harder to use in most. The case for short-rate models is not nostalgia, and it rests on four properties.

They price American and Bermudan exercise naturally. A tree gives the continuation value at every node, so an early-exercise decision costs nothing extra. A forward-rate model does not, and needs the regression machinery of Section 20.11. For the callable and Bermudan books, which are among the largest users of interest-rate models by notional, this is decisive.

They are fast. A 44-step tree prices a Bermudan in milliseconds, which matters when a desk revalues thousands of positions and computes their sensitivities every night.

They have few parameters, which is a defect when the market is rich and a virtue when it is not. Example 31.9 showed that two parameters were already one too many for the available data.

And they handle path dependence in the short rate directly, which Section 31.15 illustrated with compounded fixings.

The case against is Section 31.12 and nothing else, and Section 32.17 turns it into the rule for when to reach for something else.

31.17 WHO USES THEM, AND FOR WHAT

Callable-bond desks and the investors who buy their paper. The OAS of Example 31.10 is produced by a short-rate model at essentially every institution that quotes one, and Section 26.3's mortgage prepayment models sit on top of the same machinery: a prepayment model needs a rate path to respond to, and this is what generates it.

Bermudan swaption desks, where Example 31.9's calibration question is a daily decision rather than a textbook point.

Insurers and pension funds valuing liabilities with embedded guarantees: a minimum crediting rate is an interest rate floor with a fifty-year tenor, and it must be valued on a model that can generate a coherent curve that far out.

Structured-note issuers, for the level-dependent coupons. *Not for the spread-dependent ones*, which is the division Section 31.13 drew and the reason a single desk often runs both a one-factor and a two-factor model and routes trades between them.

And risk managers, for scenario generation, where, unusually, the equilibrium models of Section 31.2 come back. A real-world scenario generator wants a curve that is derived from an economically coherent process, not one fitted to today's quotes, because the question being asked is what could happen rather than what a hedge costs.

31.18 WHERE THIS CAME FROM

The apparatus Chapter 15 built for equity options does not transfer to bonds, because a bond's volatility must vanish at maturity and because the whole curve, not one price, is the state variable. Sorting that out took twenty years.

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Vasicek (1977) is the first serious answer: a mean-reverting short rate from which the entire curve follows in closed form. *Its defect is fundamental and instructive*—being an equilibrium model it produces a curve, and that curve is not today's. Cox, Ingersoll and Ross (1985) gave the approach its general-equilibrium foundation and a square-root diffusion that keeps rates positive, and the same defect remained.

Ho and Lee (1986) inverted the problem: take today's curve as given and fit the model to it. *That is the no-arbitrage turn*, and everything since sits on the far side of it. Hull and White (1990) combined the two, adding mean reversion to a fitted model, which is the formulation this chapter uses; Black, Derman and Toy (1990) and Black and Karasinski (1991) did the same in lognormal form. Jamshidian (1989) showed that an option on a bond portfolio decomposes into a portfolio of options in a one-factor model, which is why European swaptions had closed forms in these models from the start; Bermudans waited for the lattice.

SUMMARY

Black's model prices one option on one rate, and cannot price a Bermudan swaption or a callable bond, which need the whole curve at each future decision date. A short-rate model generates that curve from a single state variable, the source of both its tractability and its central defect. A handful of parameters cannot match a curve with hundreds of points, so the no-arbitrage generation makes the drift a solved function of time: $\theta(t)$ carries no information, and every model in this class is an interpolation device with a stochastic component. CIR's $\sqrt{r}$ keeps rates positive, which is why it could not be calibrated once rates went negative.

In Hull–White, σ sets the level of the volatility term structure and a sets its shape, forward-rate volatility decaying as σe^{-aT}. The tree reprices the curve to machine precision, because the displacement $\alpha(t)$ is solved rather than fitted, and converges at roughly first order but not monotonically.

A Bermudan exercisable at several dates is worth clearly more than the best single European and far less than the sum of all of them: a portfolio of Europeans is several options, a Bermudan is one option with several dates. Holding the calibrating European fixed and varying a leaves every European identical and still moves the Bermudan materially, because a single European observes the curve once and contains no information about a. Calibrating to one European and quoting the Bermudan to four decimals is not defensible.

On a callable bond the Z-spread exceeds the OAS by enough that much of the apparent credit compensation pays for a written option, but the OAS itself moves widely across ordinary parameter choices. The Z-spread is a fact about the price; the OAS is a fact about the price and a model, so comparing OAS across houses compares models rather than bonds.

In one factor every rate is an increasing function of one state variable, so all rates are perfectly correlated—asserted, not approximated. A CMS spread option is forced to $\rho = 1$ and is worth a fraction of its value at a realistic correlation; no calibration repairs it, and the error always has the same sign. Use one factor for level products and two for shape products, and test a payoff by what it is worth at $\rho = 1$.

Gaussian rates go negative, with a probability far from negligible at long horizons. Cited as the model's principal flaw before rates went negative, that became the reason it survived: a model's implausible region is where its next test comes from. Short-rate models remain right for early exercise, for speed, and when the data cannot support more parameters—and wrong whenever the payoff depends on the curve's shape.

FURTHER READING

Vasicek's and Cox, Ingersoll, and Ross's original papers are short and state what an equilibrium model is trying to do; reading them after this chapter is more useful than before it, because the

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thing they are trying to do is the thing Section 31.4 says a desk cannot use. Hull and White's papers on the extended model and on the trinomial construction are the primary sources for Sections 31.6 and 31.8, and the tree paper describes the two-stage procedure precisely enough to code from. Brigo and Mercurio's book is the standard reference for the whole family, including the G2++ form of Section 31.13 and the shifted variants of Section 31.14, and it is where to go for the algebra this chapter states without deriving. Andersen and Piterbarg's three volumes are the practitioner's reference and treat the calibration question of Example 31.9 at much greater length, reaching the same conclusion. On OAS, the investor literature is where the parameter dependence of Example 31.10 is set out from the holder's side (what a quoted spread assumes about volatility and the exercise policy), while the dealer literature documents the models the quotes come from; read the first to interrogate a quoted OAS, the second to reproduce one.

PRACTICE QUESTIONS

31.1 Explain why Black's model cannot price a Bermudan swaption, and be specific about what it fails to produce.

31.2 In Vasicek's model with $a = 0.10$, $b = 5\%$, $\sigma = 1\%$, compute the limiting long rate and explain why it is below b. Then explain why an equilibrium model cannot match an observed term structure, and why this is arithmetic rather than calibration.

31.3 Explain what $\theta(t)$ does in Ho–Lee, why it carries no information, and state the defect that rules the model out in practice together with the quantity that exposes it.

31.4 Using σe^{-aT}, compute the ten-year forward-rate volatility at $\sigma = 100$ bp for $a = 0.05$ and $a = 0.20$, and explain which feature of the volatility term structure is set by a and which by σ.

31.5 Describe the two stages of the Hull–White tree construction, state which quantities are solved rather than fitted, and explain why the tree reproduces the initial curve to machine precision while still mispricing options.

31.6 Explain why the branching pattern must change at $\pm j_{\max}$, and what would go wrong if it did not.

31.7 In Example 31.8, explain why the Bermudan is worth more than any single European and much less than their sum, and why the tree's convergence to the Jamshidian value is not monotone.

31.8 Explain why a single European swaption contains no information about the mean-reversion rate, use Example 31.9 to state the range of Bermudan values consistent with an exactly matched European, and describe three defensible responses and one that is not.

31.9 Explain the difference between a Z-spread and an option-adjusted spread, and what the difference between them measures.

31.10 Using Example 31.10, state the OAS at $a = 0.08$ with $\sigma = 1.0\%$ and with $\sigma = 1.5\%$, and explain what this implies about comparing OAS across dealers.

31.11 Explain why a one-factor model forces perfect correlation between all rates, and why this is an assertion rather than an approximation.

31.12 Using Example 31.11, state the CMS spread option's value at $\rho = 1$ and at $\rho = 0.70$, and explain why the one-factor error always has the same sign and why that sign is the dangerous one.

31.13 State the decision rule for choosing between a one-factor and a two-factor model, and give one product on each side of it.

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31.14 Explain why Gaussian short rates were regarded as defective before rates went negative and as necessary afterward, and name four properties that keep short-rate models in use together with the single property that rules them out.

31.15 Using Example 31.5, state the caplet's value under the forward measure and under the risk-neutral measure and comment on their agreement, then explain what the 2.52% error in part (c) consists of and its sign.

31.16 Using Example 31.6, state the range of $E^{Q}[S_T]/F(0,T)$ across the correlations shown, explain what quantity produces it, and name the two other results in this book that are the same quantity.

FURTHER QUESTIONS

31.17 Example 31.1 shows three curves converging to $R_\infty = b - \sigma^2/2a^2$. Derive this limit, explain the convexity argument that produces the second term, and determine the parameter values at which the correction becomes large enough to matter for a thirty-year bond.

31.18 Example 31.9 shows the Bermudan spanning 13.1% of its value with the European held fixed. Extend the calculation to a co-terminal strip: determine how many co-terminal swaptions must be matched before the Bermudan's range falls below 1%, and explain what this implies for a currency in which only two of them are liquid.

31.19 Example 31.10 shows the OAS moving from 20.9 to 71.5 basis points. Propose a reporting convention that would make OAS comparable across institutions, state what each party would have to disclose, and explain why the market has not adopted one.

31.20 Example 31.11 prices the one-factor defect at a factor of 4.4. Derive the spread's volatility as a function of ρ, show that $\rho = 1$ minimizes it for any $\sigma_{10} \neq \sigma_2$, and determine the conditions under which a one-factor model could be made to price a single spread option correctly, and then explain what it would then get wrong.

31.21 Section 31.13 says a two-factor model costs five parameters and a two-dimensional lattice. Compare this trade-off with Section 21.4's account of Heston, where additional parameters also failed to move features independently. State the general principle governing when an extra factor is worth its calibration cost, and apply it to both cases.

31.22 Section 31.14 argues that a model's implausible region is where its next test comes from, citing 1987 and 2007. Identify a constraint currently imposed on rate models because the alternative "cannot happen," argue both sides, and propose how a desk should hedge the possibility that it can.

31.23 Section 31.16 claims short-rate models remain right for Bermudan exercise while Chapter 32's models are more realistic. Construct the strongest case that a forward-rate model with American Monte Carlo should replace them for the callable book, state what would have to be true about accuracy and computation for that case to win, and give your own judgment with reasons.

31.24 Example 31.5 finds a 2.52% error from ignoring the payoff-discount covariance. Derive that covariance term analytically for a caplet under Hull–White, determine its dependence on rate volatility and option maturity, and identify the configuration in which it would exceed 10%.

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Chapter 32

HJM, LMM, and Multi-Curve Term Structure Models †

Section 31.12 put a number on the one-factor defect: a one-factor short-rate model prices a CMS spread option at 56.2 basis points when it is worth 245.0. This chapter is about the class of models built to fix that, and about the price of the fix.

There is a second and less often stated motivation. A short-rate model models a quantity no one trades. The instantaneous rate $r(t)$ is not observable, not quoted, and not hedgeable; every market instrument is a function of it, and the model's parameters must be inferred backward through those functions. The models in this chapter model the rates that are actually quoted, and that turns out to matter more for calibration than for realism.

The chapter runs from the general framework to the model that is actually implemented. Sections 32.1 to 32.3 are HJM: the whole forward curve at once, the drift no-arbitrage forces on it, and why no one implements it in that form. Sections 32.4 to 32.8 are the LIBOR market model: the measure and the drift that follows, how many factors are needed, and the shapes instantaneous volatility and correlation have to take. Sections 32.9 to 32.13 are practice: calibration, simulation, an incompatibility with the swap market model whose size depends on the trade, early exercise, and why sensitivities cost more than prices. Sections 32.14 to 32.16 are what requires this machinery, the smile, and multi-curve after the transition; Sections 32.17 to 32.19 choose among the models and say who runs them.

32.1 HJM: MODEL THE WHOLE FORWARD CURVE

The object being modeled is a family of curves. Section 31.5's $f(0,t)$ was today's forward curve; HJM needs $f(t,T)$, the curve as it stands at any future date t. It is defined as a limit of quoted forward rates. The simply compounded forward rate for $[T, T+\tau]$ seen from t is

$$F(t;T,T+\tau) = \frac{1}{\tau}\left(\frac{P(t,T)}{P(t,T+\tau)} - 1\right)$$

and letting the accrual shrink to nothing gives the instantaneous forward rate,

$$f(t,T) = \lim_{\tau \downarrow 0} F(t;T,T+\tau) = -\frac{\partial \ln P(t,T)}{\partial T} \tag{32.1}$$

Two relations follow, and the chapter uses both throughout. Integrating (32.1) returns the bond price,

$$P(t,T) = \exp\left(-\int_t^T f(t,u)\,du\right) \tag{32.2}$$

so the forward curve and the discount curve carry the same information: either determines the other, and a model of one is a model of both. Evaluating the forward curve at its lower endpoint returns the short rate,

$$r(t) = f(t,t) \tag{32.3}$$

That identity carries the chapter's argument. A short-rate model specifies the single point $f(t,t)$ and derives the rest of the curve from it, which is why the shapes it can produce are constrained. HJM specifies $f(t,T)$ for every T simultaneously, and derives nothing.

Heath, Jarrow, and Morton proceeded directly. Instead of a process for one rate, specify a process for every instantaneous forward rate:

$$df(t,T) = m(t,T)\,dt + \sum_{k=1}^{K} s_k(t,T)\,dW_k \tag{32.4}$$

The HJM processes are written under the risk-neutral measure Q, and each W_k is a Wiener process under it; the market model later changes numeraire, and Section 32.5 says what that does to the drift. No market price of risk appears anywhere in what follows and none is needed: a price is an expectation under Q, and the calibration of Section 32.9 fits Q-parameters to quoted prices directly. That is also what makes the next section's result possible.

The state is now the entire curve $\{f(t,T) : T \geq t\}$, driven by K independent shocks. Today's curve enters as the initial condition, so the fitting problem of Section 31.4 does not arise: it is satisfied by construction rather than solved.

The number of factors is now a modeling choice rather than a consequence. Section 32.6 makes that choice with data.

32.2 THE DRIFT IS NOT A CHOICE

Once the volatility functions $s_k(t,T)$ are specified, the drift $m(t,T)$ is determined. No-arbitrage forces

$$m(t,T) = \sum_{k=1}^{K} s_k(t,T) \int_t^T s_k(t,\tau)\,d\tau \tag{32.5}$$

which is the **HJM drift condition**.

It follows from expressing the bond price in terms of the forward curve and identifying its drift. Under Q a zero-coupon bond earns the short rate, so the bond's drift is determined before any modeling choice is made, and the drift of the forward curve must be the one consistent with it.

Take one factor for the derivation and drop the subscript; the K-factor result follows at the end. Take logs in (32.2) and name the result,

$$Y(t,T) \;\equiv\; \ln P(t,T) \;=\; -\int_t^T f(t,\tau)\,d\tau$$

Differentiation produces two terms. The integrand evolves, contributing $-\int_t^T df(t,\tau)\,d\tau$; and the lower limit advances, evaluating the integrand at that limit and contributing $+f(t,t)\,dt$. That second term is where the short rate enters, by (32.3):

$$dY = r(t)\,dt - \int_t^T \big[m(t,\tau)\,dt + s(t,\tau)\,dW\big]d\tau = \big[r(t) - M(t,T)\big]dt - S(t,T)\,dW$$

where

$$M(t,T) = \int_t^T m(t,\tau)\,d\tau, \qquad S(t,T) = \int_t^T s(t,\tau)\,d\tau$$

are the drift and the volatility of the forward curve accumulated across the bond's remaining life. That accumulation is the essential feature: a bond responds to the entire strip of forward rates it spans, not to any single one of them.

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Apply Itô's lemma to $P = e^Y$ to get

$$\frac{dP}{P} = \left[r(t) - M(t,T) + \frac{1}{2} S(t,T)^2 \right] dt - S(t,T)\, dW$$

A zero-coupon bond is a traded asset and (32.4) is written under Q, so the bond's expected return is the short rate, which leads to

$$M(t,T) = \frac{1}{2} S(t,T)^2$$

That equation is the entire content of the drift condition: the accumulated drift is not a free input, but is determined by the accumulated volatility.

M and S are both integrals over $[t,T]$, so differentiating in T gives $\partial M/\partial T = m(t,T)$ and $\partial S/\partial T = s(t,T)$, with the chain rule on the right:

$$m(t,T) = S(t,T)\, s(t,T) = s(t,T) \int_t^T s(t,\tau)\, d\tau$$

which is (32.5) with one factor. With K shocks, S becomes a vector and the variance term becomes $\frac{1}{2} \sum_k S_k^2$; the shocks are independent, so no cross terms appear, each factor contributes the one-factor result, and the contributions add.

The derivation used only that bond prices are traded and that $f(t,t) = r(t)$. No equilibrium argument and no utility function appear, which is why the condition binds every arbitrage-free term-structure model rather than a family of them.

Three consequences follow, and the third is the one that shapes practice.

The model has exactly one free input. Choose the volatility structure and everything else follows. That is a remarkably clean statement about a term-structure model, and it is the analog of Chapter 13's result that a derivative's price does not depend on the underlying's expected return: drift is not an independent degree of freedom once you insist on no arbitrage.

Every arbitrage-free term-structure model is an HJM model. Chapter 31's two Gaussian models are (32.4) under particular choices of $s(t,T)$:

- **Ho–Lee**: $s(t,T) = \sigma$, the same volatility at every maturity, so a shock shifts the whole curve in parallel;
- **Hull–White**: $s(t,T) = \sigma e^{-a(T-t)}$, a volatility decaying with the forward rate's distance, so a shock moves the short end more than the long.

Chapter 31 was therefore working inside HJM, with a volatility structure chosen for tractability rather than for fit. The two chapters are not competing frameworks but one framework and a family of choices within it.

And the drift depends on an integral of the volatility, which is why the framework is hard to implement. Section 32.3 takes that up.

32.3 WHY HJM IS NOT IMPLEMENTED DIRECTLY

Two obstacles stand in the way, and they are not of the same severity. The first removes every numerical method of Chapter 31; the second only requires the model to be restated in other variables.

The process is generally non-Markov. The drift (32.5) depends on the volatility structure along the whole curve, so the future evolution depends on the path, not just on the current state. A non-Markov process cannot be put on a recombining tree, which removes every technique of Chapter 31 and leaves only Monte Carlo—and Monte Carlo, by Section 20.11, is awkward for early exercise.

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There is an important exception, and it explains Chapter 31. If $s(t,T) = \sigma e^{-a(T-t)}$ (volatility decaying exponentially in time to maturity), the model *is* Markov in a single state variable, and that state variable is the short rate. Hull–White is exactly the HJM model that happens to be Markov, which is why it can be put on a tree while its generalizations cannot. The tractability that made Chapter 31 useful was never an accident of the short-rate formulation; it was a restriction on the volatility structure.

The exponential case is worth following through, because it is the one that collapses. With $s(t,T) = \sigma e^{-a(T-t)}$ the integral gives $S(t,T) = \sigma B(t,T)$, Hull–White's own function, and integrating the forward-rate dynamics leaves

$$f(t,T) = f(0,T) + \phi(t,T) + e^{-a(T-t)}x(t), \qquad x(t) = \int_0^t \sigma e^{-a(t-u)}\,dW(u)$$

with ϕ deterministic. The entire curve is a deterministic function of the single scalar $x(t)$, an **Ornstein–Uhlenbeck process** (a Gaussian process pulled back at rate a toward a fixed level, here zero, of the kind Vasicek's model makes of the short rate), and setting $T = t$ shows x is the short rate up to a deterministic shift.

The step that collapses it is a factorization: $e^{-a(T-u)} = e^{-a(T-t)}e^{-a(t-u)}$ separates the volatility's dependence on calendar time from its dependence on maturity. That is the precise sense in which Hull–White is the simplest Markov HJM model; a second exponential buys a second state variable, and the pattern continues.

The exception is a family, not one model. An HJM volatility that separates as $s(t,T) = g(t)h(T)$ is Markov in a small state vector, and the Cheyette and linear Gaussian Markov (LGM) models are of this kind. They are used when a Bermudan needs more than one factor but not a full LMM.

The second obstacle concerns calibration, and it is not that instantaneous forward rates are untraded. The short rate is not traded either, and Chapter 31 put it on a trinomial tree and priced Bermudan swaptions there; an unobservable state variable is the normal case in this subject, not an objection to it. The difficulty is one of dimension. HJM's free input $s_k(t,T)$ is a function of two variables (a volatility for every pair of calendar date and maturity), while the market quotes a finite set of volatilities, on rates that accrue over three or six months rather than instantaneously. No market instrument pays off on a single $f(t,T)$, so there is no quote to read any one of those volatilities from. A model written in terms of $f(t,T)$ must therefore be discretized along the maturity axis before it can touch a quote at all, and that discretization is a modeling choice the framework does not supply, which is to say the modeling happens in the discretization rather than in (32.4).

Both obstacles point the same way: work with rates that are quoted, over the accrual periods they are quoted over. That is the market model.

32.4 THE LIBOR MARKET MODEL

Fix the tenor structure $T_0 = 0 < T_1 < \cdots < T_N$ of an actual instrument (the reset dates of a cap or a swap) and model the forward rates $F_k(t)$ for the periods $[T_k, T_{k+1}]$ directly, each fixing at the start of its own period, T_k. Taking $T_0 = 0$ has one consequence used throughout: F_0 fixes today, so it is known rather than modeled, carries no caplet, and still supplies the discount factor for the first period. The caplets of a cap on this structure are therefore on F_1 through F_{N-1}, and the one on F_k has k whole periods of life. The convention costs no generality, because the grid and the instrument are separate things: a structure that starts later (a cap beginning in five years, or the swap underlying a swaption) keeps this same grid from today and simply carries no caplets on the forwards before it starts. Those forwards are still modeled, since the discount

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factors to the later payment dates run through them. Assume each forward is lognormal:

$$\frac{dF_k(t)}{F_k(t)} = \mu_k(t)\,dt + \sigma_k(t)\,dW_k \tag{32.6}$$

with $dW_i\,dW_j = \rho_{ij}\,dt$.

The point of the assumption is that it makes the model consistent with the market's own quoting convention. Section 29.3 priced a caplet with Black's model, which assumes exactly this lognormality. So the LMM reproduces the market's caplet prices by construction rather than by calibration: the model is built on the convention instead of being fitted to it.

That is the historical significance of the model and it is easy to understate. Before it, every model priced caps approximately and the market quoted them in a formula the models did not contain. Afterward, the quoting formula was a special case of the model. The name is now an anachronism (the rates are SOFR, €STR and EURIBOR forwards, not LIBOR), but the construction is unchanged, and Section 32.16 says why.

32.5 WHICH MEASURE, AND THE DRIFT THAT FOLLOWS

Under the T_{k+1}-forward measure $Q^{T_{k+1}}$ (numeraire $P(t,T_{k+1})$), the forward F_k is a martingale and $\mu_k = 0$. But different forwards are martingales under different measures, and a simulation needs all of them under one.

Pick a single numeraire and the drifts follow. Under the terminal measure (numeraire $P(t,T_N)$),

$$\mu_k(t) = -\sigma_k(t) \sum_{j=k+1}^{N-1} \frac{\rho_{kj}\,\tau_j\,F_j(t)\,\sigma_j(t)}{1+\tau_j F_j(t)} \tag{32.7}$$

The derivation is a change of numeraire performed once per accrual period, and one object does all of the work. For each j write

$$\Lambda_j(t) \;=\; \frac{P(t,T_{j+1})}{P(t,T_j)}$$

for the ratio of two consecutive numeraires. By the definition of F_j that ratio is a function of a single forward rate and of nothing else,

$$\Lambda_j(t) \;=\; \frac{1}{1+\tau_j F_j(t)}$$

It carries the letter this book uses for a Radon–Nikodym derivative because it is one up to a constant:

$$\left.\frac{dQ^{T_{j+1}}}{dQ^{T_j}}\right|_{\mathcal{F}_t} \;=\; \frac{P(0,T_j)}{P(0,T_{j+1})}\,\Lambda_j(t)$$

and a constant multiple leaves a volatility unchanged.

Since F_j has volatility σ_j, applying Itô's lemma to $\Lambda_j = 1/(1+\tau_j F_j)$ gives

$$\frac{d\Lambda_j(t)}{\Lambda_j(t)} \;=\; \mu_{\Lambda_j}(t)\,dt \;-\; \lambda_j(t)\,dW_j, \qquad \lambda_j(t) \;=\; \frac{\tau_j F_j(t)\,\sigma_j(t)}{1+\tau_j F_j(t)}$$

The drift μ_{Λ_j} is never used and is not evaluated. The volatility of Λ_j is λ_j: F_j's own volatility, damped by the factor $\tau_j F_j/(1+\tau_j F_j)$, the damping that makes the drift small, as the end of this section shows. The minus sign in the display is no part of that volatility, since $-W_j$ is a Wiener process as much as W_j is; it records that Λ_j falls when F_j rises, and that is where the minus sign on the covariance below comes from. Which measure dW_j belongs to does not matter here, since the rule stated next says a change of numeraire leaves volatilities alone.

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Section 28.9 derived this as Girsanov's theorem: a change of numeraire adds to a process's drift its covariance with the numeraire ratio, whatever that drift already is, and leaves the volatility unchanged. For consecutive bond numeraires the ratio is Λ_j, and because Λ_j moves against F_j, a forward F_k of volatility σ_k and correlation ρ_{kj} with F_j has covariance $-\rho_{kj}\,\sigma_k\,\lambda_j$ with it. Stepping from Q^{T_j} to $Q^{T_{j+1}}$ therefore adds exactly that to μ_k.

The walk goes one numeraire at a time, from the forward's own measure to the terminal one. Under $Q^{T_{k+1}}$ the forward F_k is a martingale and $\mu_k = 0$; this is the measure the caplet on F_k is priced in. Take one step, to $Q^{T_{k+2}}$. The numeraire ratio for that step is Λ_{k+1}, and with $dW_k\,dW_{k+1} = \rho_{k,k+1}\,dt$ from (32.6) the rule gives

$$\mu_k(t) \;=\; -\,\sigma_k(t)\,\rho_{k,k+1}\,\lambda_{k+1}(t)$$

The correlation enters for one reason: the covariance the rule asks for is between the diffusion of F_k and that of Λ_{k+1}, and F_{k+1} is the only forward rate Λ_{k+1} contains.

Take the next step, to $Q^{T_{k+3}}$. The ratio is now Λ_{k+2}, the identical calculation runs with $j = k+2$, and a second term $-\sigma_k\rho_{k,k+2}\lambda_{k+2}$ is added to the drift F_k already carries. In general the step from Q^{T_j} to $Q^{T_{j+1}}$ uses Λ_j and contributes $-\sigma_k\rho_{kj}\lambda_j$.

The contributions add rather than interact, because consecutive ratios multiply:

$$\frac{P(t,T_N)}{P(t,T_{k+1})} \;=\; \prod_{j=k+1}^{N-1}\Lambda_j(t)$$

The logarithm of a product is the sum of the logarithms, so the diffusion terms add (not the volatilities, since the factors are driven by different Wiener processes), and the whole change from $Q^{T_{k+1}}$ to Q^{T_N} carries $-\sum_{j=k+1}^{N-1}\lambda_j\,dW_j$. Walking all $N-k-1$ steps and collecting,

$$\mu_k(t) \;=\; -\,\sigma_k(t)\sum_{j=k+1}^{N-1}\rho_{kj}\,\lambda_j(t)$$

which is (32.7). Only forward rates longer than F_k appear, because the walk crosses exactly Λ_{k+1} through Λ_{N-1} and no others.

Simulating under the **spot measure** instead (whose numeraire is a rolling investment, rolled at each reset date into the bond maturing at the next) gives sums over $j \le k$, with the opposite sign. Neither choice is more correct, and implementations pick whichever puts the zero drift where the payoff is most sensitive.

Three observations matter for implementation.

The drift is state-dependent, since F_j appears on the right. That is what makes the simulation of Section 32.10 non-trivial and is the source of its discretization error.

The drift is a no-arbitrage consequence, not a parameter. It is the same statement as (32.5), in the discretized language.

It is also small. At a 3% rate with annual accrual, $\tau F/(1+\tau F) \approx 0.029$, so each term in the sum is roughly three percent of a product of volatilities. That smallness has a consequence worth stating in advance of Section 32.10: the predictor–corrector scheme exists because a plain Euler step freezes this state-dependent drift across the interval, and switching to it moves the answer by 0.15 basis points, while raising the step count from four to two hundred moves it by about 6. The discretization error is not where the state dependence would put it.

32.6 HOW MANY FACTORS

The whole motivation was Section 31.12's factor count, so it should be settled with data rather than convention.

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Example 32.1 The curve's factor structure

Model the correlation between forward rates as $\rho_{ij} = e^{-\beta|T_i - T_j|}$, take thirty annual forwards, and compute the eigenvalues of the correlation matrix. The cumulative share of variance explained is:

β	1 factor	2	3	4	5
0.02	82.8%	92.5%	95.3%	96.6%	97.4%
0.05	64.8%	82.2%	88.6%	91.7%	93.5%
0.10	46.6%	68.1%	78.3%	83.9%	87.3%
0.20	28.9%	49.0%	61.9%	70.2%	75.9%

This is **principal component analysis**: the eigenvectors are the factors, the eigenvalues their shares of variance. They carry the reading Litterman and Scheinkman found in bond returns and every major market has reproduced since: a roughly parallel shift (**level**), one sign change (**slope**), two (**curvature**). Further Reading has the reference.

Empirical estimates of β for major currencies fall around 0.05 to 0.10, which puts the first factor at 47–65% and the first three at 78–89%.

The form is an assumption, not a finding. What recommends it is that it is a valid correlation matrix for every $\beta \geq 0$, so no admissibility condition has to be checked (not true of every parameterization one might write down), and that it decays monotonically in tenor distance, the robust qualitative feature of measured correlation. Only β is estimated; the shape is imposed.

Neither the volatility structure nor the drift can move these percentages, since the matrix is a correlation matrix. The number of forwards can, by more than one would guess: at $\beta = 0.05$ the first factor explains 85.5% of ten annual forwards, 64.8% of thirty, 57.5% of forty. A factor count describes a strip, not a curve.

The form also sends ρ_{ij} to zero far out, where measured correlation flattens to a floor; the standard repair is $\rho_{ij} = \rho_\infty + (1 - \rho_\infty)e^{-\beta|T_i - T_j|}$. A floor is itself a common factor, so it lifts every entry: at $\rho_\infty = 0.3$ one factor explains 75.2% and three 92.0%. The table is conservative, and 92% is still not 99%.

The same table supports two readings, and both are correct.

Against one factor the case is decisive: it leaves a third to a half of the variance unexplained, and Example 31.11 priced what that omission does to a spread option.

Against the often-repeated "three factors explain 99% of the curve," it is a corrective: that figure comes from principal components of *yield levels*, dominated by a common trend. On the correlation structure a spread option depends on, three factors explain closer to 80%—and the residual is exactly the relative movement the option is written on.

The working answer is three factors for most books and two for many. The marginal factor is cheap here: it adds a column to the Cholesky factor and one more Wiener increment per step, so the cost of a factor grows linearly under simulation, where on a lattice it would grow exponentially with the dimension.

32.7 INSTANTANEOUS VOLATILITY AND ITS SHAPE

Section 29.5 distinguished flat volatility (one number per cap) from spot volatility, the piecewise-constant caplet volatilities that reproduce it. *The LMM needs a third object*: the *instantaneous* volatility $\sigma_k(t)$, a function of calendar time for each forward.

They are related by the requirement that total variance match:

$$\sigma_k^{\text{Black},2}\, T_k = \int_0^{T_k} \sigma_k(t)^2\, dt \tag{32.8}$$

The market determines the integral and not the integrand, so the shape of $\sigma_k(t)$ is a modeling choice with no caplet price to discipline it.

The standard choice is Rebonato's parametrization, in which volatility depends only on the time left to maturity, $T_k - t$, and not on where the calendar happens to stand. A specification

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with that property is **time-homogeneous**, and the term recurs below:

$$\sigma_k(t) = \big[a_R + b_R(T_k - t)\big]e^{-c_R(T_k - t)} + d_R \tag{32.9}$$

Four parameters produce a humped shape, which is what the data show: a forward rate two years out is more volatile than one thirty years out and also more volatile than one settling next week.

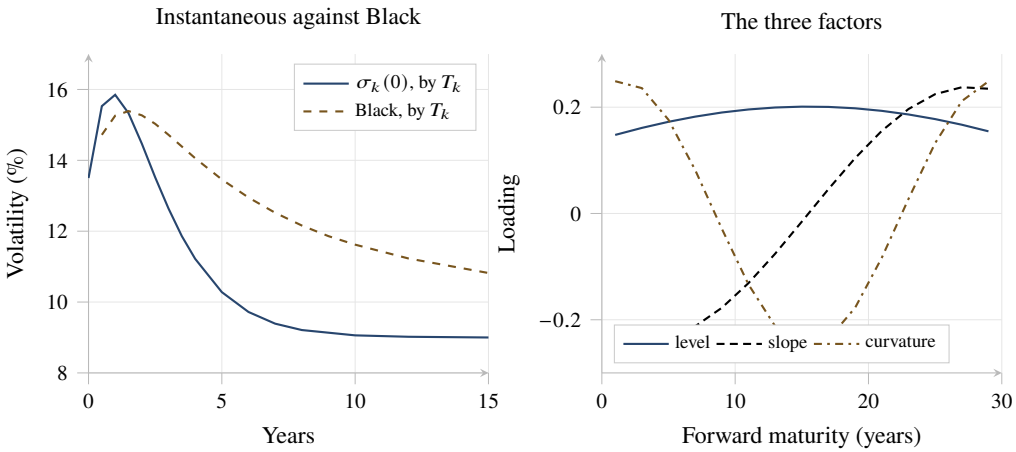


Figure 32.1: Left: equation (32.9) at $a_R = 0.045$, $b_R = 0.10$, $c_R = 0.75$, $d_R = 0.09$, and the Black volatilities it implies through (32.8). Right: the first three eigenvectors of Example 32.1 at $\beta = 0.05$. Illustrative parameters, not a calibration.

Figure 32.1's left panel shows why the distinction matters. Both curves are today's, and both are indexed by the caplet's own maturity: at T_k the solid curve is that forward's instantaneous volatility now, the dashed one its Black volatility. That the two can share an axis at all is owed to time homogeneity. As t runs from 0 to T_k the argument $T_k - t$ sweeps $[0, T_k]$, which is exactly the interval the solid curve covers, so the instantaneous volatilities a caplet meets over its life are the values the solid curve takes to the left of its own maturity, and the dashed curve is the running root-mean-square of the solid one. Were $\sigma_k(t)$ to depend on calendar time separately, the two would not be two views of one object, and the panel could not be drawn.

The instantaneous volatility peaks at 15.87% about ten and a half months out (at $1/c_R - a_R/b_R$) and falls to $d_R = 9\%$ in the tail; the Black volatilities it implies peak at 15.39% and are still at 10.8% fifteen years out, where the panel ends. Both curves tend to d_R in the end (a running root-mean-square of a function approaching 9% approaches 9%), but the dashed one takes far longer to get there, reaching 9.95% only by thirty years. Being a running average, the dashed curve is smoother, flatter, and shifted. A desk that reads a quoted Black volatility as though it were the instantaneous volatility of that forward has the wrong number by more than three volatility points around five years, and still by nearly two at fifteen.

The right panel shows the eigenvectors of Example 32.1: the first is positive throughout, the second changes sign once near fifteen years, the third twice.

A second specification is in wide use, and it is worth knowing because it makes the freedom in (32.8) concrete. Instead of a smooth function, take $\sigma_k(t)$ **piecewise constant** (one level per accrual period) and time-homogeneous. Write $\hat{\sigma}_i$ for the level in the i-th period counting backward from the reset, so that $\hat{\sigma}_1$ is the level over the last period before the rate fixes.

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Then (32.8) is a sum rather than an integral,

$$\sigma_k^{\text{Black},2}\,T_k \;=\; \sum_{m=1}^{k} \hat{\sigma}_{k-m+1}^2\,\tau_m \tag{32.10}$$

counting forward: period m runs for τ_m and carries the level with $k-m+1$ periods still to run. When the accruals are all of one length τ, the right-hand side is $\tau\sum_{i=1}^{k}\hat{\sigma}_i^2$, and differencing consecutive caplets inverts it:

$$\hat{\sigma}_k^2\,\tau \;=\; \sigma_k^{\text{Black},2}\,T_k \;-\; \sigma_{k-1}^{\text{Black},2}\,T_{k-1} \tag{32.11}$$

Each caplet in turn supplies one more level. No optimizer is involved, and the fit to the caplet strip is exact by construction. That is the attraction: the instantaneous volatilities come off the quoted strip with a subtraction, and a reader can do it by hand.

Only equal accruals reduce to that subtraction, and the condition is not a technicality: on a strip alternating one-year and six-month periods, using (32.11) anyway gets $\hat{\sigma}_2$ wrong by a third and $\hat{\sigma}_6$ by more than half. The general case costs one more line rather than an optimizer. The level with k periods to run appears in (32.10) for caplet k exactly once, in the $m=1$ term, so each caplet still brings exactly one new unknown and the bootstrap still runs one caplet at a time; what gets subtracted is the remainder of the sum rather than the previous caplet's total variance:

$$\hat{\sigma}_k^2\,\tau_1 \;=\; \sigma_k^{\text{Black},2}\,T_k \;-\; \sum_{m=2}^{k} \hat{\sigma}_{k-m+1}^2\,\tau_m$$

The cost is the mirror image. There are as many free levels as caplets, so nothing smooths the result, and nothing prevents (32.11) from returning a negative $\hat{\sigma}_k^2$, which it does whenever the quoted strip rises steeply enough that a later caplet's total variance falls short of an earlier one's. The parametric form cannot fail that way, and cannot fit exactly either.

The two specifications look very different as functions even when they agree on every caplet. On the strip implied by (32.9) at the parameters above, the bootstrap gives

Periods to reset	1	2	3	4	5	6	7	8	9	10
$\hat{\sigma}_i$, instantaneous	15.26	15.28	13.54	11.89	10.72	9.98	9.54	9.29	9.16	9.08
Caplet Black vol	15.26	15.27	14.72	14.06	13.46	12.95	12.52	12.16	11.86	11.62

in percent. The instantaneous levels fall to 9.1% where the caplet volatilities they reproduce fall only to 11.6%, because each caplet volatility averages the instantaneous one over the whole life. Figure 32.1's left panel is the same relationship for the smooth form.

The reason to prefer a time-homogeneous form is not fit but stability: the volatility structure such a model implies for one year from now looks like today's. A model that is not time-homogeneous predicts that the market's volatility surface will change shape by tomorrow—not because anything happened, but because the calendar advanced—and a desk hedging vega on such a model is hedging an artifact. Section 21.3 made the same criticism of local volatility's forward smile, and it is the same defect.

What the choice between the two forms is worth in money is a question with an answer, and Example 32.2 computes it on three products built to see the difference.

Example 32.2 What the shape is worth: three nonstandard caps

Two specifications that price every caplet identically must differ somewhere, and the products that find the difference are the ones whose payoff depends on the path of the forward-rate vector rather than on one rate at one date. Three standard ones, each a cap whose strike or exercise rule is not fixed in advance:

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- a **ratchet cap** resets each strike to the rate just observed plus a spread, $K_{k+1} = R_k + s$;
- a **sticky cap** resets it to the previous *capped* rate plus a spread, $K_{k+1} = \min(R_k, K_k) + s$, so the strike follows the rate down but rises by at most the spread each period;
- a **flexi cap** is an ordinary cap on which only the first N caplets that finish in the money pay.

Take annual periods on the grid of Section 32.4, so that the cap's ten caplets are the ones on F_1 through F_{10} and the terminal measure is $Q^{T_{11}}$. Put the forwards flat at 5%, take the correlation of Example 32.1 at $\beta = 0.05$, a spread of 25 basis points, and a flexi cap limited to five exercises. Calibrate both specifications to the same caplet strip (the parametric form to (32.9)'s parameters, the piecewise-constant form by the bootstrap (32.11) from the volatilities that form implies), so that every vanilla caplet price agrees to eight decimals. Then simulate both on the same draws:

	Parametric	Piecewise	Difference	Std. err.
Ratchet cap	1.8169	1.8065	−0.57%	0.04%
Sticky cap	3.5472	3.5369	−0.29%	0.03%
Flexi cap	2.5339	2.5333	−0.02%	0.07%
Vanilla cap	3.6729	3.6733	0.01%	0.02%

Per 100 of notional, from 12,000 paired paths at sixteen steps a year with common random numbers.

The vanilla cap is a control rather than a result. Both specifications reproduce its price by construction, so any difference there is numerical, and 0.01% against a standard error of 0.02% is what says the experiment is sound. The ratchet's −0.57% is therefore shape and not noise, at more than a dozen standard errors; the sticky cap's −0.29% likewise; the flexi cap's −0.02% is inside its own error: the flexi cap does not see the shape at all.

The control is not decoration. At four steps a year instead of sixteen, the vanilla cap reads +0.14% against the same 0.02% standard error (discretization bias, which is larger for a step volatility than for a smooth one, and now large enough to be detected), and the ratchet reads −1.60%. That is nearly three times the true figure, and nothing inside the ratchet's own number reveals it. This is Example 32.4's lesson met from the other side: a quantity computed as a difference between two models inherits both models' discretization error, and the only defense is an instrument whose answer you already know.

The size carries a lesson of its own. A ratchet at 0.57% of premium is inside most desks' bid–ask on a ten-year structure but not far inside it, and the exposure is one-directional: the smooth form is dearer on both path-dependent products. The reason is not where the volatility sits: each bootstrapped level is the root-mean-square of the smooth curve over its own period, so the two forms put the same variance into every period. It is how neighboring forwards move together within a period. The step function moves them in lockstep, so it correlates successive fixings slightly more, and a ratchet or sticky cap, whose strike is set by the previous fixing, is an option on the change from one fixing to the next. A desk that calibrates one way and hedges against a counterparty who calibrates the other keeps that difference as an unexplained mark.

32.8 CORRELATION: INSTANTANEOUS AND TERMINAL

Two distinct quantities carry the same name and are routinely confused.

Instantaneous correlation ρ_{ij} describes how the two forwards move together over an instant. *Terminal correlation* describes how their values at some horizon are related, and that is what a spread option pays on.

Example 32.3 Do the two differ enough to matter?

With the Rebonato form (32.9) at $a_R = 0.045$, $b_R = 0.10$, $c_R = 0.75$, $d_R = 0.09$, terminal

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correlation over a horizon H is

$$\frac{\rho_{ij}\int_0^H \sigma_i\sigma_j\,dt}{\sqrt{\int_0^H \sigma_i^2\,dt\int_0^H \sigma_j^2\,dt}}$$

Set the instantaneous correlation to 1 and see what the volatility shapes alone produce:

T_i	T_j	**Horizon**	**Terminal corr. at $\rho = 1$**
2	3	2y	0.9968
2	5	2y	0.9970
2	10	2y	0.9993
5	10	5y	0.9936
10	20	5y	0.9994

At most 0.6% of decorrelation, which invites being ignored. It should not be. A spread's variance is $V_i + V_j - 2\rho\sqrt{V_iV_j}$ with $V_i = \int_0^H \sigma_i^2\,dt$, and the subtraction nearly cancels at high ρ, so the surviving error is amplified: treating the instantaneous ρ as the terminal one understates the 5y/10y spread volatility by 1.27% at $\rho = 0.85$ and 4.18% at $\rho = 0.99$. An at-the-money premium moves with its volatility, so that is at or above the bid–ask—and worst where spread options are actually struck.

So in the LMM the correlation you calibrate is close to the correlation you get, which makes the correlation matrix the object to spend calibration effort on and the model's spread-option prices interpretable. It does not make the shape correction droppable: a multiplier of 0.994 is not a multiplier of one once the cancellation levers it.

The contrast with Chapter 31 sharpens too. A one-factor short-rate model has $\rho_{ij} = 1$ by construction and no offsetting shape effect either, so its terminal correlations are essentially 1 as well. The defect of Section 31.12 cannot be repaired with clever volatility structures; it takes more factors.

32.9 CALIBRATION IN PRACTICE

Caplet volatilities come first and are exact. Equation (32.8) determines the total variance of each forward from the market's caplet quotes, and the LMM reproduces them by construction. *No optimization is involved*, which is the model's chief practical attraction.

The volatility shape is then fitted subject to that constraint. The four parameters of (32.9) are chosen to fit swaption volatilities (which, unlike caplets, depend on the shape and on correlation), with the total variances held at their caplet values.

Correlation is fitted last and is the hard part. It can be estimated historically, parametrized (the $e^{-\beta|T_i-T_j|}$ form of Example 32.1 is standard), or implied from spread options where those are liquid. Historical and implied correlations differ systematically, and by more than the volatility analogs of Section 19.10. That should be expected. A **correlation risk premium** (a persistent gap between the correlation priced into options and the correlation the rates go on to show) arises for the same reason as a volatility risk premium, but far less trading can close it: volatility is bought and sold directly, through variance swaps and a deep listed options market, whereas correlation trades only through the dispersion trade of Chapter 22 (which ties up capital and carries other exposures) or through correlation swaps that barely trade. A gap between historical and implied correlation can therefore persist with no one able to arbitrage it away.

Then the correlation matrix is reduced to K factors, which changes it. A rank-K approximation to a full correlation matrix does not reproduce the original entries, and Example 32.1's table is exactly the statement of how much is lost.

LMM calibration is therefore one exact step, one well-posed fit, and one genuinely underdetermined estimate. A desk that reports the three separately is doing better than one that

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reports a single fitting error.

32.10 SIMULATION

The LMM has no lattice, so everything is Monte Carlo, and the drift (32.7) is state-dependent.

That state dependence rules out the exact scheme of Section 20.7, which is exact only because a geometric Brownian motion's drift is deterministic. What is used instead is an Euler step taken on the logarithm. Itô's lemma turns (32.6) into

$$d\ln F_k = \left(\mu_k(t) - \frac{1}{2}\sigma_k(t)^2\right) dt + \sigma_k(t)\, dW_k$$

and stepping that over $[t, t+\Delta t]$ gives

$$F_k(t+\Delta t) \;\approx\; F_k(t)\exp\left[\left(\mu_k(t) - \frac{1}{2}\sigma_k(t)^2\right)\Delta t + \sigma_k(t)\,\varepsilon\sqrt{\Delta t}\right]$$

It is an approximation in both coefficients: each is held at the value it has when the interval opens, and neither is constant across it, σ_k least of all, since Section 32.7 gives it a hump.

The two are not on the same footing, though. Because σ_k is deterministic, its approximation is removable: use the variance the model actually accumulates over the step, $\int_t^{t+\Delta t} \sigma_k(u)^2\, du$, in place of $\sigma_k(t)^2\Delta t$, and the diffusion becomes exact. The drift admits no such treatment, because μ_k depends on the forwards themselves and is not known until the step has been taken. One approximation can be integrated away and the other has to be corrected for, which is why a predictor–corrector exists for the drift and nothing analogous is needed for the diffusion.

Stepping the logarithm rather than F_k itself keeps the forward positive, since it is an exponential, so the failure of a direct Euler step never arises. With the variance integrated exactly, the frozen drift would be the only error left anywhere, and it is what the predictor–corrector removes. The comparison below keeps the step as displayed, with both coefficients frozen, so that it can measure the two errors side by side: the gap between the two schemes is the drift's error, and the movement as the step count rises is mostly the volatility's.

The standard concern is that freezing the drift introduces bias, and that the predictor–corrector of Section 20.7 removes it.

That concern deserves to be measured rather than repeated, and measuring it needs a product the model cannot price in closed form. A cap will not do: the LMM reproduces every caplet exactly by construction, so simulating one tests the simulation against an answer already in hand. A swaption is the simplest instrument whose value depends on the forwards jointly (on their correlations as well as their volatilities), so the model has no formula for it and it has to be simulated. It is also the trade Section 32.11 needs, which is why the same one appears there.

Example 32.4 Where the simulation error actually is

A one-year option on a five-year swap, five annual forwards flat at 3%, instantaneous correlation 0.85, Rebonato volatility as in Example 32.3. Four hundred thousand antithetic pairs were used.

Each simulated price is inverted through Black's swaption formula of Section 29.6 to give the Black volatility the third column reports. That assumes nothing about the swap rate: Black's value is monotone in volatility, so the inversion is a change of units, and it states the error in basis points of volatility rather than in currency. The last column then measures each figure against Rebonato's approximation, which Section 32.11 sets out; the approximation serves only as a fixed yardstick here, so what the table is about is the movement down the column rather than the distance across to it.

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Steps	Scheme	Implied vol	vs. approximation
4	Euler (frozen drift)	12.52%	−6.2 bp
4	Predictor–corrector	12.52%	−6.3 bp
12	Euler	12.53%	−5.2 bp
12	Predictor–corrector	12.53%	−5.3 bp
50	Euler	12.58%	−0.2 bp
50	Predictor–corrector	12.58%	−0.2 bp
200	Euler	12.58%	−0.1 bp
200	Predictor–corrector	12.58%	−0.1 bp

The Monte Carlo standard error is 2.2 basis points of volatility on each level, and rows at different step counts do not share draws, so a difference between two of them carries about 3.1. The difference between the two schemes at one step count is another matter: it shares its draws, and is measured to a thousandth of a basis point.

Two findings follow, and the second contradicts the usual advice.

Discretization bias is real and it goes away quickly. Four steps a year leaves the answer about 6 basis points low. That is only two standard errors of the difference, but the shortfall is not noise: it is what holding σ_k at its value at the start of each step does across the hump, and computing that effect directly, with no simulation, gives 7.7 basis points at four steps and 0.6 at fifty. Fifty steps closes the gap to inside the Monte Carlo error, and two hundred does not improve on fifty. A scheme that needs fifty steps a year is not an obstacle.

The predictor–corrector makes no measurable difference here. On shared draws the two schemes differ by 0.15 basis points at four steps a year and by 0.003 at two hundred—smaller than the bias it is supposed to remove, at every step count.

The reason is Section 32.5's third observation. Every term of (32.7) carries a product $\sigma_k\sigma_j$, so the drift is second order in volatility where the diffusion is first order, and the factor $\tau F/(1+\tau F) \approx 0.029$ damps it further. Over this option's one-year life the largest drift moves $\ln F$ by 0.0019, against a diffusion standard deviation of 0.153. Correcting a term that small cannot move the answer, and the residual bias is coming from the frozen volatility (the error that integrating the variance removes) rather than from the frozen drift.

The drift correction earns its cost when the drift is large (high rates, and many forwards between the simulated rate and the numeraire) and not otherwise. The accrual convention matters less than it looks: halving τ halves each term of (32.7) but doubles the number of terms over the same stretch of curve. A desk running a five-forward book at 3% that pays for a predictor–corrector has bought nothing, while one running a thirty-forward book in a high-rate currency needs it. This is Section 20.7's rule: a bias-reduction technique should be validated on the problem at hand, not adopted on the strength of the case that motivated it.

32.11 THE SWAP MARKET MODEL, AND WHAT THE INCOMPATIBILITY COSTS

Section 29.6 priced swaptions with Black's model applied to the forward swap rate, which assumes the *swap rate* is lognormal. Section 32.4 assumes the *forward rates* are lognormal.

Behind the first assumption is a model built exactly as the LMM is. Section 28.8 showed the forward swap rate $S(t)$ to be a traded asset divided by the annuity, and therefore a martingale under the annuity measure Q^A. Give it a deterministic volatility and

$$\frac{dS(t)}{S(t)} = \sigma_S(t)\, dW^A$$

which is the **swap market model**. A swaption is then priced by Black's formula exactly, in the same sense and for the same reason that the LMM prices a caplet exactly: each model takes the rate its own market quotes, makes that rate lognormal under the measure in which it is already a martingale, and so reproduces the quoting convention by construction rather than by calibration.

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That is the one-rate case, and it is all the argument below needs. The swap market model in full carries a family of *co-terminal* swap rates (swaps starting on different dates and all ending on the same one) under a single annuity measure, with drift conditions derived exactly as Section 32.2's were and with swap rates in place of forwards; it is Jamshidian's, and this chapter states it without developing it, because the derivation would repeat Section 32.2 with the letters changed while the LMM is the model more commonly implemented. Its natural application is the Bermudan swaption, whose exercise dates are precisely a co-terminal strip, so there the model's state variables are the exercise decisions themselves, which is the other route to the problem Section 32.12 takes in LMM terms.

Both cannot be exactly true. A swap rate is a weighted sum of forwards with stochastic weights, and a weighted sum of lognormals is not lognormal. This is a theorem, and it means the market's two standard quoting conventions are mutually inconsistent.

That statement usually ends the discussion. The useful question is how large the inconsistency is, and it has a number.

Example 32.5 Measuring the inconsistency

Rebonato's approximation expresses the swaption's Black volatility as a weighted average of forward volatilities, with weights

$$w_k = \frac{\partial S}{\partial F_k} \cdot \frac{F_k}{S}$$

For the swap of Example 32.4 these are 0.2120, 0.2058, 0.1998, 0.1940, 0.1884, *summing to 1.0000*, which is the check that the approximation is a genuine weighted average and not merely a fitted formula.

The approximation gives a Black volatility of **12.5818%**. Simulating at 800 steps gives **12.58%**, which is under a basis point away against a Monte Carlo error of about two.

So for this trade the incompatibility has no measurable price, and the effort spent on it in the literature looks out of proportion to its size.

Now change the trade, since a short option on a short swap is where an approximation has the least to do. A five-year option on a ten-year swap, ten annual forwards flat at 6%, instantaneous correlation 0.70:

	1y into 5y at 3%	**5y into 10y at 6%**
Rebonato approximation	12.5818%	9.0048%
Simulation, 800 steps	12.58%	9.00%
Error	under 1 bp	under 1 bp

Within the noise, the longer trade is no worse than the shorter one. Both errors are under a basis point of volatility, both are well inside the simulation's own error of about two, and neither is visible against a bid–ask measured in half-points. Four separate estimates were taken of each (two step counts under each of two numeraires), and they bracket the approximation rather than sitting to one side of it.

That the weights sum to one is an identity, not a normalization. The forward swap rate is homogeneous of degree one in the forwards (scaling every F_k by λ scales S by λ), so Euler's theorem gives $\sum_k F_k \, \partial S/\partial F_k = S$, hence $\sum_k w_k = 1$. This is why the check in Example 32.5 is informative: a weight vector that misses one signals an error in the discount-factor recursion rather than a modeling choice.

The approximation looks cruder than it is. It freezes the weights w_k at today's values, and the weights do move: more with more forwards, with longer expiration, and with larger τF, all three of which grow between the two columns. But the swap rate is a weighted average, so a weight that drifts up is offset by others drifting down, and the error enters at second order in those displacements rather than at first. Freezing a set of numbers that must keep summing to one is a much smaller sin than freezing a single one of them would be.

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So the swap/forward incompatibility is a theorem with no measurable price, at either end of the range tested. A desk may calibrate to caps and price swaptions with Rebonato's formula and not think about it again—which is what desks in fact do, and this is the calculation that says they are entitled to.

The general lesson is one this book has drawn twice before, and this is the first time it comes out the other way. Section 21.3 found local volatility's forward-smile error structural and large; Section 31.12 found the one-factor correlation error structural and a factor of four. Here the structural error is real as mathematics and negligible as a price—which is the most comfortable of the three answers and the rarest.

32.12 EARLY EXERCISE

A Bermudan swaption needs a continuation value at every exercise date, and Monte Carlo runs forward while the exercise decision needs backward induction. Section 20.11's Longstaff–Schwartz regression is the standard answer: regress realized continuation values on state variables at each exercise date, and use the fitted function as the exercise rule.

Three practical points follow, since this is where implementations go wrong.

The regression basis must span the state. With a high-dimensional forward-rate vector, the natural choices are the swap rate and the annuity, sometimes with a second swap rate to capture the curve's shape. Choosing too small a basis produces a suboptimal exercise rule and therefore a downward-biased price, which looks like conservatism and is not.

The estimate is biased in both directions and the biases are separable. Using the same paths to fit the rule and to price gives an upward foresight bias; using a suboptimal rule gives a downward bias. The standard discipline is to fit on one set of paths and price on another, which removes the first and leaves the second, so the reported price is a genuine lower bound. Upper bounds via the dual formulation (which values exercise with perfect foresight, less a martingale chosen to charge for that foresight) exist and are expensive.

And the whole apparatus is slower than a tree by orders of magnitude. Example 31.8's Bermudan takes milliseconds on a trinomial tree; the same product in a three-factor LMM with regression takes seconds to minutes. For a book of thousands of Bermudans revalued nightly with sensitivities, that difference decides the model, which is the reason short-rate models still run the callable book.

32.13 SENSITIVITIES, AND WHY THEY COST MORE THAN PRICES

A price is one number; a book needs its derivatives with respect to every calibration input, and in an LMM there are many.

The obvious method is **bump-and-revalue**: move one input by a small amount, revalue, and take the difference against the unperturbed price. It scales badly. With M curve points and V volatility points, a full sensitivity run is $M + V + 1$ simulations (the base, plus one for each input). At fifty curve points and thirty volatility points, that is eighty-one Monte Carlo runs for one product. That is why an LMM book's overnight risk run, not its pricing, is the binding computational constraint.

Common random numbers are not optional here. If the bumped and base simulations use different draws, the difference of two noisy estimates carries the noise of both, which can exceed the bump's whole effect, so the delta comes out smaller than its own standard error. Section 20.10 made the point for equity Greeks, where sharing the draws improves an estimate; in an LMM risk run it decides whether there is an estimate at all, and Example 32.6 measures the gap.

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Example 32.6 Why the draws must be shared

A swaption prices at 100.0000 with a Monte Carlo standard error of 0.0500 at 100,000 paths, and a one basis point bump to the curve moves it by 0.0480.

With independent draws the difference of the two runs has standard error $0.0500\sqrt{2} = 0.0707$:

$$\frac{\text{signal}}{\text{noise}} = \frac{0.0480}{0.0707} = 0.68$$

The delta is smaller than the uncertainty in the delta—the run has produced no information at all, and a risk report built this way is reporting sampling noise with a sign.

With common random numbers the two runs differ only through the bump, most of the sampling error cancels, and a standard error near 0.0004 is routine:

$$\frac{\text{signal}}{\text{noise}} = \frac{0.0480}{0.0004} = 120$$

The cost of not sharing the draws is not a factor of two; it is a factor of 31,250 in paths. Matching that precision with independent draws requires

$$\left(\frac{0.0707}{0.0004}\right)^2 \times 100{,}000 \approx \textbf{3.1 billion paths}$$

per bump, and there are eighty bumps. One line of implementation—seed the bumped run identically—buys what no amount of hardware can.

The same arithmetic explains why the input count is what kills the method rather than the path count. Adding a currency to the book does not make the simulation slower; it makes the number of simulations larger:

Curve points	Vol points	Bump-and-revalue	AAD
50	30	81 runs	~4 runs
80	45	126 runs	~4 runs
120	70	191 runs	~4 runs

At twelve minutes per pricing run, eighty-one runs is **16.2 hours** and does not fit in an overnight window; four is **48 minutes**. *The right column does not move*, which is the whole of the argument for adjoint methods.

Where the payoff permits it, Section 20.10's pathwise differentiation beats bumping: differentiating the simulated payoff with respect to the input along each path gives an unbiased estimator out of a single simulation, with no bump size to choose. *It fails for discontinuous payoffs* (a digital, or a barrier), where the payoff has no derivative to take, and the repairs are to smooth the payoff or to fall back on the likelihood-ratio method.

And adjoint algorithmic differentiation changed the economics of the whole exercise. AAD (Section 20.10) computes all $M + V$ sensitivities in a fixed multiple (typically three to five times) of the cost of one price, independently of how many inputs there are. *Eighty-one simulations become four*, which is a factor of twenty and is the reason models of this complexity are viable on a trading desk at all.

This deserves a place in a chapter on models rather than a footnote in one on computation. Section 9.1 noted that cost, not method, is the binding constraint on an XVA desk's exposure calculation; the same logic operates here. A model that cannot produce its own risk overnight will not be used, however correct it is, and AAD is what moved several models across that line after 2010. It is also why the LMM's cost relative to a tree, which Section 32.12 put at orders of magnitude, is smaller for sensitivities than for a single price.

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32.14 PRODUCTS THAT REQUIRE IT

The abstract case for more factors is Section 31.12. Here is the concrete list, because the model exists for these and for very little else.

CMS spread options and the steepener notes built on them. A coupon paying $\max(S_{10} - S_2 - K, 0)$, sold to investors who want yield and will be long the curve's slope to get it: the coupon pays only while the ten-year rate exceeds the two-year by more than K, which a merely flat curve already fails, let alone an inverted one. Example 31.11 measured what one factor costs here. These are the largest single use of the model.

Range accruals on two rates. A coupon accruing only on days when a rate (or the spread between two) sits inside a band. The payoff depends on the joint distribution at many dates, not on one date's marginal.

Ratchets and target redemption notes. A ratchet's coupon is capped relative to the previous coupon, which makes it path-dependent in the forward-rate vector; a **target redemption note (TARN)** terminates once cumulative coupons reach a threshold, which makes the termination date itself a functional of the path. Neither is a function of a single state variable at a single date, which is precisely the condition under which Chapter 31's tree stops being applicable.

Callable structured notes, which combine the above with the exercise problem of Section 32.12, and are the hardest products in the rate market for exactly that reason.

Who is on each side is worth stating, because it explains the market's shape. The buyers are yield-seeking investors: private banks, insurers, regional banks. In a callable note or a range accrual they are short optionality and are paid a coupon for it, but the steepener's buyer is long a CMS spread option, and the steepener is what sets the correlation market. The dealer is therefore short spread optionality and must hedge it, which means trading swaptions and CMS in the correlations of Section 32.8. A spread option is worth more the less the two rates are correlated, so the large and persistent one-way demand that structured-note issuance creates bids spread volatility up, and that is why dealers' implied correlations sit systematically below historical ones. It is the same asymmetry Section 19.10 identified in equity volatility, arising the same way.

32.15 SMILE

The LMM as stated is lognormal, so it produces no smile, and Section 29.8 established that rate markets have a pronounced one. Two repairs are in use.

The shifted or displaced LMM models $F_k + s$ as lognormal, which produces a skew and, by the same mechanism as Section 31.14's shifted model, accommodates negative rates. It is cheap and it is what most books use for the skew alone.

SABR-LMM gives each forward its own stochastic volatility with the SABR dynamics of Section 29.9, correlated across forwards. This is the state of the art and it is expensive: the parameter count multiplies, and the correlation structure now has to cover volatility-volatility and rate-volatility correlations as well as rate-rate.

The market's actual practice is worth recording because it is inconsistent and works. Swaptions are quoted through SABR per expiration-tenor cell (the volatility cube of Section 29.9), and the LMM is calibrated to those quotes. So SABR does the smile interpolation and the LMM does the term-structure dynamics, and the two are not a single consistent model. Section 21.12's judgment applies: a model used for interpolation and a model used for dynamics are being asked different questions, and requiring one object to do both has costs that are usually larger than the inconsistency.

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32.16 MULTI-CURVE, AFTER THE TRANSITION

Chapter 7 rebuilt the framework around a discount curve and one or more forecast curves. What does that do to a model whose state variable is a forward rate?

Less than it did to the short-rate models of Chapter 31, and for the opposite reason. A short-rate model models a risk-free rate and had to have the spread added to it. An LMM models whatever forward rates you tell it to, so a multi-curve LMM simply carries more of them (the forwards of each index's forecast curve, plus the discount curve), with the drift condition applying to each set under the chosen numeraire.

Two genuine changes follow.

The numeraire is now the discount curve's bond, so the drift (32.7) is taken with respect to the discounting curve while the forwards being modeled come from forecast curves. That separation is precisely what (32.2) ruled out: a forward curve and a discount curve carrying the same information, either determining the other. The identity still holds within each curve, but it no longer runs between them, so the two must be carried and calibrated separately. The correction is a covariance term between the two curves, and most implementations set it to zero by treating the basis between those two curves as deterministic. That is reasonable while the basis is small and stable, which the move to overnight rates largely made it (Section 33.9 traces the tenor basis shrinking to term premia and compounding mechanics once the credit component went), and it is wrong for anything whose payoff depends on the basis itself, such as the currency option inside a two-currency CSA that Section 9.6 prices as ColVA.

And backward-looking rates change what a forward is. Section 29.4's compounded-in-arrears rate is not known at the start of its period, so the "forward rate" for a SOFR period does not fix at the start of its accrual period: it keeps moving, as a martingale, until the period ends. The clean formulations of this (the forward-market model with backward-looking rates) were developed after 2018, and they show that the received LMM machinery survives with the accrual convention changed. The transition altered what is compounded, not how the model is built, which is a fair summary of its effect on this whole chapter.

32.17 CHOOSING A MODEL

The reader who skips the rest of the chapter should take this section.

Use a one-factor short-rate model when the payoff depends on the level of rates and the product has early exercise: callable bonds, Bermudan swaptions, mortgage books. Chapter 31's speed and natural treatment of exercise decide it, and Example 31.9's calibration warning is the price.

Use a two-factor short-rate model when the payoff has mild shape dependence and still needs exercise. It keeps the lattice, at the cost of a two-dimensional one.

Use an LMM when the payoff depends on several rates at once and on their joint distribution: CMS spread options, ratchets, range accruals on more than one rate, anything where Section 32.8's correlation matrix is the answer to the pricing question. Accept that early exercise will cost you regression and time.

Use SABR alone when the question is interpolation across strikes at one expiration, which is most of what a swaption desk does most days.

And the general principle, which outlasts all four: a model should be chosen by which of its errors you can afford, not by which is most realistic. Chapter 31's one-factor model is unrealistic in a way that costs a factor of four on a spread option and nothing on a callable bond. The LMM is realistic in a way that costs seconds instead of milliseconds. Neither fact is a defect; both are the specification.

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32.18 WHO USES THESE

Exotic rate desks at the large dealers, which is where the LMM lives. The products it exists for (CMS spread options, structured coupons on more than one rate) are dealer-intermediated and issued into structured notes, so the model's user and the product's ultimate holder are several steps apart.

Model validation groups, disproportionately. An LMM implementation has more places to be wrong than any other model in this book (drift, measure, factor reduction, regression basis, discretization), and the validation effort is correspondingly large. Example 32.4 is the kind of test that belongs in it.

Insurers with variable annuity books (investment-linked savings policies that carry a guaranteed minimum benefit), where the guarantee depends jointly on equity and rate paths and a consistent multi-asset simulation is unavoidable.

And academics and vendors, more than trading desks. *This should be said plainly*: the LMM is the most-written-about interest rate model and not the most-used one. Chapter 31's models price more notional every day. The LMM's importance is that it defines what the simpler models are approximating, and Example 32.5 is an instance—you cannot know that a structural inconsistency is worth under a basis point without a model in which it is absent.

32.19 WHERE THIS CAME FROM

Heath, Jarrow and Morton (1992) generalized Chapter 31's models by dropping the short rate altogether: model the whole forward curve, and derive the condition its drift must satisfy for no arbitrage. *HJM is a framework rather than a model*, and its practical difficulty is the one Section 32.3 identifies: no instrument is quoted on an instantaneous forward rate, so the object being modeled is not the object being quoted.

Brace, Gatarek and Musiela (1997), Jamshidian (1997), and Miltersen, Sandmann and Sondermann (1997) fixed that within a year of each other by modeling the rates the market actually quotes. The result was not only a usable model but a justification: it showed that the Black formula the market had been applying to caps and swaptions for years was consistent with an arbitrage-free term structure, which until then no one had established.

SUMMARY

HJM models every instantaneous forward rate at once, with today's curve as the initial condition, so Chapter 31's fitting problem is built in rather than solved. Once the volatility functions are chosen, no-arbitrage determines the drift, and every arbitrage-free term-structure model is an HJM model. The framework is not implemented directly because the process is generally non-Markov and so has no recombining tree; Hull–White is the volatility structure that happens to be Markov, which is why Chapter 31 could use a lattice at all.

The LIBOR market model fixes a tenor structure and makes each forward lognormal, so Black's caplet formula is reproduced by construction rather than by calibration. The number of factors it needs depends on the correlation structure a product is written on: one factor leaves much of that variance unexplained, and the often-quoted 99% comes from principal components of yield levels, not of the correlations a spread option depends on.

The market fixes each forward's total variance and not the shape of its instantaneous volatility, so Rebonato's humped time-homogeneous form is chosen for stability rather than fit. Terminal and instantaneous correlation differ by little, so the correlation you calibrate is essentially the correlation you get, and Chapter 31's one-factor defect needs more factors rather than cleverer volatility.

In simulation, the predictor–corrector is worth its cost only when rates are high. Forwards and swap rates cannot both be exactly lognormal, so the market's two quoting conventions are

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inconsistent, but the inconsistency is smaller than the simulation's own error: it is a theorem without a price. Smile enters through a displaced or SABR-based LMM, and in market practice SABR interpolates across strikes while the LMM supplies the dynamics. The move to overnight rates changed what is compounded, not how the model is built.

Sensitivities, not prices, are the binding constraint: bump-and-revalue costs a simulation per input, while adjoint algorithmic differentiation delivers all of them at a small multiple of the cost of one price. The products that require the model (CMS spread options, two-rate range accruals, ratchets, and TARNs) depend on several rates at several dates, and dealers' one-way issuance flow is why their implied correlations sit below historical ones. Early exercise costs the LMM its main advantage, which is why short-rate models still price the callable book. Choose a model by which of its errors you can afford, not by which is most realistic.

FURTHER READING

Heath, Jarrow, and Morton's paper is the origin of Section 32.2 and the original statement of why the drift is not free; it is harder going than the market-model papers and worth the effort once. Brace, Gatarek, and Musiela, and independently Jamshidian and Miltersen, Sandmann, and Sondermann, established the market model and its consistency with Black's formula, and the original papers are the place to see the historical point of Section 32.4 as it was first made. Litterman and Scheinkman is the source of the principal-component reading of Example 32.1, and is where level, slope and curvature were first named as the factors of a curve; it is short. Rebonato's books are the practitioner's reference on everything in Sections 32.7 through 32.9, he treats the volatility-correlation trade-off in calibration at length, and the parametric correlation forms of Example 32.1, including the long-end floor, are his. Hull and White take the other route through the same material: the piecewise-constant instantaneous volatility of Section 32.7, bootstrapped period by period off the caplet Black volatilities, is theirs, and the same paper relates forward-rate to swap-rate volatilities, which is the approximation Section 32.11 measures the simulation against. Brigo and Mercurio cover the algebra of the drift conditions and the SABR-LMM of Section 32.15. Andersen and Piterbarg are the reference for simulation and for the early-exercise machinery of Section 32.12, including the dual upper bounds this chapter only mentions. On backward-looking rates and the post-transition formulations of Section 32.16, the literature is recent and still settling; the paper of Lyashenko and Mercurio is the standard entry point.

PRACTICE QUESTIONS

32.1 Explain what the state variable is in an HJM model, why today's curve poses no fitting problem, and state the HJM drift condition in words together with what it says about the model's degrees of freedom.

32.2 Explain in what sense Ho–Lee and Hull–White are HJM models, why a general HJM model cannot be put on a recombining tree, and identify the volatility structure that makes one Markov.

32.3 Explain why the LIBOR market model reproduces Black's caplet formula by construction rather than by calibration, and why the forward rates cannot all be martingales under one measure.

32.4 Compute $\tau F/(1+\tau F)$ at $F = 3\%$ and at $F = 8\%$ with annual accrual, and explain the relevance to the drift.

32.5 Using Example 32.1, state the share of variance explained by one and by three factors at $\beta = 0.05$, and explain why the often-quoted figure of 99% for three factors is misleading in this context.

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32.6 Distinguish flat, spot, and instantaneous volatility, state which the market quotes, and explain why the market determines the integral of instantaneous variance but not its shape.

32.7 Explain why a time-homogeneous volatility parametrization is preferred, and relate this to Section 21.3.

32.8 Distinguish instantaneous from terminal correlation, state which one a spread option pays on, use Example 32.3 to give the maximum decorrelation produced by volatility shape alone, and explain why that result strengthens rather than weakens Chapter 31's criticism.

32.9 Describe the three stages of LMM calibration, state which is exact and which is underdetermined, and explain what is lost when a full correlation matrix is reduced to K factors.

32.10 Using Example 32.4, state the change in implied volatility between 4 and 200 time steps, explain why the predictor–corrector makes no difference there, and state the conditions under which it would.

32.11 Explain why forward rates and swap rates cannot both be exactly lognormal, explain the significance of Rebonato's weights summing to one, and use Example 32.5 to state the size of the inconsistency in each of the two cases and compare it with a typical bid–ask.

32.12 Compute the number of simulations a bump-and-revalue risk run requires for a book with forty curve points and twenty-five volatility points, and explain why common random numbers are essential rather than merely helpful.

32.13 State the condition under which pathwise differentiation fails and name one repair, then explain what adjoint algorithmic differentiation delivers and why its cost does not grow with the number of inputs.

32.14 Name four products that require a multi-factor model, state for each what makes a single state variable insufficient, and explain why dealers' implied correlations sit systematically below historical ones.

32.15 Explain why using the same Monte Carlo paths to fit and to apply an exercise rule biases the price upward, and why a suboptimal rule produces a lower bound rather than an error of unknown sign.

32.16 State which model you would use for a callable bond, for a CMS spread option, and for quoting a single swaption, with reasons.

FURTHER QUESTIONS

32.17 Derive the HJM drift condition for a one-factor model by imposing that discounted bond prices are martingales. Then verify that $s(t,T) = \sigma e^{-a(T-t)}$ reduces the resulting dynamics to a Markov process in a single state variable, and identify that variable.

32.18 Example 32.1 shows three factors explaining 78–89% of variance on a correlation structure, against the often-quoted 99% on yield levels. Reconcile the two figures precisely, determine which is relevant for a CMS spread option and which for a duration hedge, and state the general principle for deciding which decomposition a given problem needs.

32.19 Example 32.3 finds the volatility-shape channel contributes at most 0.6% of decorrelation under Rebonato's parametrization. Determine whether this is a property of the parametrization or a general feature: construct a volatility structure that maximizes the gap between instantaneous and terminal correlation, state how large it can be made, and judge whether such a structure is economically plausible.

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32.20 Example 32.4 shows discretization bias dominating drift-correction error. Identify the source of the residual bias, propose a scheme that would remove it, and estimate the number of time steps at which the remaining bias falls below the Monte Carlo standard error for a book of one million paths.

32.21 Example 32.5 finds the swap/forward incompatibility worth under a basis point of volatility on both a short and a long trade, although the weights w_k that Rebonato's approximation freezes demonstrably move over the option's life. Derive the approximation's error term explicitly, show that it is second order rather than first order in the weights' displacements, and explain what the constraint $\sum_k w_k = 1$ has to do with it.

32.22 Section 32.15 notes that the market lets SABR do smile interpolation and the LMM do term-structure dynamics, which is inconsistent. Construct the argument that this should be replaced by a single consistent model, construct the argument that it should not, and state what evidence would settle it.

32.23 Section 32.17 argues a model should be chosen by which of its errors you can afford. Apply this to a specific book (a portfolio of Bermudan swaptions and CMS spread options in one currency) and design the model configuration you would run, stating explicitly which errors you are accepting on which trades and how you would monitor them.

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Chapter 33
Swaps Revisited

Chapter 7 priced the plain vanilla swap and built the curves it needs. It also deferred almost everything else, because the non-standard swaps require the adjustments of Chapter 30 and the measure machinery of Chapter 28, and those had not been developed.

They have now, and this chapter is where the deferred instruments are priced. The organizing observation is that a non-standard swap is almost always a standard swap plus a convexity adjustment, and that the adjustment is computable, has a sign that can be reasoned about, and is large enough to matter.

There is a second reason to return to swaps here. The transition to risk-free rates changed some of these instruments profoundly and others not at all, and the pattern of what changed is instructive. Section 33.4 contains the sharpest case: a convexity adjustment the market spent twenty years computing became roughly sixty times smaller overnight, not because anyone solved it but because the underlying convention changed.

The chapter runs from the variations that need no new theory to the ones that need all of it. Sections 33.1 to 33.3 change the notional or the payment dates and price by discounting alone. Sections 33.4 to 33.6 are where convexity enters: in-arrears swaps before and after the transition, CMS swaps, and range accruals, which are a strip of digitals in disguise. Sections 33.7 to 33.10 are the swaps whose payoff references something else: an embedded option, a commodity, a basis, an inflation index. Sections 33.11 to 33.13 are the equity and total return swaps whose funding leg Chapter 27 made expensive, how a position is unwound, and who uses these.

33.1 AMORTIZING, ACCRETING, AND ROLLER-COASTER SWAPS

The simplest variation is a notional that changes over time. *An amortizing swap* steps the notional down, typically to match a loan being repaid; *an accreting swap* steps it up, matching a construction drawdown; *a roller-coaster swap* does both.

Pricing needs nothing new. The schedule is known in advance, so each leg is discounted as it stands.

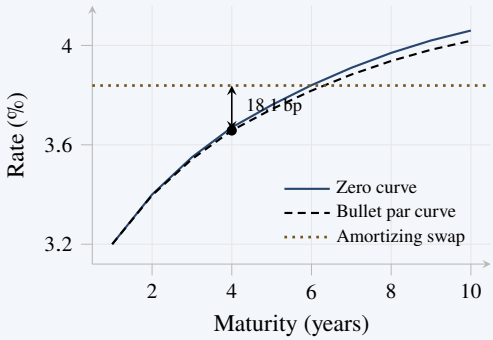
Example 33.1 Why the average-life approximation fails

Take a ten-year annual swap amortizing linearly from 100 to zero (notional 100 in the first year, 90 in the second, down to 10 in the tenth). The zero curve is upward-sloping and flattening, at 3.20, 3.40, 3.55, 3.67, 3.76, 3.84, 3.91, 3.97, 4.02 and 4.06 percent for one through ten years, annually compounded. The par rate is

$$\frac{\sum_k N_k \left[P(0,t_k) - P(0,t_{k+1})\right]}{\sum_k N_k \, P(0,t_{k+1})} = \mathbf{3.8386}\%$$

For comparison, against **bullet swaps**, ordinary swaps whose notional stays at 100 until maturity:

Bullet 10-year par rate	4.0187%
Bullet 5-year par rate	3.7432%
Bullet 4-year par rate	3.6576%
Amortizing par rate	**3.8386%**



The tempting shortcut is to price the amortizing swap as a bullet swap at its weighted average life. Weighting each payment date by the notional outstanding over the period ending on it, that life is exactly 4.0 years here, and the bullet par rate at four years is 3.6576%, an error of −18.1 **basis points**.

Which definition is used matters. Weight instead by principal repaid, which is what *weighted average life* means in the bond market, and the answer is 5.5 years, where the bullet rate is 3.7775% and the error is 6.1 basis points. Two standard summaries of one schedule differ by eighteen months, so the shortcut's error is not even a well-defined number until the convention is fixed.

Weighted average life summarizes the schedule by its first moment, and the par rate is a nonlinear function of the discount factors. Replacing a schedule by its mean is exact only when the function being applied is linear, and the par rate is not. The error grows with the curvature of the curve, which is why the shortcut is at its worst precisely when the curve is most interesting.

A desk that uses it is making the same class of error as Section 20.9's naive expectation: replacing a distribution by its mean inside a nonlinear function.

33.2 FORWARD-STARTING AND DELAYED SWAPS

A forward-starting swap (also called a delayed or deferred-start swap) begins at a future date. It requires no new machinery: the forward swap rate is read off the curve as in Section 7.3, and the value is zero at the forward par rate by construction.

What it requires is care about which curve. A ten-year swap starting in five years is priced from the five- and fifteen-year points, and an error of one basis point in the fifteen-year rate moves its rate by about *one and a half*, because the forward rate is a difference of two long rates divided by a shorter span, and differences amplify errors.

This is why forward-starting swaps are the instrument that exposes curve-construction disagreements between desks, and why the interpolation choices of Section 7.6, which look like housekeeping, are argued about.

33.3 COMPOUNDING SWAPS

In a compounding swap, the floating payments are not paid as they accrue; they are reinvested at a specified rate and paid as a single amount at maturity.

The reinvestment rate is the whole of the pricing question. Three conventions exist and they price differently.

Flat compounding reinvests at the floating rate itself. The leg's value is then the same as an ordinary floating leg's: the payments are simply rolled up at the rate that would have been earned by receiving and reinvesting them, so no adjustment arises. This is the common convention and the reason compounding swaps are usually treated as an operational rather than a valuation matter.

Compounding at the floating rate plus a spread is the standard convention where the leg

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carries a credit spread. Here the leg's value is *not* the same, because the spread is being compounded at a rate that would not have been earned. The correction is small but real, and it is the reason ISDA's compounding definitions run to several pages.

Compounding at a fixed rate makes the leg a portfolio of forward rate agreements with deterministic weights: straightforward, and rare.

The general point is the one Chapter 28 supplies. A payment moved from its natural date to another date is a change of measure, and whether an adjustment arises depends on whether the rate remains a martingale under the measure that goes with the new payment date. Flat compounding happens to preserve it. The other conventions do not.

33.4 IN-ARREARS SWAPS, BEFORE AND AFTER

This is the instrument the transition changed most.

The classic LIBOR-in-arrears swap pays the rate for $[T_k, T_{k+1}]$ at T_k, at the *start* of the period it applies to, rather than the end. Since LIBOR was set in advance, this was operationally possible and was done.

But the rate is a martingale under the T_{k+1}-forward measure, and the payment is under the T_k one. Section 30.4 derived the resulting adjustment, and Example 33.2 puts numbers on it:

$$E^{T_k}[F_k] \approx F_k + \frac{F_k^2\,\sigma^2\,\tau\,T_k}{1 + F_k\tau} \tag{33.1}$$

where F_k is the forward rate for the period, τ its length in years, and σ the rate's volatility.

Example 33.2 A convexity that the transition mostly removed

$F = 3.5\%$, $\tau = 0.5$, $\sigma = 25\%$:

T_k	Adjustment	Adjusted rate
0.5y	0.19 bp	3.5019%
1y	0.38 bp	3.5038%
2y	0.75 bp	3.5075%
5y	1.88 bp	3.5188%
10y	3.76 bp	3.5376%

The transition removed the mismatch rather than the volatility. A compounded-in-arrears SOFR rate is determined at the end of its accrual period and paid then, so the payment is no longer early and (33.1) has nothing to correct. What is left is the short delay between the end of the observation period and the payment date, typically two business days:

T_k	Adjustment at a 2-day lag
1y	0.0061 bp
5y	0.0304 bp
10y	0.0607 bp

At ten years the adjustment falls from 3.76 basis points to 0.06, about sixty times smaller.

That changes how the product should be classified. The rate replacing LIBOR is the overnight rate compounded in arrears, which is what the floating leg of Section 7.1's overnight indexed swap pays, so the instrument this section files as non-standard is now the standard one. Paying in arrears is the convention, and it is a rate fixed in advance that needs a special name and a separate construction, which is what Chapter 4's Term SOFR is. The convention traded places with its own exception.

It also means the old product is not extinct, only currency-specific. Paying a period's rate at that period's start requires knowing it at the start, so the structure needs a forward-looking term rate, and one survives wherever the transition left it standing. The Euro Interbank Offered

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Rate (EURIBOR) was never retired, so a EURIBOR-in-arrears swap carries the classic adjustment unchanged: 3.76 basis points at ten years, not 0.06. Term SOFR would serve in dollars, but the ARRC's recommendation limits it to business loans and the derivatives hedging them, not the swap market at large.

A convexity adjustment is a property of a convention rather than of a market: nothing about interest rate volatility changed, the timing convention did, and the adjustment followed it—into the currencies that kept a term rate. Twenty years of literature on in-arrears convexity was solving a problem the market later removed by changing the contract.

The new convention has a problem of its own, in a different place. Section 29.4's backward-looking caplet needs an adjustment the forward-looking version did not, because the rate is unknown until the period ends. The complexity moved rather than vanished, which is true of most convention changes in this book.

33.5 CMS SWAPS

A constant-maturity swap exchanges a fixed rate for a long swap rate (the ten-year rate, say) observed and paid periodically. It is the largest non-standard swap market and the one where the adjustments of Chapter 30 are unavoidable.

Example 33.3 What ignoring the adjustment costs

Pay the ten-year swap rate annually for ten years against a fixed rate. Forward ten-year rate flat at 3.40%, swap-rate volatility 25%, discounting at 3.2%.

Section 30.5's adjustment applies to each payment, growing linearly in its date:

Payment	Adjustment	Adjusted rate
1	2.72 bp	3.4272%
3	8.17 bp	3.4817%
5	13.62 bp	3.5362%
7	19.06 bp	3.5906%
10	27.23 bp	3.6723%

The fair fixed rate is the annuity-weighted average of the adjusted rates:

Fair fixed rate, with adjustments	3.5426%
Fair fixed rate, ignoring them	3.4000%
Difference	**14.26 bp**

The difference is fourteen basis points on the fixed leg, every year for ten years. A ten-year swap trades at a bid–ask of well under a basis point, so this is not a refinement—a desk quoting a CMS swap without the adjustment is not slightly wrong, it is not in the market.

And the adjustment depends on a volatility, which is the structural point. An ordinary swap is priced off the curve alone; a CMS swap requires a swaption volatility, so a linear-looking product carries a vega. That has three consequences.

The CMS desk is a volatility desk. Its hedges are swaptions, not swaps, and its risk report has a vega column.

The adjustment uses the whole smile, not one volatility. The replication of Section 30.5 integrates over strikes (the same $1/K^2$-flavored argument as Section 22.3), so the number depends on the smile's wings, which Section 29.9 noted are the least reliable part of the volatility cube.

And the sign is always the same. The adjustment is positive, so a payer of CMS always pays more than the naive forward. An investor buying a CMS-linked note is being sold that convexity, and whether they are being sold it at a fair price is not something the note's headline coupon reveals.

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33.6 RANGE ACCRUALS

A range accrual coupon accrues only on days when a reference rate sits inside a band. It is a strip of digital options, one per observation day, and it is the most widely sold rate-linked structured coupon after CMS.

Example 33.4 The enhancement is the short vega

A coupon accruing only when the three-month rate is inside $[2.50\%, 4.50\%]$. Forward rate 3.50%, volatility 25%, observation at two years.

Under the forward measure the rate is lognormal, so each digital is a Black-style probability. For a strike K,

$$P(R < K) = N(-d_2), \qquad d_2 = \frac{\ln(F/K) - \sigma^2 T/2}{\sigma\sqrt{T}}$$

$P(\text{rate} < 2.50\%)$	0.2192
$P(\text{rate} < 4.50\%)$	0.8126
P(**inside**)**—the expected accrual fraction**	**0.5934**

So a coupon of $3.50\%/0.5934 = 5.8980\%$, paid only on in-range days, has the same value as a vanilla 3.50% coupon. The headline enhancement is **240 basis points, 69% above the vanilla coupon**, which is what sells the note.

Now see what the investor gave up:

Realized volatility	Accrual fraction	Value of the 5.8980% coupon
15%	0.8322	4.9082%
20%	0.7012	4.1357%
25%	0.5934	**3.5000%**
35%	0.4422	2.6079%
50%	0.3095	1.8254%

At the pricing volatility the coupon is worth exactly the vanilla 3.50%, by construction. Above it the investor receives less, below it more.

The enhancement is the short vega, priced. A range accrual investor is selling volatility, and the 240 basis points is the premium for that sale spread over the coupon rather than paid up front. Nothing is being given away.

Two structural points follow.

The dealer is long volatility and hedges with a strip of digitals or, since digitals are hedged as tight call spreads, with a great many options near the barriers. Section 12.8's building-an-arbitrary-payoff argument is the machinery, and the practical consequence is that the dealer's hedge is concentrated exactly at the range boundaries, which is where the note's value is most sensitive and where the smile is least reliable.

And the investor's loss is asymmetric in a way the table understates. Volatility rising is correlated with the rate moving away from its forward, so the accrual fraction and the reinvestment opportunity deteriorate together. The coupon fails in the states where its holder most needs income, which is Section 27.9's wrong-way structure arriving in a retail product.

Figure 33.1's left panel is the picture a range accrual prospectus does not contain. The coupon is a decreasing function of the very quantity the investor is unlikely to be forecasting, and the curve is steepest near the pricing volatility.

33.7 CANCELLABLE AND EXTENDABLE SWAPS

A cancellable swap may be terminated early by one party; an extendable swap may be prolonged. Both are a vanilla swap plus a Bermudan swaption, and Chapter 31 priced exactly that option.

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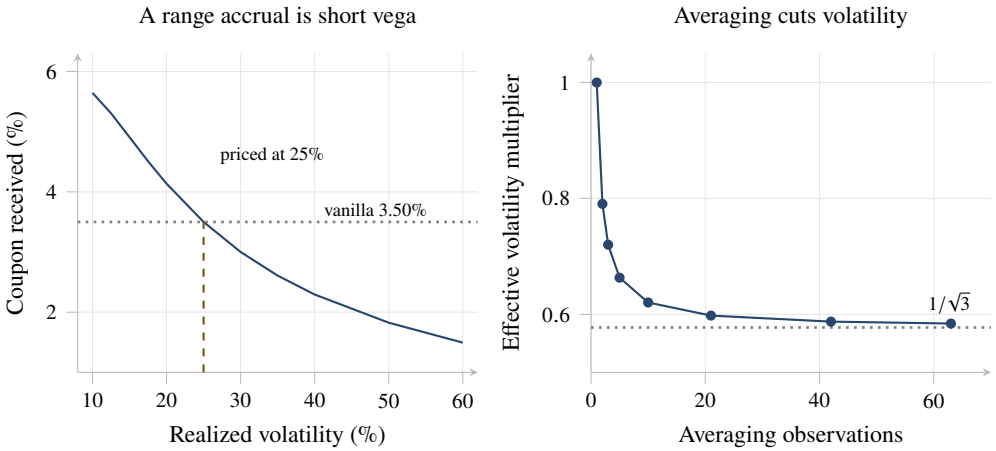


Figure 33.1: Left: Example 33.4's coupon against realized volatility. Right: the multiplier $\sqrt{(n+1)(2n+1)/6n^2}$ of Example 33.6.

Example 33.5 carries its value onto the swap by dividing it by the swap's annuity, which turns a one-off option premium into a running spread on the fixed leg.

Example 33.5 The Bermudan, annuitized

Take Example 31.8's six-year swap with annual cancellation rights from year one, whose Bermudan value was **1.6995** per 100 of notional.

Annuity of the six-year swap	5.272406
Bermudan option value	1.6995
Spread on the fixed leg	**32.23 bp**
Vanilla six-year par rate	3.9862%
Cancellable swap fixed rate	**4.3085%**

To first order: Example 31.8's option is struck at 4%, not at the cancellable rate of 4.3085% that this calculation produces, so the spread slightly overstates the option's value, and a full solution would iterate the strike to a fixed point.

A borrower whose debt may be prepaid or refinanced is the natural buyer, and it pays 32 basis points a year for the right to walk away—the whole economics of the instrument.

The reason to put this here rather than in Chapter 31 is what it does to the calibration problem. Example 31.9 showed the Bermudan's value spanning 13.1% across mean-reversion assumptions that all reproduce the calibrating European exactly. Annuitized, that becomes

$$0.131 \times 32.23 = \mathbf{4.2} \text{ basis points}$$

of ambiguity in a quoted swap rate, to first order.

Four basis points is well outside the bid–ask on a six-year swap, which is a fraction of a basis point. So the model ambiguity of Chapter 31 is not a theoretical concern that gets absorbed in spreads—it is several times the spread, and two dealers quoting the same cancellable swap from the same European volatilities can differ by more than the market they are quoting into.

Model risk is therefore a pricing problem and not only a risk-management one.

33.8 COMMODITY SWAPS AND THE COST OF AVERAGING

A commodity swap exchanges a fixed price for a floating one, and the floating one is almost always an average of daily prices over the delivery month rather than a single fixing, the one

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published price observed on a stated date. That convention exists because the underlying physical exposure is itself an average (a refiner buys crude every day), and it has a large valuation consequence.

Example 33.6 What averaging does to volatility

The average of a lognormal price over n equally spaced dates $T/n, 2T/n, \ldots, T$ has, in logs, variance

$$\frac{(n+1)(2n+1)}{6n^2}\,\sigma^2 T$$

which tends to $\sigma^2 T/3$ as n grows. This is exact for the geometric average and a close approximation for the arithmetic average a contract actually settles on. The effective volatility multiplier is therefore

n	Variance ratio	Vol multiplier
1	1.0000	1.0000
5	0.4400	0.6633
10	0.3850	0.6205
21	0.3575	**0.5979**
63	0.3413	0.5842
∞	0.3333	$1/\sqrt{3} = 0.5774$

Twenty-one fixings, at one every two and a half weeks over a year, already take the multiplier to within four percent of its continuous limit, so a modest number of fixings does essentially all the work continuous averaging would. What the multiplier does assume is that the averaging spans the option's whole life; an average over only the last month of a longer option cuts far less.

The pricing consequence is large. An at-the-money one-year option on an \$80 commodity with a 35% volatility, settled either on one fixing or on 21 fixings spread across the year:

	Effective volatility	Premium
On a single fixing at expiration	35.00%	11.1136
On the 21-fixing average	20.93%	6.6669

The averaging alone removes 40% of the option premium, with no change to the underlying's volatility.

Two implications follow, and the second is a trap.

A commodity option quoted on an average is not comparable with one quoted on a fixing, and comparing their implied volatilities without adjusting is a common and expensive error.

And the swap itself is not affected while the option is. A commodity *swap* settles against the average, but its fair fixed price is the average of the forwards, a linear function, so averaging changes nothing. It is only when optionality is attached, as in a commodity cap or a swaption on the average, that the multiplier bites. The distinction between what averaging does to a linear payoff and to a convex one is the same distinction Example 33.1 drew about weighted average life.

33.9 BASIS SWAPS, AND WHAT THEY PRICE

Section 7.7 introduced tenor basis swaps (three-month against six-month) and Section 7.8 the cross-currency basis. Here is what a basis swap is, in this Part's language.

A tenor basis swap prices the difference between two forecast curves. Before 2022 that difference was largely credit and liquidity: three-month LIBOR embedded less bank credit risk per unit time than six-month. After the transition to overnight rates compounded over the relevant period, the credit component is gone, and what remains is a much smaller basis reflecting term premia and the mechanics of compounding.

The tenor basis market therefore shrank rather than transformed, which is a rare case of a market being genuinely simplified.

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The cross-currency basis did not. It prices the relative scarcity of funding in two currencies, which is not a convention and does not go away. Section 7.8's account stands, and Section 9.6's ColVA is its valuation-side counterpart: a CSA permitting collateral in either currency contains an option on the basis, and that option is worth something.

The resettable notional

One feature of the cross-currency market deserves naming because it solves a problem the rest of this chapter would otherwise leave open.

A fixed-notional cross-currency swap exchanges principal at inception and at maturity at the same rate. If the exchange rate moves, the present value of the final exchange moves one-for-one with it, so the swap's mark-to-market grows with the move. Variation margin does cover that mark, but a mark that large means large and growing collateral flows, a large exposure over the margin period of risk, and a large SA-CCR add-on, all on a swap whose principal is paid once at the end.

A mark-to-market or resettable cross-currency swap fixes this by resetting one leg's notional at each payment date to the current spot equivalent, with the difference paid in cash. The mark is then reset to zero periodically rather than accumulating, and the swap's exposure profile of Section 9.1 is dramatically flatter.

Two consequences follow.

The resettable version prices differently, and the difference is a convexity adjustment. Resetting the notional means the interest paid on it depends on the exchange rate at the reset date, so the payment is a product of a rate and an FX level, which is the quanto configuration of Section 30.6. The correction is the covariance between the exchange rate and the interest rate, and it is why resettable and fixed-notional swaps quote at different bases.

And the choice is driven by Chapter 27, not by pricing. The resettable structure is now the market standard for interbank cross-currency swaps precisely because it cuts exposure at default and therefore capital. A convention adopted for capital reasons carried a valuation consequence with it, which is the same pattern as Section 33.4's in-arrears adjustment, running in the opposite direction: there a convention change removed a convexity; here one introduced it.

Example 33.7 puts both halves of that trade-off in numbers, because the exposure saving is large and obvious while the valuation cost is small and easy to omit.

Example 33.7 What the reset saves and what it costs

A five-year $100 million cross-currency swap paying semiannually, spot 1.10, a 3% euro forward rate with 25% rate volatility and 10% FX volatility. The table shows three of the ten payment dates; the average is over all ten.

The saving in margin and exposure is the reason it exists. Under a fixed notional, a move in spot changes the swap's mark by the move times the notional. Variation margin covers the mark, but the amount to be margined, and the exposure over the margin period of risk, grow with it. The table shows the size of the mark:

FX move	Mark, fixed notional	Mark after reset
+10%	$10.0m	≈ $0
+20%	$20.0m	≈ $0
+30%	$30.0m	≈ $0

Thirty percent of notional is a very large mark to margin daily, and Section 27.3's add-on prices it accordingly.

The valuation cost runs the other way and is not zero. Interest accrues on a notional that is

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itself a function of spot, so each payment is a product of a rate and an FX level and the forward rate must be adjusted by $-F\rho\,\sigma_S\,\sigma_F\,T_i$, where ρ is the correlation between the exchange rate and the euro rate, σ_S and σ_F are their volatilities, and T_i is the payment date:

Payment date	$\rho = 0$	$\rho = 0.3$	$\rho = 0.6$
1 year	0.00 bp	−2.25 bp	−4.50 bp
3 years	0.00 bp	−6.75 bp	−13.50 bp
5 years	0.00 bp	−11.25 bp	−22.50 bp
Average	**0.00 bp**	**−6.19 bp**	**−12.38 bp**

At a correlation of 0.3 the reset feature is worth about 6 basis points on the floating spread, the same order as the year-on-year inflation adjustment of Section 33.10 and two orders of magnitude larger than the post-transition in-arrears correction. It grows linearly in maturity and linearly in the correlation, and it vanishes only if rates and the exchange rate are independent, which for a currency pair whose basis is a funding variable is not a natural assumption.

So the two effects are not comparable in kind and both are real: tens of millions of mark to margin against single-digit basis points of value. The market chose the exposure saving, correctly, and then had to learn to price the residue—which is the general shape of every convention change in this chapter.

33.10 INFLATION SWAPS

Inflation swaps are the largest swap market this book has not yet priced, and one where the convexity structure is unusually clean.

A zero-coupon inflation swap exchanges a fixed compounded rate for the realized index ratio $I_T/I_0 - 1$, settled once at T. One payment, one observation ratio, no timing mismatch—no convexity adjustment. The quoted rate is the **breakeven inflation rate** directly: the fixed inflation rate at which the swap is worth zero, which is the market's price for inflation over the term.

A year-on-year swap pays $I_k/I_{k-1} - 1$ each year. Each payment is a ratio of index levels observed at two different dates and paid at the later one, and that is exactly the configuration Chapter 28 says produces an adjustment. Example 33.8 sizes it.

Example 33.8 Why the two inflation markets quote separately

Breakeven inflation flat at 2.50%, inflation volatility 1.2% in normal terms.

Year	Year-on-year convexity adjustment
1	0.00 bp
2	1.44 bp
5	5.76 bp
10	12.96 bp

Averaged over ten years: 6.48 basis points.

So a ten-year year-on-year swap does not strike at the ten-year zero-coupon breakeven, and the gap is not a quoting error. It is the accumulated convexity of ten ratio payments.

This is why the two inflation markets are quoted independently rather than one being derived from the other. Whenever a market quotes two instruments that look like transformations of each other, the difference is usually a convexity adjustment and is usually worth locating.

Two features of inflation swaps have no analog in rate swaps.

Indexation lag. The index used for a payment is typically the one published two or three months earlier, because the statistics office needs time. The swap therefore references a rate that is already partly known, which reduces its volatility near each reset and complicates the interpolation

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between monthly index points.

Seasonality. Consumer price indices have a pronounced and stable annual pattern. A year-on-year swap spanning exactly twelve months is nearly seasonality-neutral; anything else is not, and a curve built without a seasonal adjustment will misprice every off-anniversary payment in a predictable direction.

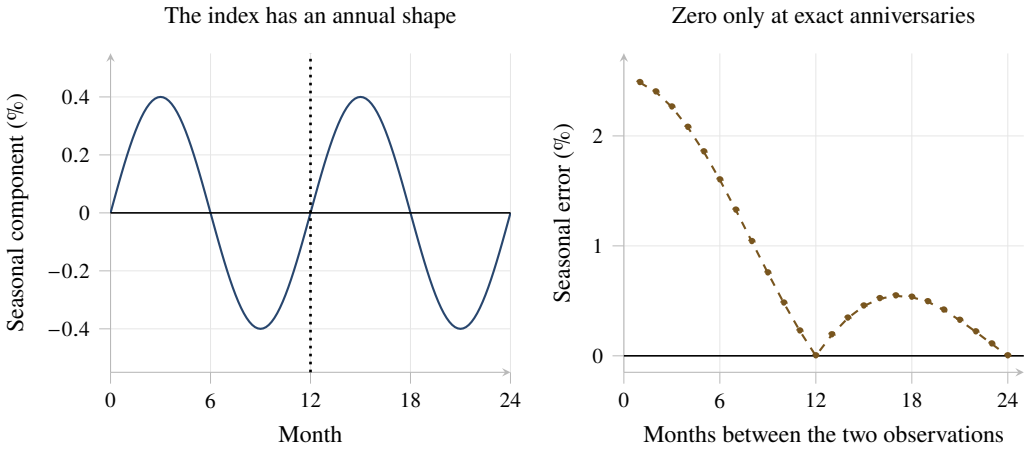


Figure 33.2: Illustrative seasonal amplitude of 0.4%, not market data. Right panel: worst case over the ending month, annualized.

Figure 33.2 is why an inflation curve is quoted by calendar month rather than by tenor. The left panel's pattern cancels exactly when the two observations are twelve months apart, which is what makes a standard year-on-year swap clean. The right panel is what happens when they are not: a swap observing three months apart carries a seasonal error of over two percentage points, annualized, and it is a *predictable* error rather than a risk: the same months produce the same sign every year. An unadjusted curve therefore does not misprice off-anniversary payments at random; it misprices them in a direction a counterparty can look up.

The users are worth naming, because the flow is one-directional and it shows in the price. Pension funds and insurers with inflation-linked liabilities are structural receivers of inflation, and there is no natural payer of comparable size; inflation-linked bond issuance by governments is the main one, and it is smaller. So breakeven inflation from swaps has tended to sit above breakeven from bonds (the gap between a nominal government bond's yield and an inflation-linked bond's real yield) in markets such as the sterling market, where pension demand is concentrated, and the gap is a demand imbalance rather than a forecast disagreement. This is the same structure as Section 7.9's pension demand compressing swap spreads, in a different market. In sterling, the retail prices index (RPI) is also to be aligned with CPIH in February 2030, as the government and the UK Statistics Authority announced in November 2020; the change reprices long-dated RPI swaps and index-linked bonds alike, and it falls on the pension schemes that hold them.

33.11 EQUITY AND TOTAL RETURN SWAPS

Section 7.12 introduced these and Section 27.14 showed what they made possible. The pricing is the easy part and the funding is not.

An equity swap exchanges the total return on an index or single stock for a floating rate plus a spread. Its fair spread is not a forecast of the equity's return (the equity leg's expected return under the pricing measure is the financing rate, by Chapter 13) but a function of the dealer's

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cost of holding the hedge:

$$\text{spread} \approx \text{funding} + \text{borrow cost} + \text{dividend leakage} + \text{balance sheet charge}$$

The borrow cost is the fee for borrowing stock to hedge a client's short; **dividend leakage** is the part of the dividends the dealer loses to withholding tax and so cannot pass on; the balance sheet charge is the cost of the capital the hedge position consumes. Not every term applies to every trade. A dealer hedging a client's long buys the stock, so it pays funding and loses the leakage but borrows nothing; one hedging a client's short borrows the stock and sells it, so it receives cash rather than needing it and holds no dividends to lose, and only the borrow cost and the balance sheet charge remain.

Each term is observable to the dealer and none is to the client, which is why equity swap spreads vary far more across dealers than swap rates do, and Example 33.9 shows how little a client learns from the quote.

Example 33.9 Four costs behind one number

A one-year total return swap on a $50 million basket yielding 2.0%. The dealer's costs:

Component	bp	Determined by
Funding above the risk-free rate	25	the dealer's own credit
Dividend leakage: 15% withholding on 2.0%	30	the dealer's tax residence
Balance sheet under the leverage ratio	15	Section 27.10's constraint
Quoted spread, synthetic long	**70**	
Stock borrow, if the client is short	40	securities lending supply
Balance sheet under the leverage ratio	15	Section 27.10's constraint
Quoted spread, synthetic short	**55**	

The short-side spread comes off the floating rate the client receives, and it is not the long-side spread plus the borrow: of the long side's three costs only the balance sheet charge survives.

The client sees one number and cannot decompose it. The 25 bp between dealers quoting 70 and 95 basis points can be assembled in more than one way (the cheaper dealer funding 10 bp below the other and a tax treaty sparing it 15 bp of dividend leakage, say, or the treaty plus a leverage ratio that binds it less), and the client cannot tell which, because none of the three is disclosed and all three move independently of the equity market.

The practical consequences run in both directions. The dividend term is a tax fact presented as a market price, so a client that could hold the shares directly and reclaim withholding is paying 30 bp for a service it does not need. And the balance sheet term is the one that moves without warning: it is a function of the dealer's leverage ratio headroom rather than of anything about the trade, so a spread can widen at a quarter-end for reasons the client will never observe. That is Section 27.10's constraint reaching a customer who has never heard of it.

Three uses account for most of the market, and they explain the spread's components.

Synthetic long exposure with leverage, which is the Archegos configuration of Section 27.14. Here the balance-sheet charge is the component that grows with the client's book: on a single trade it is the smallest of the three, as in Example 33.9, but once one client's exposure is large enough to move the dealer's leverage ratio it becomes the cost that limits how much the dealer will extend.

Synthetic short exposure, where the borrow cost dominates and Section 5.2's hard-to-borrow stocks carry spreads of hundreds of basis points.

And dividend and tax positioning, where the swap's treatment of dividends differs from direct ownership. This is the least discussed and historically the most controversial use, and the regulatory attention it has received in several jurisdictions is the reason many dealers now

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restrict it.

33.12 UNWINDS, NOVATIONS, AND COMPRESSION

A swap can be ended in three ways, and they are not equivalent.

Unwind: the two parties agree on a termination payment equal to the mark-to-market. Simple, and it requires the original counterparty's cooperation and pricing.

Novation: a third party steps into one side. This preserves the trade and changes the counterparty, and it requires consent because the remaining party is acquiring a new credit exposure, which is Chapter 27's subject appearing as a documentation clause.

Offsetting: enter an opposite trade. The market risk is flat and the counterparty risk is doubled, which is exactly the wrong trade under Chapter 27's framework and is done anyway when an unwind cannot be agreed.

Compression exists because of the third. Multilateral compression services tear up offsetting trades across many participants while preserving each one's net market risk. Section 25.18 recorded compression, alongside clearing and a genuine decline in activity, as one of the three forces that cut the CDS market's gross notional by roughly an order of magnitude; it has done the same in rates, and the notional statistics that headline the derivatives market are largely a story about compression rather than about activity.

The point for this chapter is that a swap's life does not end when its risk does, and the gap between the two is a capital cost under Chapter 27 and an operational cost always.

33.13 WHO USES NON-STANDARD SWAPS

Corporate borrowers take amortizing and accreting swaps because their debt amortizes: the non-standard swap exists to make the hedge match the exposure, and a bullet swap against an amortizing loan leaves a residual that grows over time.

Insurers and pension funds are the structural receivers in inflation swaps and the largest users of long-dated forward-starting swaps, for the reason Section 7.9 gave: their liabilities are long and they hedge duration they cannot buy in bond markets deep enough.

Structured note investors are the ultimate holders of the CMS convexity of Example 33.3, usually without seeing it separated from the coupon.

Hedge funds use equity and total return swaps for leverage, shorting, and, as Section 27.14 documented, for the position invisibility that the format provides.

And dealers sit in the middle of all of it, running the vega that CMS creates, the basis that cross-currency creates, and the balance sheet that equity swaps consume. The recurring pattern of this chapter is that a non-standard swap moves a risk from a client who does not want it to a dealer who prices it, and the price is a convexity adjustment the client usually cannot compute.

SUMMARY

A non-standard swap is almost always a standard swap plus a convexity adjustment, and the adjustment is computable, has a reasoned sign, and is large enough to matter. It arises wherever a payoff is nonlinear in the rate it depends on, or a rate is paid at a time other than its natural one. An amortizing swap priced at its weighted average life shows the first case, because the par rate is not linear in the schedule.

The classic LIBOR-in-arrears adjustment grows with maturity, while a compounded-in-arrears SOFR rate is determined and paid at the end of its period, so the adjustment all but vanishes: a convexity adjustment is a property of a convention, not of a market. The complexity moved rather than vanished, since Section 29.4's backward-looking caplet needs an adjustment its forward-looking predecessor did not. On a CMS swap the adjustment is many times the

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bid–ask spread, and because it depends on a volatility, a linear-looking product carries vega hedged off the whole smile.

Other products carry options inside them. A range accrual coupon is a strip of digitals whose enhancement is short vega, and it fails in the states where its holder most needs income. A cancellable swap is a vanilla plus a Bermudan, and Example 31.9's calibration ambiguity, quoted as a rate, makes model risk a pricing problem rather than only a risk-management one. A commodity swap settles against an average, which removes much of an option's premium and is neutral only inside a linear payoff.

Basis and inflation swaps show what a spread prices. Tenor basis priced bank credit before 2022 and prices much less now, while the cross-currency basis prices funding scarcity and did not shrink. Zero-coupon and year-on-year inflation swaps differ by a convexity adjustment, which is why the two markets quote independently, and pension and insurance demand can put swap breakevens above bond breakevens. An equity swap's spread is built from funding, dividends, borrow and balance sheet, every term observable to the dealer and none to the client.

Unwind, novation, and offsetting are not equivalent: offsetting flattens market risk and doubles counterparty risk, which is why compression exists and why headline notional describes compression more than activity. The recurring pattern is a non-standard swap moving a risk from a client who does not want it to a dealer who prices it, at a convexity adjustment the client cannot compute.

FURTHER READING

The ISDA definitions are the primary source for every convention in this chapter; the compounding and rate-cut-off provisions of Section 33.3 in particular repay direct reading, because the summaries in textbooks (including this one) elide distinctions that matter operationally. Hull's chapter on swaps is a short treatment of the non-standard varieties. On CMS, Hagan's papers on convexity conversion are the practitioner standard and are where Example 33.3's replication argument is developed properly. On inflation, Deacon, Derry, and Mirfendereski is the standard book and covers the seasonality and indexation-lag questions of Section 33.10 in the detail they require. On compression and the notional statistics of Section 33.12, the BIS semiannual survey's methodology notes explain what the headline notional counts. For the post-transition conventions, the market-practice recommendations published by the currency-specific working groups are the record of what was adopted, and they differ across currencies in ways the academic literature tends to ignore.

PRACTICE QUESTIONS

33.1 Using Example 33.1, state the amortizing par rate and the error from the weighted-average-life shortcut, and explain why the error grows with the curve's curvature.

33.2 Explain why a forward-starting swap amplifies curve errors, and why compounding at a spread produces an adjustment while flat compounding does not.

33.3 Using Example 33.2, state the classic in-arrears adjustment at ten years and the post-transition adjustment at ten years, and explain why the ratio between them says nothing about rate volatility.

33.4 Using Example 33.3, state the fair fixed rate with and without convexity adjustments and the difference, and explain why a product with a linear-looking payoff carries a vega.

33.5 Explain why a CMS convexity adjustment always has the same sign, and what that implies for the investor in a CMS-linked note.

33.6 Using Example 33.4, state the accrual fraction, the enhanced coupon, and the coupon's

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value if realized volatility is 50%, then explain what the enhancement actually is.

33.7 Explain why a range accrual's coupon fails in the states where its holder most needs income, and name the earlier section whose structure this repeats.

33.8 Using Example 33.5, state the ambiguity in the quoted cancellable swap rate implied by Example 31.9 and compare it with a typical bid–ask, then explain why this makes model risk a pricing problem.

33.9 Derive the variance ratio for an average of n observations, evaluate it at $n = 21$, and state its limit. Explain why averaging changes an option's value and not the swap's.

33.10 Explain why an implied volatility quoted on an average is not comparable with one quoted on a single fixing.

33.11 Explain why the mark-to-market of a fixed-notional cross-currency swap grows with the exchange rate, what that does to collateral flows and exposure, and what the resettable structure does about it.

33.12 Explain why a resettable cross-currency swap prices differently from a fixed-notional one, and identify the configuration from Chapter 28 that produces the correction.

33.13 Using Example 33.8, state the year-on-year adjustment at ten years and its average, and explain why the zero-coupon market needs no adjustment.

33.14 Explain why, where pension demand for inflation protection is concentrated, inflation swap breakevens can sit above bond breakevens, and why this is a demand imbalance rather than a forecast disagreement.

33.15 Distinguish unwind, novation, and offsetting, state what each does to market risk and to counterparty risk, and explain why headline notional statistics describe compression more than activity.

FURTHER QUESTIONS

33.16 Example 33.1 finds an 18.1 basis point error from the average-life shortcut. Derive the error's dependence on the curve's shape, determine the curve under which the shortcut is exact, and state how large the error can become on a realistically steep curve.

33.17 Example 33.2 shows the in-arrears adjustment falling by a factor of about sixty. Derive both adjustments from the change-of-numeraire argument, verify the factor, and determine the payment lag at which the post-transition adjustment would again become material.

33.18 Example 33.3 finds 14.26 basis points on a ten-year CMS swap. Determine how this scales with the CMS tenor, the swap's maturity, and the volatility level; then identify the configuration in which the adjustment exceeds 100 basis points and judge whether such a product would be sellable.

33.19 Section 33.5 argues that a CMS-linked note investor is being sold convexity without seeing its price. Design a disclosure that would make the convexity component visible, state what the issuer would have to reveal, and discuss why such disclosure is not standard.

33.20 Example 33.8 attributes the zero-coupon/year-on-year gap to convexity. Construct the arbitrage that would exist if the two markets quoted at the same breakeven, determine what would have to be traded to capture it, and explain what practical frictions prevent the two from being kept in line.

33.21 Section 33.10 says pension demand makes the inflation swap market one-directional. Model the resulting price impact, compare it with Section 7.9's swap-spread compression

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and Section 26.3's mortgage convexity flows, and state the general conditions under which a structural demand imbalance persists rather than attracting offsetting supply.

33.22 Example 33.4 shows the range accrual coupon falling from 4.91% to 1.83% as volatility rises from 15% to 50%. Derive the coupon's vega, determine the volatility at which it is largest, and explain why that volatility is close to the one the note is priced at rather than far from it.

33.23 Example 33.5 converts a 13.1% model ambiguity into 4.2 basis points on a quoted rate. Work out what a dealer should do about an ambiguity several times its bid–ask: consider reserving, widening, hedging the parameter, and declining the trade, and state the conditions under which each is the right response.

33.24 Section 33.9 says the resettable cross-currency structure was adopted for capital reasons and carried a pricing consequence. Derive the covariance correction between the resettable and fixed-notional versions, estimate its size for a major currency pair, and judge whether it is large enough that the capital saving might be partly illusory.

33.25 This chapter argues that a non-standard swap moves risk from a client who cannot price it to a dealer who can. Assess this as a description of the market: identify where the client is compensated for the information asymmetry, where they are not, and what a regulator concerned with the second could usefully require.

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